Prevention, treatment and lifecycle health types

Paper skeleton — first full draft, bullet form

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1 Introduction

[**TODO:** Deferred.]

1.1 Related literature

- Include a discussion here of the earnings dynamics literature and how they use the covariance matrix
- Also a discussion of health measurement in economics.
- On the stochastic specification we are close to Hosseini et al. (2022), who model log frailty in the PSID as a fixed effect plus an AR(1) plus white noise, with persistence 0.991. Our log(frailty + 1/31) on UKHLS reproduces their hump-shaped variance profile, and it behaves like θ throughout (Section 3.2).

2 Conceptual framework

[Condensed from `conceptual_framework.tex`, which holds every derivation, proof and simulated figure. Result and equation numbering here is local; cross-reference the note for proofs.]

2.1 Setup: deterministic type and stochastic noise

- Health of person i of latent type j at age a (age counted from a base age, not birth):

$$h_{i,a} = \bar{h}_{j,a} + u_{i,a}, \qquad \bar{h}_{j,a} \equiv H_j + \sum_{a'=1}^{a} d_{j,a'}. \tag{1}$$

  $(H_j, d_{j,a})$ is the *deterministic* component: the type's level at the base age and its expected change at each age. $u_{i,a}$ is the *stochastic* component, mean zero within type, otherwise unrestricted (could be transitory, persistent, or measurement error).
- Three kinds of information in a health history:
  1. what *stays* (the level, including permanent shocks that have landed),
  2. what is *in the moment* (transitory variation and measurement error),
  3. how one is *evolving* ($d_{j,a}$, the only component that predicts future *changes*).

Prevention targets the third; the first two are what is paid for, as treatment, once realised.

- Only what is carried from one wave to the next produces covariance between waves: the three components leave different traces on the covariance matrix, which is the basis for our identification.
- Two-by-two matrix of possibilities: dispersion is deterministic (across types) or stochastic (within type), and actionable or not. Deterministic and actionable is *targetable* prevention; stochastic and actionable is untargeted prevention, or treatment once the shock has arrived. Predictability of levels $\neq$ predictability of changes.
- Health types are thus the object of interest: they are the support of (H, d), and clustering estimates it. Between/within variance maps to type/individual.
- Link to other life course: With the base age at birth, the life-course accounts are second moments: critical period = $\mathrm{Var}(H)$, accumulation = $\mathrm{Var}(d)$, pathway = $\mathrm{Cov}(H, d)$; at a later base age the triple's evolution with the base age separates them.
- Throughout this section, we will use a linear illustration (constant type slopes, $d_{j,a} = d_j$): a random-coefficients growth model with V_H, V_d, C_{Hd} and noise variances v_a,

$$\begin{aligned} \bar{h}_a &= \bar{H} + a\bar{d}, \\ \mathrm{Var}(h_a) &= V_H + 2aC_{Hd} + a^2 V_d + v_a, \\ \mathrm{Cov}(h_s, h_t) &= V_H + (s+t)C_{Hd} + st\, V_d + \mathrm{Cov}(u_s, u_t). \end{aligned} \tag{2}$$

- A key issue that will come up again in relation to costs and when we bring to the data is what the units of health are. The covariance matrix treats health as cardinal, so transformations of a health index that are rank neutral can change the covariance matrix. We report the unbounded latent scale θ in the main text; its bounded expected-score twin h and two frailty indices are in the appendix, and these are all discussed in more detail in Section 3.

2.2 Policy: the treatment bill, and prevention as avoiding it

- Cost of poor health $c(h) = (1-h)^2$, full health normalised to one. The quadratic is the second-order approximation to any smooth increasing convex cost and makes the first two moments of health sufficient for every policy object below. It is the analytic illustration: the prevention exercises in Section 5 use the cost curve estimated in pounds on θ (Figure 1), with the general-cost versions of the results given below.
- Population cost at age a: mean gap plus between-type plus within-type variance,

$$\mathbb{E}[c(h_{i,a})] = (1 - \bar{h}_a)^2 + \mathrm{Var}_j(\bar{h}_{j,a}) + \textstyle\sum_j \pi_j \mathrm{Var}(u_{i,a} \mid j). \tag{3}$$

- Cost is treatment: $c(h)$ is what the health system spends on a person in health h, and the cost index of Section 3 measures exactly that (utilisation at unit costs). The object a policy acts on is the **treatment bill still to come** from age r,

$$C(r) = \sum_{a \geq r} \mathbb{E}[c(h_{i,a})] = \sum_{a \geq r} \Big[\underbrace{c(\bar{h}_a)}_{\text{mean path}} + \underbrace{\textstyle\sum_j \pi_{j,a} c(\bar{h}_{j,a}) - c(\bar{h}_a)}_{\text{type differences}} + \underbrace{\mathbb{E}[c(h_{i,a})] - \textstyle\sum_j \pi_{j,a} c(\bar{h}_{j,a})}_{\text{realised deviations}} \Big], \tag{4}$$

  which for the quadratic is (3) summed over ages: the mean gap, the between-type variance and the within-type variance. The third part is paid as treatment but is not preventable: it is the in-the-moment variation, including the "spike".
- One instrument. **Prevention** is ex ante and scales every future type change by $(1 - \mu)$ from age r, so health gained is μ times the type's preventable decline

$$\bar{h}_{j,r-1} - \bar{h}_{j,a},$$

and its return is the treatment cost that avoids.

Result 1 (The treatment bill; note Result 1)**.**

$$C(r) = \sum_{a \geq r} \left[(1 - \bar{h}_a)^2 + \mathrm{Var}(h_{i,a}) \right]$$

under the quadratic. Cross-sectional moments only; no identification problem. (The note's "return to treatment" $T(r)$ is $2C(r)$: restoring a share τ of every realised gap is worth τ times twice the bill.)

Result 2 (Return to prevention; note Results 2 and 3)**.** *For a uniform policy,*

$$P(r) = \textstyle\sum_j \pi_j P_j(r) = 2 \sum_{a \geq r} \Big[\underbrace{(1 - \bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a)}_{\textit{mean channel}} + \underbrace{\mathrm{Cov}_j\big(1 - \bar{h}_{j,a},\, \bar{h}_{j,r-1} - \bar{h}_{j,a}\big)}_{\textit{between-type channel}} \Big], \tag{5}$$

The between-type channel splits as $\mathrm{Cov}_j(1 - H_j,\, \bar{h}_{j,r-1} - \bar{h}_{j,a}) + \mathrm{Cov}_j(H_j - \bar{h}_{j,r-1},\, \bar{h}_{j,r-1} - \bar{h}_{j,a}) + \mathrm{Var}_j(\bar{h}_{j,r-1} - \bar{h}_{j,a})$*: the pathway, persistence and accumulation margins.*

By type, $P_j(r) = 2 \sum_{a \geq r} (1 - \bar{h}_{j,a})(\bar{h}_{j,r-1} - \bar{h}_{j,a})$*: gap times preventable decline.*

[**TODO:** Reformulate this result around the returns by type - gap times preventable decline. Explains why we want the evolution of the mean by type.]

- For a general increasing convex cost $c(\cdot)$ the bill is (4) as written and the prevention return keeps its form with the derivative as the weight: $P_j(r) = \sum_{a \geq r} \mathbb{E}[-c'(h_{i,a}) \mid j]\, (\bar{h}_{j,r-1} - \bar{h}_{j,a})$, to first order in the policy intensity; the quadratic case reproduces Results 1 and 2. $P(r)/C(r)$ is the share of the remaining bill a unit of prevention avoids.
- Prevention averts future gaps but must be assigned ex ante; the uniform policy pays the population-average preventable decline and its between-type channel is bounded by deterministic heterogeneity. What it can reach at most is the first two parts of (4): the realised deviations are treated, not prevented.
- Measurement error sits in the third part of the bill and in no prevention return; Case 4 below removes it from the type dynamics.

Result 3 (Targeting; note Results 5–7 and Lemma C.1)**.**

- *Any allocation $\mu(z)$ with the same average intensity saves* $\mathrm{Cov}(\mu(z), \mathbb{E}[P_i \mid z])$ *over the uniform policy, at most* $\mathrm{sd}(\mu)\, \mathrm{sd}(\mathbb{E}[P_i \mid z])$*: a signal is worth the return dispersion it reveals.*
- *We can consider two forms of targetting:*
- *Targeting on initial health $h_{i,0}$ has a gap channel and a compounding channel, both cells of the first row of the covariance matrix, attenuated by the level/noise share λ_0.*
- *Targeting on type* $\max_j P_j(r) - \sum_j \pi_j P_j(r)$.

In the linear case of Equation 2, $\bar{h}_{j,a} = H_j + a\, d_j$ the between-type channel is $(a - r + 1)(C_{Hd} + aV_d)$, so

$$P(r) = 2 \sum_{a \geq r} \left[(1 - \bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a) + (a - r + 1)\, C_{Hd} + a(a - r + 1)\, V_d \right] \tag{6}$$

We can (loosely?) think of the policy objects in terms of the moments that they require. *[In Leuven, we labelled these P(i)-P(iii), but I think I maybe misremembered things a bit here as I couldn't quite make things match up. There's a lot more on this in the conceptual framework document, so I need to properly revisit that.]*

	Policy object	Moments it needs
P(i)	$C(r)$, and the mean channel of $P(r)$	the mean path $\bar{h}_a$ and the cross-sectional variance $\mathrm{Var}(h_a)$: Figure 2 panels (a) and (b)
P(ii)	the accumulation margin of $P(r)$; the timing profile $r \mapsto P(r)$	V_d (with C_{Hd}): the curvature of the surface
P(iii)	the pathway margin; targeting on a level or type	C_{Hd}, locally $C_{Hd}(a) \equiv \mathrm{Cov}_j(\bar{h}_{j,a}, d_j)$, whose sign says diverging or converging: Figure 2 panel (c). For type targetting also need $P_j(r)$ across types, i.e. the clusters of Section 4

- If we've already shown/foreshadowed Figure 2, we could explain that through (6): an accelerating mean raises the mean channel for later starts; a V_d that falls with age puts the accumulation margin early; a $C_{Hd}(a)$ that changes over time works through the pathway margin, and the sign of C_{Hd} is crucial for targetting on initial health.
- *[In the data the mean channel dominates $P(r)$ and the between-type channel is small . The between-type moments matter for* targeting *far more than for the uniform return, so we could organise this section more around that: the bill and uniform prevention priced by the cross-section, targeting priced by the types.]*

2.3 Identification

- Every observed covariance is a deterministic (i.e. between-type) part plus a noise part, $\mathrm{Cov}(h_s, h_t) = \mathrm{Cov}_j(\bar{h}_{j,s}, \bar{h}_{j,t}) + \mathrm{Cov}(u_{i,s}, u_{i,t})$. The returns need the deterministic part but we only observe the sum.
- Start with some broad assumptions:
  - A1 window linearity ($d_{j,a} = d_j$ over the waves used),
  - A2 orthogonality of the noise to (H, d),
  - A3 a noise class that puts some restrictions on the covariance matrix
- In the linear case
$$\mathrm{Cov}(h_s, h_t) = V_H + (s+t)C_{Hd} + st\,V_d + \mathrm{Cov}(u_s, u_t). \tag{7}$$
which means that on the diagonal $\mathrm{Var}(h_a) = V_H + 2aC_{Hd} + a^2V_d + v_a$
- If the noise variances v_a are unrestricted the variance profile can never rule out pure noise.
- With stationary noise, $\mathrm{Var}(h_a)$ can identify (C_{Hd}, V_d) but never split V_H from σ_u^2. Identification thus needs restrictions on how the noise variance moves with age and at least some off-diagonals.

[I am not super happy with "spike" here (Claude came up with is) but I am also not really happy with "measurement error" either as the shock could also be a genuine transitory health shock...]

Proposition 1 (Identification within the AR-plus-"spike" class; note Proposition 1)**.**
Assume A1–A2 and $\mathrm{Cov}(u_s, u_t) = \rho^{t-s}v_s + \mathbf{1}\{s = t\}\sigma^2_{\varepsilon,s}$ *for* $s \le t$, $\rho \in [0, 1)$*: a carried component with memory* ρ *and variances* v_s*, plus a one-period spike. Then*
(i) five waves identify V_H, C_{Hd}, V_d, ρ *and both noise variances at the waves used;*
(ii) four waves are sufficient without the spike (Case 3) or with both noise variances stationary

(Case 4);
(iii) three waves are sufficient without the spike absent and stationary variance, for any ρ (Cases 1–2);
(iv) two waves are sufficient if in addition $C_{Hd} = 0$ and the noise is memoryless.

- The forces underlying , one per region of the matrix. This is the precise content of I(i) and I(ii):

I(i) **Diagonal** $V(a)$: every component adds variance. Stationary noise cancels in differences and leaves C_{Hd} (slope) and V_d (curvature). Fails when v_a drifts.

I(ii) **First row** $C_0(a)$: the deterministic part is the line $V_H + aC_{Hd}$, carried noise a decay $\rho^a v_0$; a line cannot mimic a decay, which splits the level from the noise and measures the memory; the row's slope beyond the noise is C_{Hd} itself, locally $C_{Hd}(s)$ for the row anchored at s. A one-period spike is invisible to the row and is read as the anchor's excess over the row's back-extrapolation. Limit: $\rho \to 1$ merges permanent shocks with the level.

I(iii) **Interior**: $st\, V_d$ needs both indices to move; delivers V_d when the diagonal cannot be read, overidentifying tests otherwise.

- *[There are a whole bunch of cases in the conceptual framework note, but we certainly don't want to go through all of them here, so we should think about what we want to include here. At least one or two are probably useful to help illustrate the identification (but maybe they can also wait until Section 3... In the illustrative cases on real data, the sign of the small C_{Hd} does depends on the noise treatment (although the gradient from positive to negative is relatively consistent.]*

- The linearity assumption is local, so A1 holds only over a window. But given this local linearity, $C_{Hd}(a) \equiv \mathrm{Cov}_j(\bar{h}_{j,a}, d_j)$ and so the deterministic variance increases by $C_{Hd}(a) + C_{Hd}(a+1)$ per year: a variance profile that flattens requires $C_{Hd}(a)$ to change sign, which the global linear model cannot produce.
- The between-type channel of $P(r)$ is the exposure-weighted sum $\sum_{a \geq r}(a - r + 1)C_{Hd}(a)$ and so depends crucially on C_{Hd}.
- Everything can be represented as person-level OLS of h on $(1, a)$, except for the noise correction which is case-specific (note Appendix E).

- [**TODO:** We should decide which noise class to focus on. The data generally pushes us towards a spike and pretty high ρ (maybe the high rho is also making up for some individual fixed effect though?). This would be Case 4 getting close to its $\rho \to 1$ limit. We do have a random walk case in the conceptual note (Proposition F.1 random walk plus spike), but here the diagonal becomes useless which sort of undercuts our identification forces...]
- [**TODO:** For the covariance surface, a permanent shock and a type level are the same thing, all bundled into H. The clustering models don't really have this as they require everyone in the cluster to have transitory noise around a shared intercept (related to your point about insufficient power to identify all clusters)... Is the AR(1) a "noise within class", or is it part of "what stays"?]
- [**TODO:** Should maybe already mention selection here as every row conditions on being observed later (and return to this with the multidimensional model).]

3 Data and motivating descriptives

3.1 Data and the health measure

Data:

- Understanding Society (UKHLS), waves 1–15, 2009–2024 (University of Essex, Institute for Social

and Economic Research, 2025); adults aged 20–90. The SF-12 is asked every wave; the impairment questions every wave; the choronic conditions are asked at a person's first interview and updated new diagnoses (but do not register if the condition is no longer present).

- We choose these variables as they are consistently reported over the UKHLS and they all focus on physical health (mental health displays quite different lifecycle dynamics so we look at that separately in Section 4.3).[1]
- 498,424 person-waves on 84,759 people aged 20–90 are complete on the SF-12 physical items. Of these, 387,599 person-waves on 69,958 people (5.5 per person) are also complete on the condition inventory, and the graded-response model is estimated and scored on that sample. The lifecycle analysis restricts further to people with at least three observed ages between 20 and 89, one row per person-age: 334,194 person-ages on 38,963 people.[2]

[**TODO:** I have (obviously with Claude's help) replicated most of this in the PSID as well, so we can look at those as well. I will send this as a separate document. Generally most things carry over - θ looks very similar on both, but h has variance slowing between 60 and 70 but then accelerating after 70...]

Table 1 shows the underlying health variables that we use for our main health measure. The first four are the SF-12 physical compoments of the commonly used PCS metric; the next three are grouped counts the UKHLS impairment areas that relate to physical limitations; the last counts diagnoses of chronic conditions. None of the items is tied to employment: the role-limitation items ask about problems "with your work or other regular daily activities", and the others make no reference to work. Two checks confirm that the measure is not picking up employment status.[3]

Table 1: Underlying health variables

Item	What it records	Categories
GH	self-rated general health	5
PF	moderate activities and climbing stairs	5
RP	accomplished less, limited in kind of work or other activity	9
BP	pain interference with normal activity	5
LIM_FL	functional limitations: mobility, lifting, dexterity, co-ordination	5
LIM_SELF	self-care: personal care, continence	3
LIM_SENS	sensory: hearing, sight	3
COND	ever-diagnosed physical conditions, 0–5 and 6 or more	7

[Maybe here (or back in the introduction) we can already discuss that there is no single cardinal health

[1] We test for local dependence with Yen's Q_3 and find no evidence that any of these variables are principally capturing a second factor.

[2] UKHLS interview dates within a wave spread over up to two years, so a person can be interviewed in two consecutive waves at the same integer age. 20,114 such person-age rows are collapsed to one by averaging the scores and keeping the earlier wave. The K-means types and the Bayesian mixtures are fitted on exactly these rows.

[3] First, at the state pension age of 65 every item improves by about 0.05 pooled standard deviations (a linear trend either side of 65 on ages 58–72, person-clustered t between 2 and 5), and the role-limitation testlet's jump, 0.047, sits in the middle of the range, so retirement does not move the work-related items specifically. Second, at ages 25–64 the employed cost about a third less than the non-employed at every level of health, but the shape of the cost–health curve is the same: in a quadratic in θ interacted with an employed indicator, the interactions with the slope and the curvature have $|t| \leq 2.6$ against everyone else and $|t| \leq 1.2$ once the long-term sick are excluded from the comparison (`descriptives/38_paper_employment_checks.py`). We find no evidence that retirement changes the way people answer these questions.

measure and so, arguably, any monotonic transformation of a health measure is a valid health measure [see Cunha and Heckman (2008) on test scores; Cunha et al (2021) on "Cardinality"; Bond and Lang (2013, 2019) for test scores and happiness scales]. However, if we want to use the covariance matrix to identify dynamics, then the scale does matter as these covariance are cardinal. So what we would need is some cardinal anchor and fortunately our framework gives us two obvious candidates: the mapping to costs and the mapping to interventions. Given that the focus of this paper is the potential returns to prevention rather than an evaluation of specific interventions it probably makes sense to focus more on the costs (and we can construct a proxy measure for these from the data on utilisation). If we had constructed the framework around mortality or QALYs or employment/productivity then this would imply an alternative anchoring.]

To construct a continuous health metric from these underlying variables we use a graded-response model Samejima (1969) - a one-factor model for ordinal indicators (with continuous and unbounded indicators this would be equivalent to the linear factor models common in economics, e.g. Cunha et al., 2010; Attanasio et al., 2020; Agostinelli and Wiswall, 2025). Let $Y_j \in 1, ..., k_j$ be the response to item j, which has k_j ordered categories. Given the respondent's underlying health state is θ, the probability of answering at or above category c is logistic in θ,

$$P(Y_j \geq c \mid \theta) \;=\; \frac{1}{1 + \exp\{-a_j(\theta - b_{j,c-1})\}} \tag{8}$$

where a_j and the $k_j - 1$ thresholds $b_{j1} < \cdots < b_{j,k_j-1}$ are estimated item-specific parameters. The latent variable θ thus capture the respondent's underlying health that determines their expected answers across all items. The item parameters are estimated by marginal maximum likelihood. Latent health is assumed standard normal in the pooled sample, which fixes the scale; θ is integrated out of the likelihood of each answer pattern over a grid of 121 points, and the discriminations and thresholds are chosen to maximise the resulting likelihood by EM, over the 28,360 distinct answer patterns. Scoring is a second step with the item parameters held fixed: each person-wave's posterior over θ is the standard normal prior times the likelihood of its eight answers, and the reported θ is the mean of that posterior, with its standard deviation kept alongside. Each person-wave is scored on its own, with the same prior, so a respondent's other waves and age do not inform their score.

[we never take the posterior uncertainty around θ into the clustering models, which it might be useful to do (and I think relatively straightforward).]

The model allows thresholds the be unevenly spaced and for different items to separate observations in different parts of the distribution. In our case PF, RP and LIM_FL discriminate the most (i.e. have the highest values of a_j) and their thresholds sit between -2.5 and 0, so they separates people in the unhealthy half of the distribution. GH and BP discriminate less but their thresholds are further up in the distribution so they separate out the healthy half. LIM_SENS and COND have the lowest values of a_j and their thresholds go down below -4.0 so they prevent the very sick from piling up at a floor.

The underlying health factor θ is designed to be continuous, unbounded and distributed $N(0, 1)$. However, we can map it back into an expected share of the measure's total scale, i.e. summing overall all J items as a share of the maximum possible score(34). We define this as

$$h(\theta) \;=\; \frac{\sum_{j=1}^{J} \sum_{c=2}^{k_j} P(Y_j \geq c \mid \theta)}{\sum_{j=1}^{J} (k_j - 1)}, \qquad h_i = h(\hat{\theta}_i), \quad \hat{\theta}_i = E[\theta \mid \mathbf{y}_i]. \tag{9}$$

The map is monotone, so θ and h order people identically, but not linear so the variance profile and cost function differ across the two. In practice the $h()$ transformation compacts the health scores at the healthy end of the distribution and spreads them out more at the unhealthy end. Figure B.1 compares the measures and illustrates how they relate to one another.

The h score is close to a weighted sum of the coded answers: regressing h on the standardised codes gives $R^2 = 0.993$. The parallel is not perfect for two reasons: the uneven gaps between the categories of an ordered item; and an item's best and worst categories are open-ended so they only tell us that a person lies beyond the last threshold.[4] A linear one-factor model fitted to the same eight codes, treated as continuous, gives a score with a Pearson correlation of 0.995 with h and 0.998 with that weighted sum, so on the $[0, 1]$ scale the graded-response model's practical content is the weighting rather than the functional form.

In addition to θ and h we construct a Rockwood-style frailty index (Mitnitski et al., 2001; Searle et al., 2008): the share of 31 equal-weight deficits a person has, namely the six SF-12 physical items graded 0 to 1 by response step, each of the eight physical impairment areas, and each of the 17 ever-diagnosed conditions (clinical depression among them), so that 0 is no deficit and higher is frailer (Appendix B.3). 7% of person-waves have none. Following Hosseini et al. (2022), who model the log of frailty, we also report $\log(\text{frailty} + 1/31)$: adding one deficit's worth keeps the zeros in the sample, where Hosseini et al. (2022) drop them into a separate probit.[5] Both indices rank people much as θ and h do (Spearman 0.92 with each), and the two pairs line up: h behaves like the frailty index and θ like its log (Section 3.2).

Relation of our health metric to the economics literature.

- Why this measure - the underlying data are ordinal and bounded, so our model respects this.
- Factor methods are common in health economics: Poterba et al. (2013, 2017) take the first principal component of 27 HRS items, and Blundell et al. (2023) of the three subjective health measures common to HRS and ELSA.
- In the skill-formation literature, Cunha et al. (2010) estimate a closely related dynamic factor model, on continuous test scores that enter linearly in the log of the latent skill.
- Item-response models are standard in health measurement: the PROMIS item banks, now widely used in health research and clinical care, were calibrated with the graded-response model (Cella et al., 2010), including the 124-item physical function bank (Rose et al., 2014).
- Item-response models themselves are rare in health economics: Graeber (2023) scores latent health from five SOEP items with the graded-response model, Wallace et al. (2017) build an item-response index from 21 HRS items, and Meijer et al. (2011) estimate the same ordinal factor model on SHARE without calling it one. They are common in the economics of education, for scaling and linking test scores (Das and Hammer, 2005; Das and Zajonc, 2010; Singh, 2015; Muralidharan et al., 2019; Jacob and Rothstein, 2016), and in the measurement of poverty and deprivation (Moffitt and Ribar, 2016; Corsini and Martelli, 2020; Farcomeni et al., 2025; Davis et al., 2026).
- The closest work to ours in health dynamics uses five-category self-assessed health kept categorical and usually collapsed: to three states by De Nardi et al. (2025) and Capatina (2015), to a bad-health indicator by Halliday (2008), and kept at five by Contoyannis et al. (2004). But this would really limits our study of dynamics and in particular the timing and targetting of prevention.
- In our case, the choice of methodology relative to a more familiar principal component or linear factor model is innocuous, if we focus on h - a linear factor model estimated on the same data has a Pearson correlation of 0.995 (Spearman 0.992) with h.
- There is a precedent for modelling the log of a bounded health index, the log frailty of Hosseini

[4]The latent scale θ departs is somewhat further from a weighted sun: regressed on the same codes gives $R^2 = 0.925$. Most of the gap comes from the model not scoring an item's categories as equally spaced: letting each category carry its own score raises the R^2 to 0.979. The rest comes from the items discriminating in different parts of the scale, so that a step on one item counts for more where few others still move.

[5]Adding half the smallest positive value, 1/248, instead puts the zeros so far below the next group that they drive the variance at young ages; with 1/31 the variance profile is smooth.

et al. (2022); on our data $\log(\text{frailty} + 1/31)$ has a variance profile by age like θ's, and the frailty index in levels one like h's. Table B.2 shows the correlation with alternative metrics (UKHLS PCS, MCS, raw chronic count). All of these have theoretical disadvantages over our measure, but PCS is closest.

[I think overall conclusion here is that there is no one way to measure health, but for dynamics we kind of need cardinality which we can maybe pin down with costs.]

As our preferred case, we report θ in the main text. It is the scale the measurement model is defined on; it is unbounded, so the Gaussian dynamic models of Section 4 fit it without a ceiling (on h the healthiest class collapses under AR(1) plus "spike"); and its cost curve in pounds is convex. Appendix C repeats the main-text figures on h, the frailty index and log frailty, and the text flags where the measure changes a conclusion.

In addition to our health metric, we also construct a proxy for healthcare cost based on the healthcare utilisation information in UKHLS.

- Per-person-wave estimate of annual NHS cost from in-patient, out-patient and GP contacts, in a fixed price year, for waves 7–15.
- It is wrong in level but roughly right in *shape*, and the variation of cost is all we need.
- The construct is :

$$\widehat{C}_{it} \;=\; \underbrace{c_{\text{spell}}(k) + c_{\text{bed}} \cdot (\text{nights}_{it} - \bar{L})^{+}}_{\text{in-patient}} \;+\; \underbrace{c_{\text{op}} \cdot \widehat{n}^{\text{op}}_{it}}_{\text{out-patient}} \;+\; \underbrace{c_{\text{gp}} \cdot \widehat{n}^{\text{gp}}_{it}}_{\text{GP}} \tag{10}$$

  with k the attributed condition where there is one, and everything indexed to a single price year.
- Underlying data, consutruction and validation against pubished statistics are shown in Appendix B.4

Figure 1 shows that:

- Cost rises convexly with poor health: the quadratic term in standardised θ, person-clustered, has $t = 19.5$ (36.9 on costs capped at the 99th percentile).[6]
- The relationship is close to age-invariant. The binned curves for the three age bands lie on top of one another; in the sickest pooled tenth mean cost is £2,143 at 20–34 and £2,136 at 75–90; and the quadratic term is positive within every one of six age bands. The 75–90 band's curvature is roughly double the others' ($t = 3.6$), but this difference is driven by the top 1% of costs: when we cap at the 99th percentile the curvature–age interaction is insignificant ($t = 0.7$).[7]
- Mortality also rises steeply with poor health, roughly log-linearly: pooled across ages with a quadratic in age, the log-odds of death fall by 0.90 per standard deviation of θ.
- The health-mortality gradient is relatively constant with age, although the slight flattening after 45 is statistically significant ($t = -2.3$); the level, however, is not age-invariant. Age thus carries mortality risk that the health measure does not capture, which matters for survival but not for cost. *[This kind of relates to what we were thinking about right at the start in terms of the age-health-mortality relationship...]*
- The same holds on h, with one difference on the frailty indices (Appendix C): the curvature has $t = 10.3$ on h and 22.6 on log frailty, but only 5.8 on the frailty index in levels (11.3, 42 and 1.2 capped), which is only mildly convex in cost and close to linear once the top 1% is capped.

[6] A quadratic coefficient is driven by the top of the cost distribution, so the uncapped and capped versions are both reported here and below, uncapped first.

[7] The linear term's interaction with age is specification-sensitive: insignificant in a quadratic with a single health × age term, mildly negative ($t = -1.5$, capped) once the curvature is also allowed to vary with age.

[We can move this figure to the appendix or simplify it, but I wanted to include something for you to have a look at...]

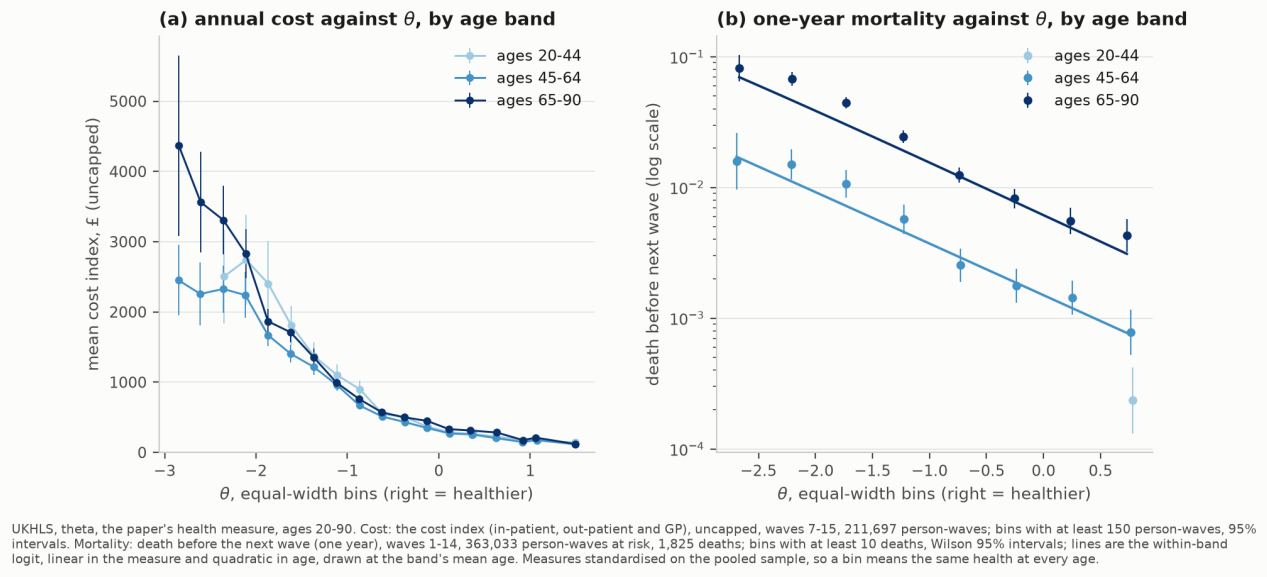


Figure 1: Cost and mortality against θ, by age band. (a) Mean annual cost index in equal-width bins of θ within three age bands, with 95% intervals. (b) The one-year death rate in the same bins on a log scale, with the within-band logit drawn through it. The measure is standardised on the pooled sample, so a bin is the same health at every age. The same figure on h and the frailty indices is in Appendix C.

3.2 The moments of health over the life course

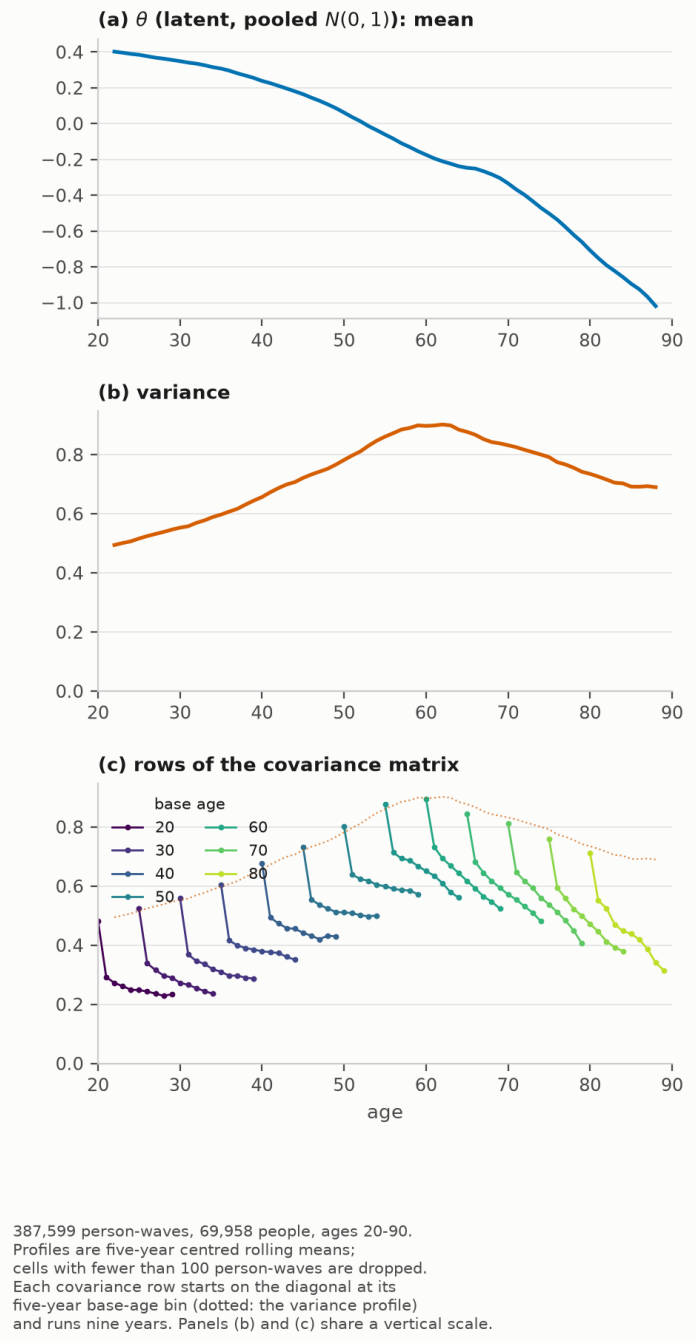


Figure 2: The measure's mean path, variance profile and covariance rows, model-free, on θ. 387,599 person-waves; five-year centred rolling means; each covariance row starts on the diagonal at a five-year base age and runs nine years. The same figure on h, the frailty index and log frailty is in Appendix C.

- From panel (a) the mean accelerates down: θ falls by about 0.014 standard deviations a year between 30 and 50, 0.020 between 50 and 70, and 0.038 after 70. $\mathbb{E}(d)$ rises with age, which on its own says intervene late: the mean channel of $P(r)$ is largest where the preventable decline is largest.

- From panel (b) the variance rises by about 60% between 30 and 60, peaks at 62, and falls by a fifth by 85. Under stationary noise this says the local covariance $C_{Hd}(a)$ has reached zero around 60 and turned negative after it: interventions on dispersion pay early.
- From panel (c) we see divergence then convergence: every row drops sharply in its first year, by 40% at base age 20 and about 20% at 50–60 (the one-year "spike", health not carried even one year). After this the younger rows run flatter while the older rows fall more steeply.
- The shape of the variance profile depends on the measure as Section 3 anticipated (Appendix C).
- On $\log(\text{frailty} + 1/31)$ the pattern is very similar, with variance rising by a factor of 1.70 from 30 to 60 and falling to 0.70 of its 60 level by 85, against 1.62 and 0.78 on θ.
- On h and the frailty index, variance continues to increase after 60, but at a markedly slower rate. The bounded measures keep dispersion growing into old age; the unbounded ones have it peak around 60 and then compress.
- Across all three alternative indices, panels (a) and (c) show the same stylised facts.
- One stationary noise process cannot produce both. Either the noise moves with age or the local C_{Hd} has changed sign: divergence among the young, compression among the old.
- This has implications for timing and targeting: intervene early and target the low early in life; later in life the covariance turns against targeting on levels so not much use targetting on low.
- Cautions carried to the next section: composition (who survives to be measured), floors and cohort pooling are all in these curves; the type models will use within-person information.
- The covariance rows in Figure 2 are based on pairwise covariances. Covariance rows on balanced samples, i.e. each row over the people observed at all of its ages, are shown in Appendix D (Figure D.20).

3.3 The linear approximation on real data

[**TODO:** Decide which cases, if any, that we should present here. We have four main cases and also a decision on whether we should use pairwise or balanced moments.]

- Figure 3 shows the V_H, C_{Hd} and V_d identified from pairwise moments on four ages four years apart (so over a 12 year span. To account for the non-linearity we show this pooled within four bands. We show Case 4 (AR(1) + spike), the most general case we've looked at in detail; Case 1 (iid noise), the simplest, is in Appendix D (Figure D.22).
- These are not computed by solving the just-identified propositions cell by cell, but by least squares on the ten pooled moments of each band.
- Given $H_a = H + a\,d$ the covariance of the measure at offsets i and j years from the base age is $V_H + (i + j)\,C_{Hd} + ij\,V_d$ plus the noise.
- Under Case1, the noise is one variance on the diagonal;
- For Case 4 the noise is $\rho^{|i-j|}$ times a carried variance plus a diagonal spike.
- The ten cells are regressed on these columns with V_H, V_d and the noise variances non-negative and $C_{Hd}^2 \le V_H V_d$ imposed, and ρ profiled over a grid.
- Each case is therefore overidentified, ten moments against four or five parameters, and the bars in Figure 3 stack the fitted layers against the observed moment in every cell, so the misfit is visible.
- $\text{Corr}(H, d) = C_{Hd}/\sqrt{V_H V_d}$ stated in each subtitle is calculated from the fitted coefficients.
- There are many different ways to represent this (i.e. the health metric, the gap between ages, pairwise or balanced moments, the noise case) and the details vary, but relatively consistently C_{Hd} gets more negative with age, see a few more in Figures D.23 and D.24.

- For Case 4, on θ, Corr(H, d) runs $+0.45$, $+0.78$, -0.56, -0.53 across the four bands, with ρ 0.85–0.94 and no constraint binding.
- For Case 1 (Figure D.22), Corr(H, d) is negative in every band (-0.22, -0.02, -0.42, -0.48), but it clearly gets more negative with age. *[I chose this specification because it works nicely for Case 4, but there are plenty that work nicely for Case 1 as well]*.
- On the other measures the sign path is the same, but whether it fits as cleanly depends on the specification (Appendix C). Under Case 1, log frailty almost reproduces θ (-0.14, 0.00, -0.44, -0.48). Under Case 4, h runs $+0.43$, $+0.63$, -0.31, and at 75–90 V_H goes to zero so nothing is identified; log frailty runs $+0.48$, $+1.00$, -1.00, -0.92, with the admissibility constraint binding at 40–60 and 60–75; the frailty index runs $+0.40$, $+0.98$, -0.22, and at 75–90 the constraint binds with V_H near zero.

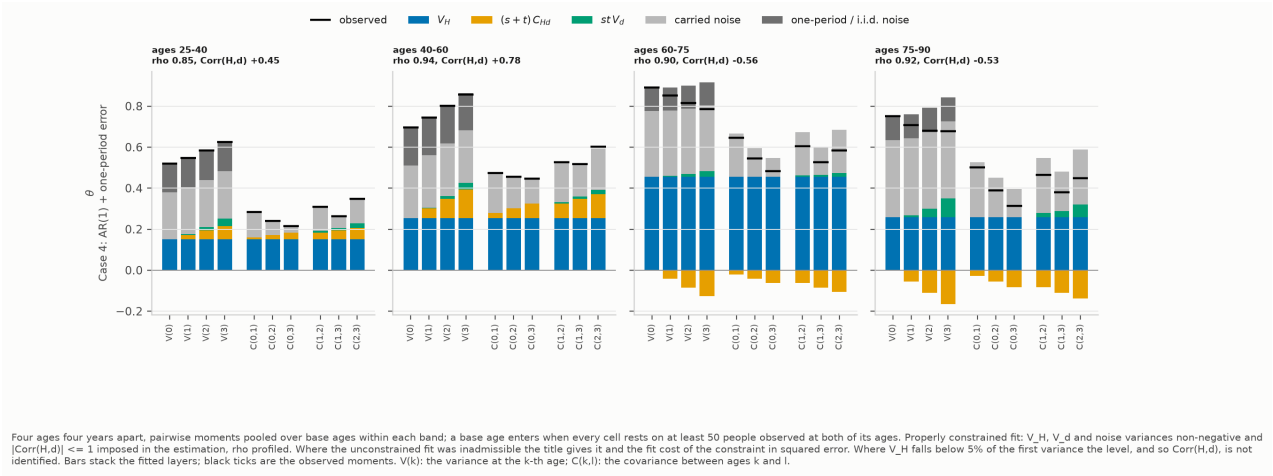


Figure 3: The linear approximation on θ: the ten moments of four ages four years apart in each band, pairwise moments, decomposed into the fitted layers under Case 4. Properly constrained fit: variances non-negative and $|\text{Corr}(H, d)| \leq 1$ imposed in the estimation. A base age enters when every cell rests on at least 50 people observed at both of its ages. Case 1 is in Appendix D, and Case 4 on h, the frailty index and log frailty in Appendix C.

4 Lifecycle types

4.1 Why and how

- The health-dynamics literature generally works with the old (HRS, ELSA) or short windows; De Nardi et al. (2025) splice the PSID early in life with the HRS later, but health is modelled only for the older people; Hosseini et al. (2022) build a frailty index on the PSID.
- The prevention framework says H_{55} is an *outcome* of the dynamics before 55, not an input. And the data say the dynamics differ across the lifecycle (Figure 2): starting in late middle age misses the divergence phase. So this is why we focus on the full lifecycle.
- **How.** Overlapping cohorts observed over 15 waves: cohort-independent types with cohort-dependent shares. First non-parametric (partial K-means), then a parametric refinement (Bayesian mixtures) that separates persistent deviations from one-period "spikes", then against observables. Everything is on θ; the h versions, and the K-means types on the two frailty indices, are in the appendix.

- **Four challenges** addressed along the way: (1) power and type categorisation (functional forms, stochastic structure, several dimensions); (2) mortality and attrition; (3) cohort effects; (4) units and bounds (ceiling, floor, cardinalisation), which is why we work on θ and repeat the analysis on the bounded measures in the appendix.

4.2 (A) Non-parametric types: partial K-means

- Method: Included respondents with at least three observed ages between 20 and 89 (38,963), the same rows as the Bayesian mixtures below. Each person is a vector over ages 20-90 with most entries missing.
- Estimate type/cluster by minimising average squared deviations over observed entries only - "partial" K-meanrs. 50 seeded starts to insure that results are not sensitive to initialisation.
- Baseline case is $K = 3$.
- "Equal penalty": every person-wave is punished equally, so the method is neutral about whether the stochastic process is transitory or persistent. *[Open: the person is assigned to one cluster for life, which does impose that the type is permanent, so that does mean that the within person dynamics drive the results.]*
- We focus on means because they are what the policy objects use and what the interventions considered shift.

Figure 4 shows the cluster results on θ; h, the two frailty indices and predicted cost are in Appendix C.

- On θ, three ordered, non-crossing paths, shares 18 / 43 / 40%. The worst type starts below the others at 20 and falls fastest to 60, by 1.2 standard deviations against 0.4 for the best type; after 60 the order reverses, the worst type falling 0.3 by 85 and the other two about 0.8. Divergence and then convergence is visible in the type paths themselves.
- The choice of health metric matters a bit (Appendix C), but the overall shape looks very similar across each of the four we consider.
- On h K-means isolates a 10% sick tail (shares 10 / 28 / 61%) instead of splitting the healthy majority, and agrees with the θ types at ARI 0.37 (71% in the same type): θ stretches the healthy end. The frailty index lands with h (ARI 0.61, 86%) and log frailty with θ (ARI 0.44, 78%, shares 21 / 41 / 38%).
- The stochastic part (middle panel): within-type deviations are persistent, with lag-1 autocorrelation 0.64 in the worst type and 0.37–0.38 elsewhere, decaying by nine years to 0.24, 0.16 and 0.03 (the middle type). Suggests that a final model should include some persistence, and that it might differ by type.
- Composition: on θ the worst type's share of observed person-waves is 22% at 20, falls to 13% at 50 and recovers to 18% by 70. This is selection into the observed sample, not people changing type. *[And for θ in the AR(1)+spike model, this pattern does somewhat reverse and we get a falling share of the unhealthy type by age.]*

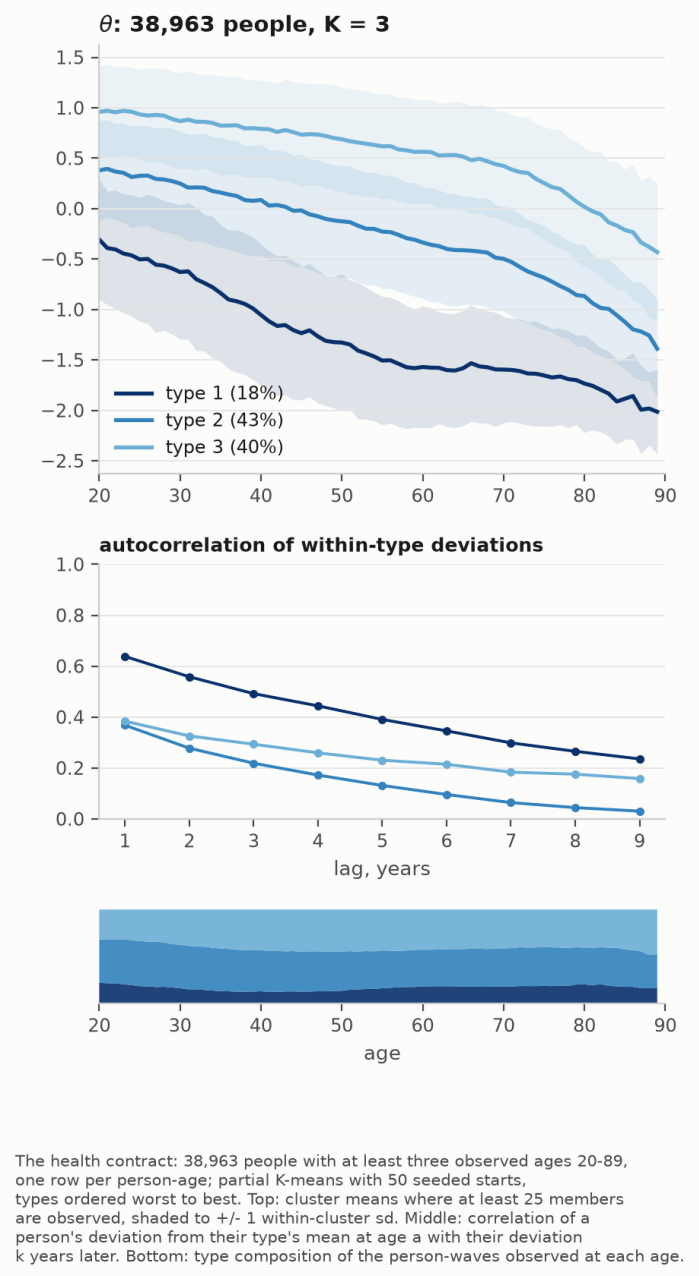


Figure 4: Partial K-means types ($K = 3$) on θ. Top: type means with the within-type ± 1 sd band; middle: autocorrelation of within-type deviations by lag; bottom: type composition of the person-waves observed at each age. The same figure on h and the two frailty indices is in Appendix C, and on predicted cost in Appendix E.1.

Figure 5 shows the key moments based on the estimated K-means clusters.

- Top panel: the between-type share of the cross-sectional variance rises from 48% at 30 to 65% at 70 and falls back to 57% by 85. *[within type variance is satisfyingly flat for θ]*
- Middle panel: roughly a U shape in $\mathrm{Var}_j(d_{j,a})$, lowest at about 48, so slopes are most consistent in middle age.
- Bottom panel: $\mathrm{Cov}_j(\bar{h}_{j,a}, d_{j,a})$ goes from positive to negative at about 59: divergence and then

convergence, so P(ii) and P(iii).

- The key messages are relatively consistent across h and θ (Appendix C).
- *[Between-type shares are partly mechanical: the clusters are chosen to maximise them. The comparison that is not mechanical is with observables in Section 4.4.]*

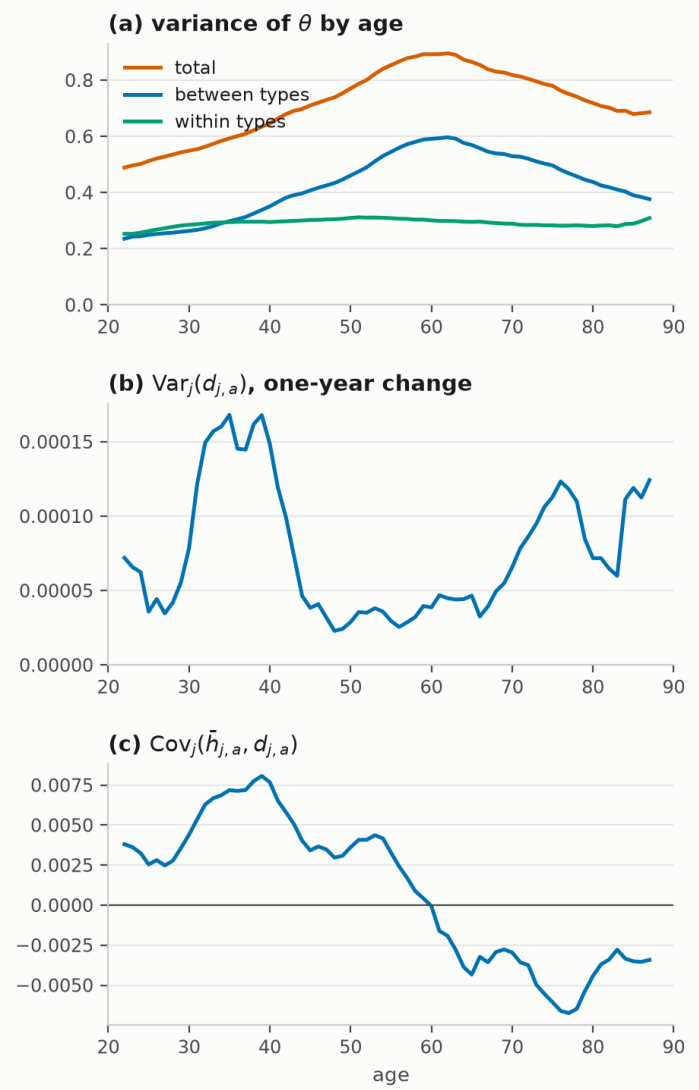


Figure 5: Between versus within by age, on θ. (a) Cross-sectional variance split by type composition; (b) between-type variance of the one-year change of the smoothed type paths; (c) the level-change covariance across types. The same figure on h and the two frailty indices is in Appendix C.

Small robustness exercise:

- Cohort adjustment within K-means (Appendix E): regresses the score on single-year age dummies and birth-decade dummies, subtracting the decade shifts (so identifies the decade effects from the overlapping ages).
- Re-clustering leaves the typology unchanged (adjusted Rand index 0.90 on θ, 0.96 on h; at least 97% of people keep their type). The shifts themselves are zero for cohorts born before 1960 and negative only from the 1970s on. *[worth noting that he Bayesian model does pick up more meaningful cohort effects, which I have not really thought about in depth yet.]*

4.3 (B) Parametric types: Bayesian mixtures

The parametric counterpart of the K-means types is a finite mixture of quadratic growth curves. Let y_{ia} be the measure of person i at age a, standardised on the contract, and $\tilde{a} = (a - 55)/10$. Person i belongs to one of K classes, $k_i \in \{1, \dots, K\}$, with $\Pr(k_i = k) = \pi_k$, and in the baseline specification

$$y_{ia} = \alpha_{k_i} + \beta_{k_i}\tilde{a} + \gamma_{k_i}\tilde{a}^2 + \varepsilon_{ia}, \qquad \varepsilon_{ia} \sim N(0, \sigma^2) \text{ i.i.d.}, \tag{11}$$

so a class is a path and a residual scale common to the classes. The class indicator is marginalised: person i's likelihood is $\sum_k \pi_k \prod_a N\big(y_{ia} \mid \mu_k(a), \sigma^2\big)$ over their observed ages, and their posterior class probabilities follow from the fitted parameters. Labels are identified by ordering the intercepts, $\alpha_1 < \cdots < \alpha_K$, so class 1 is worst health. Priors are weakly informative on the standardised scale: $\pi \sim \text{Dirichlet}(1.5)$, $\alpha_k \sim N(0, 1.5^2)$, $\beta_k \sim N(0, 1)$, $\gamma_k \sim N(0, 0.05^2)$ and $\sigma \sim N(0.7, 0.5^2)$ truncated below at 0.05. Estimation is by Hamiltonian Monte Carlo (the no-U-turn sampler) in Stan: four chains of 1,000 warm-up and 1,000 sampling iterations, with convergence judged by the split-$\hat{R}$ of the structural parameters.

Figure 6(a) shows the Bayesian equivalent of the partial K-means:

- Same data as the partial K-means, fitted on θ; the h, frailty-index and log-frailty versions are in Appendix C.
- The baseline $K = 3$ mixture (quadratic paths, independent residuals, no AR term) converged (max $\hat{R} = 1.001$, about 3 h).
- The mixture and the K-means types are the same typology. Shares agree to a few points (.15/.44/.41 against .18/.43/.40), the class paths track the type means, and person by person 95% of people are in the same class (ARI 0.85, mean maximum posterior 0.93).
- The same holds on h (ARI 0.82, 94% in the same class): there both methods isolate a 10% sick tail, where on θ both split the healthy majority, so the h–θ split is a property of the scale, not of the method.
- The composition strips move the same way under both methods: on θ the worst class is 13% of the observed person-waves at 30 and 16% at 85 under the mixture, against 16% and 18% under K-means.

The paper's stochastic specification changes only the residual. In place of the i.i.d. ε_{ia} of (11), the deviation from the class path is a persistent state plus a one-period "spike",

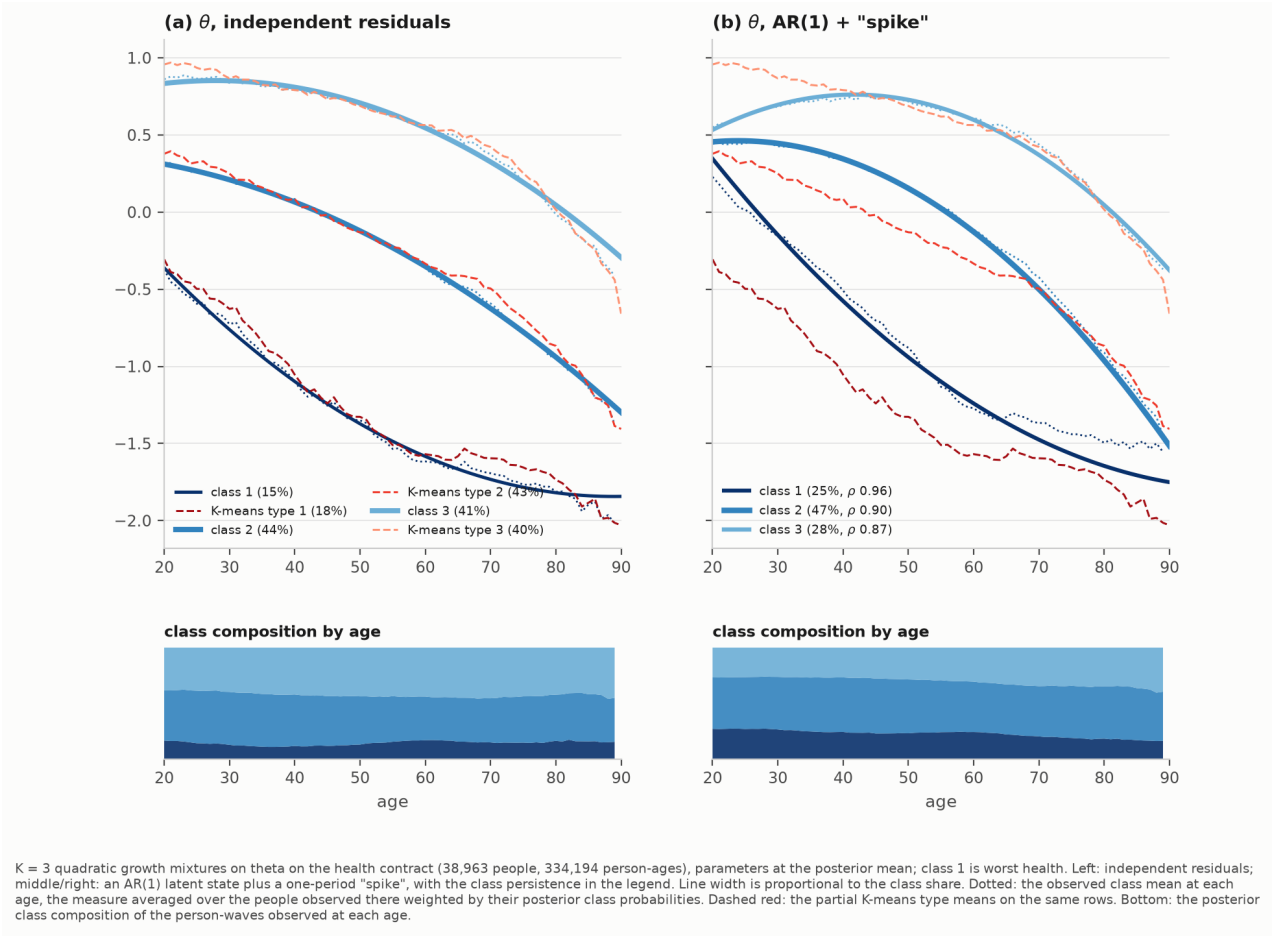
$$y_{ia} = \mu_{k_i}(a) + u_{ia} + \varepsilon_{ia}, \qquad u_{ia} = \rho_{k_i}^{g}\, u_{i,a-g} + \eta_{ia}, \quad \eta_{ia} \sim N\Big(0,\ \sigma^2 \tfrac{1-\rho_{k_i}^{2g}}{1-\rho_{k_i}^{2}}\Big), \quad \varepsilon_{ia} \sim N(0, \sigma_\varepsilon^2), \tag{12}$$

where g is the gap in years since the person's previous observation, the state starts from its stationary distribution $u \sim N\big(0, \sigma^2/(1-\rho_k^2)\big)$, and the persistence ρ_k is class-specific while the innovation scale σ and the "spike" scale σ_ε are common. The two variances are identified off the autocorrelation function,

ρ^g times the state's share of the variance, so ρ comes from the ratio of successive autocorrelations and the "spike" from their level; each person's likelihood is computed by the Kalman filter, class by class, and the rest is as in the baseline. Priors add $\rho_k \sim \text{Beta}(2, 2)$ and $\sigma_\varepsilon \sim N(0, 0.4^2)$ truncated below at 0.01.

Figure 6(b) shows the AR(1) plus "spike" specification:

- The same specification on θ (and on h for the appendix), on the same contract; converged (max $\hat{R} = 1.002$, 4–8 h).
- On θ the story carries over. Shares .25/.47/.29, persistence .96/.90/.87, a "spike" of 0.42 against a state innovation of 0.23: between a fifth and a half of the variance around each class's path is the "spike" (signal shares 0.78, 0.63 and 0.56), the worst class starts within half a standard deviation of the middle and falls away, and the composition drift reverses: the worst class is 26% of the observed person-waves at 30 and 17% at 85, against 13% → 16% under the baseline.
- Agreement with the K-means types falls to ARI 0.42 (77% in the same class), and classification is less certain (mean maximum posterior 0.69 against 0.93).
- On h (Appendix C) the healthiest class collapses: its persistence is 0.25, against 0.96 and 0.78 for the other two, and agreement with the K-means types falls to ARI 0.46, with half of K-means type 2 reassigned to the worst class. This bounded-scale problem is one reason we report θ. The frailty index, bounded at zero, does the same at its floor (Figures C.11 and C.12): under AR(1) plus "spike" its healthiest class has persistence 0.36 against 0.98 and 0.81, and agreement with the K-means types falls to ARI 0.34; the shifted log lifts the floor only partly, with a small healthiest class (10%) at persistence 0.44 against 0.96 and 0.92 for the other two and ARI 0.29. Under independent residuals all four measures reproduce their K-means types (ARI 0.81–0.89, at least 94% of people in the same type). *[when we go to $K = 4$ then we start getting a convergence again though]*
- The specification is close to that of Hosseini et al. (2022), who model log frailty as a fixed effect plus an AR(1) plus white noise. Our classes play the role of a coarse fixed effect, and their persistence of 0.991 compares with our 0.87–0.96.



K = 3 quadratic growth mixtures on theta on the health contract (38,963 people, 334,194 person-ages), parameters at the posterior mean; class 1 is worst health. Left: independent residuals; middle/right: an AR(1) latent state plus a one-period "spike", with the class persistence in the legend. Line width is proportional to the class share. Dotted: the observed class mean at each age, the measure averaged over the people observed there weighted by their posterior class probabilities. Dashed red: the partial K-means type means on the same rows. Bottom: the posterior class composition of the person-waves observed at each age.

Figure 6: $K = 3$ mixtures on θ: (a) independent residuals, (b) an AR(1) latent state plus a one-period "spike", with the class persistence in the legend. Top: class paths at the posterior mean (solid, width proportional to the share), the observed class means, the measure averaged at each age over the people observed there weighted by their posterior class probabilities (dotted), and the partial K-means type means on the same rows (dashed red). Bottom: the posterior class composition of the person-waves observed at each age. The same figure on h is in Appendix C.

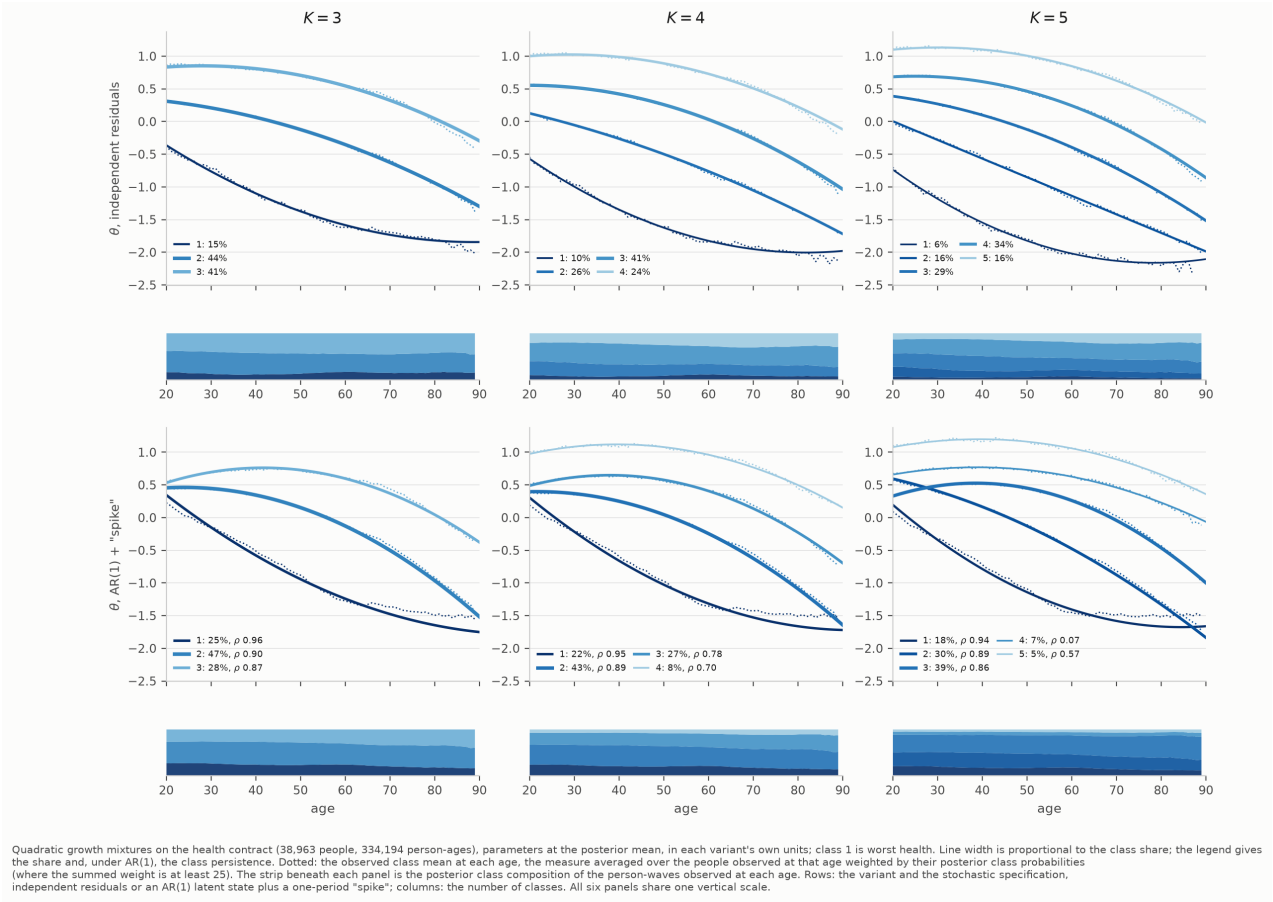


Figure 7: Fitted class trajectories and shares on θ by number of classes. Rows: independent residuals and an AR(1) latent state plus a one-period "spike"; columns: $K = 3$, 4 and 5. Line width is proportional to the class share, the legend gives the share and, under AR(1), the class persistence; dotted lines are the observed class means, the measure averaged at each age over the people observed there weighted by their posterior class probabilities; and the strip beneath each panel is the posterior class composition of the person-waves observed at each age. The same figure on h is in Appendix C.

Extensions: K, dimensions, mortality.

- $K = 4$ and $K = 5$ (Figure 7) converge cleanly on every specification (max $\hat{R} \leq 1.004$). Under independent residuals each added class is a new sickest group, 10% of people at $K = 4$ and a further 6% at $K = 5$, and the healthy majority is left alone.
- Under AR(1) plus "spike" the split goes the other way. At $K = 4$ the three $K = 3$ classes barely move and the new class is a small healthy group, 8% with persistence 0.70. At $K = 5$ the three large classes are unchanged, with persistence 0.86–0.94, and the two new classes are both small and healthy, 7% and 6%, one with persistence 0.07: a collapse like h's reaches θ too at $K = 5$, but in a small class rather than half the sample. Classification certainty falls from a mean maximum posterior of 0.69 at $K = 3$ to 0.66 at $K = 4$.
- On h (Appendix C) the healthiest-class collapse spreads rather than resolves as classes are added: the top two classes have persistence 0.52 and 0.19 at $K = 4$, and 0.39 and 0.18 at $K = 5$, covering about half the sample.
- Multidimensional model (Figure 8): the physical measure (θ; h in Appendix C) and a mental GRM

(the three SF-12 mental testlets and two GHQ-12 testlets, no depression diagnosis; Appendix H looks at it on its own) as Gaussian channels, plus mortality, one shared class structure; the health contract's people with a mental score on every row, 38,181 people and 2,045 deaths. Under AR(1) plus "spike" the persistence is class and channel specific while the innovation and "spike" scales are common to the classes; the hazard is Gompertz–Makeham, $\lambda_k(a) = c + b_k e^{\vartheta_k(a-55)}$, integrated over each row's years at risk, with class-specific level and slope and a common Makeham constant. The figure is the fit with one level shift per birth decade on both channels, common to the classes, drawn at the 1950s: without the shifts (Appendix F.1) the mental class paths rise with age, which the shifts attribute to the older cohorts' higher level, with the caveat on reading them as cohort effects set out below.

- The physical typology survives the extra channels. The classes are .18/.50/.32 with physical persistence .94/.90/.86, the same to two decimals as without the decade shifts (.19/.51/.31 and .94/.91/.87), and 76% of people land in the same class as under the single-channel fit without shifts (73% against the single-channel fit with them, Figure 9). On h (Appendix C) the healthiest class keeps its collapse, persistence 0.28, even though its mental persistence is 0.86.
- The decade shifts change nothing else either: the same "spike" scales, a class-1 to class-3 hazard ratio of 11.5 at 55 and 3.2 at 85 against 10.6 and 3.3 without them, and a held-out twin that predicts exactly as the fit without shifts does (the cohort rows of Figures 13 and 14: R^2 0.61 and 0.74 for θ_{t+1} from the classes and the forecast, against 0.61 and 0.74). The shifts themselves fall monotonically from the 1920s to the 2000s, by 0.9 sd on θ and 1.3 sd on the mental score.
- Mental health is faster and noisier than physical health in every class: persistence .93/.89/.82 against .94/.90/.86 on θ, and a "spike" of 0.54 against 0.42, so the signal share of the mental channel is 0.45–0.66 against 0.55–0.74. A shared persistence across channels, as in earlier versions, would have been wrong. At the 1950s the worst physical class sits a standard deviation below the best on mental health at 55, and every class's mental path declines with age once the decade shifts are taken out.
- Mortality is strongly graded by class and converges with age: the yearly hazard of the worst class is 11.5 times the best class's at 55 and 3.2 times at 85 (10.6 and 3.3 without the shifts; 7.1 and 2.7 under independent residuals), the sickest class having the flattest Gompertz slope (0.09 against 0.14 per year). The Makeham constant is below 10^{-4} in every fit. Mortality describes survival by class; it does not define the classes. This is where the survivor-bias discussion lives.
- Note that in the AR(1) plus "spike" case, the class composition by age now looks like it reflects mortality selection. We need to look into this a bit more, but it is certainly promising!
- Windows (Figure 10, K-means version): types refitted on twenty-year windows agree with the lifecycle typology at ARI 0.74–0.83 up to the 60–80 window and 0.61 at 70–90; backward-expanding windows from 90 reach ARI 0.76 once the window starts at 60 and 0.93 from 30. What a late window sees are the types after the divergence phase: the worst type's share of a window's people rises from 15% (20–40) to 27% (70–90) as the window moves, and from 18% to 27% as the backward window shrinks from 30–90 to 70–90. The lower row says how much of that is composition and how much relabelling. Within the ages two windows share, the worst type's share of the person-waves observed at each age is nearly the same whichever window assigned it: the backward windows starting at 30, 40 or 50 sit on the lifecycle typology's line (about 13% at 45, 18% at 65, 19% at 75), so the rise in the worst type's share of people is the window's people getting older, not a growing bad type. Only the windows confined to late life relabel: a window starting at 60 or 70 puts 21–25% of the person-waves at 65–75 in its worst type against the lifecycle typology's 18–19%, picking up people who were middle type for most of their lives and decline late. The bad type expands at the margin, then, and only when the early life that distinguishes a late decliner

from a lifelong low type is out of view. [**TODO:** Bayesian version of this figure for the main text; move the K-means version to the appendix.]

- **Cohort effects.** The AR(1) plus "spike" fits, single-channel and multidimensional, were repeated with one level shift per birth decade, common to the classes (Appendix F.1). With fifteen waves the panel observes each person over at most a fifteen-year span, so a cohort's level is identified from the overlap of neighbouring decades at the ages they share.
- **The typology is unaffected.** Shares, persistence and the "spike" scale agree with the fits without cohort effects to two or three decimals on every variant, and the class paths keep their shape; the worst class is a little steeper once the decade shifts carry part of the level. Whatever cohort variation there is, it is not what separates the classes.
- **The cohort effects themselves should not be read strongly.** The fit reports a monotone decline between the 1950s and the 2000s of about two thirds of a standard deviation on θ (a third on h). But the raw decade profiles line up almost exactly at the ages they share, and subtracting the estimated shifts opens gaps that were not there. Over a fifteen-year window age and birth year are nearly collinear, and a decade dummy is close to a linear function of age within it, so with a single quadratic age path per class the dummies absorb the curvature the quadratic cannot follow. With single-year age dummies in place of the quadratic, the pre-1960 shifts vanish and only the post-1970 cohorts decline (Appendix E.2), so the mixture's shifts should not be read as cohort effects.
- **Specification.** The same $K = 3$ fits with an AR(1) in the class residual but no "spike" give persistence of 0.79/0.28/0.40 on θ, against 0.87–0.96 once the one-period term is separated: without it ρ has to absorb the one-period noise and is pulled down.
- [**TODO:** still to do for the bounded measures in the appendix: a bounded observation model (scores at the ceiling of h or the floor of frailty treated as censored).]

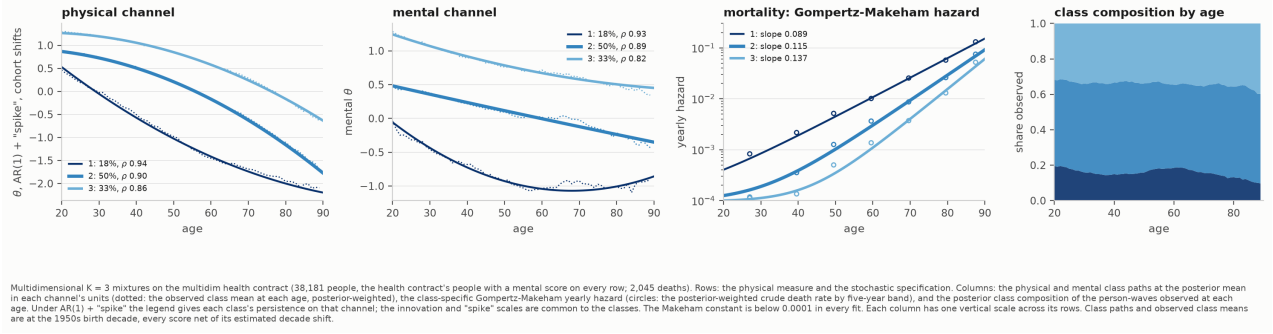
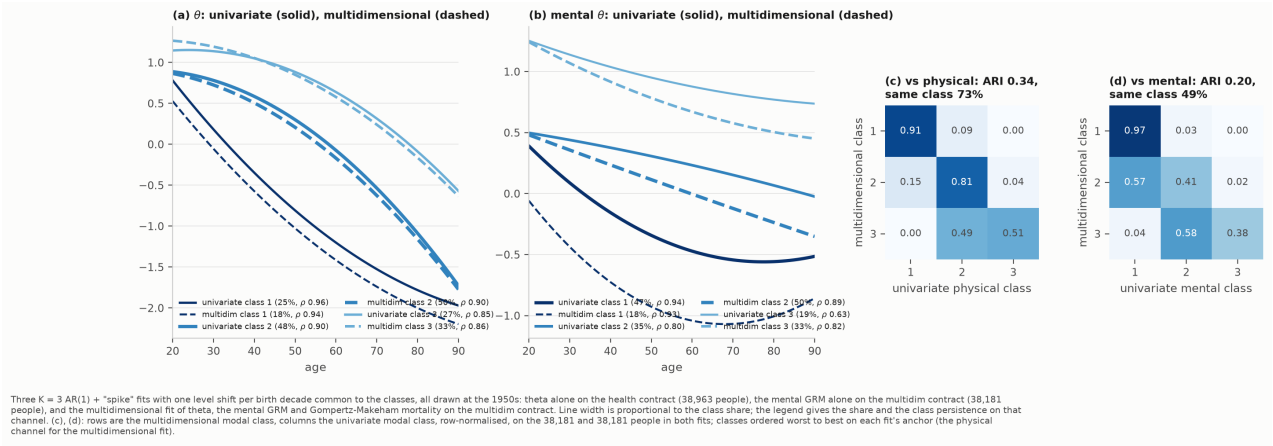


Figure 8: The multidimensional $K = 3$ mixture on θ, AR(1) plus "spike" with one level shift per birth decade on the physical and the mental channel, common to the classes. Columns: the physical and mental class paths at the posterior mean in each channel's units, drawn at the 1950s birth decade with every score net of its estimated decade shift (dotted: the observed class means, posterior-weighted); the class-specific Gompertz–Makeham yearly hazard on a log scale (circles: the posterior-weighted crude death rate by five-year band); and the posterior class composition of the person-waves observed at each age. The legend gives each class's persistence on that channel. The fits without the decade shifts, AR(1) plus "spike" and independent residuals, are in Appendix F.1; the same on h in Appendix C.



Three K = 3 AR(1) + "spike" fits with one level shift per birth decade common to the classes, all drawn at the 1950s: theta alone on the health contract (38,963 people), the mental GRM alone on the multidim contract (38,181 people), and the multidimensional fit of theta, the mental GRM and Gompertz-Makeham mortality on the multidim contract. Line width is proportional to the class share; the legend gives the share and the class persistence on that channel. (c), (d): rows are the multidimensional modal class, columns the univariate modal class, row-normalised, on the 38,181 and 38,181 people in both fits; classes ordered worst to best on each fit's anchor (the physical channel for the multidimensional fit).

Figure 9: Do the multidimensional classes differ from the univariate ones? Three $K = 3$ AR(1) plus "spike" fits with birth-decade shifts, all at the 1950s: θ alone, the mental GRM alone, and the multidimensional fit. (a) The physical class paths from the univariate fit (solid) and the multidimensional fit (dashed); (b) the same for the mental channel; (c), (d) the multidimensional modal class against the univariate physical and mental modal classes on the people in both fits, rows normalised, with the ARI and the share in the same class. Classes are ordered worst to best on each fit's anchor, the physical channel for the multidimensional fit.

- **The multidimensional classes are the physical classes with a mental path attached** (Figure 9; `descriptives/45_paper_multidim_vs_univariate.py`). The physical paths of the multidimensional fit lie on top of the univariate θ paths, with the worst class a little lower (-1.23 against -1.01 at 55) and a little smaller (18% against 25%); 73% of people take the same class in both fits and the disagreement is almost all between the best and the middle class (half of the univariate best class is the multidimensional middle class), the boundary the univariate fit draws with less to go on.
- **The mental classes are a different partition**, as Appendix H found: on its own the mental GRM separates people by how persistent their score is (shares .47/.35/.19, persistence .94/.80/.63) rather than by its level, so its largest class is its lowest. Against the multidimensional classes the agreement is ARI 0.20 with 49% in the same class: the multidimensional worst class is the univariate mental worst class (97%), but the multidimensional middle class splits between the two lower mental classes and the best between the two upper ones.
- **The mental paths the multidimensional fit learns are steeper and more spread than the univariate ones**: at 55 its classes sit at -1.00, 0.06 and 0.73 on the mental score against -0.41, 0.27 and 0.91 for the univariate classes, and the worst class's mental path falls by a standard deviation from 20 to 55 where the univariate worst class falls by 0.8. With the classes pinned by physical health and mortality, the mental channel reads as the mental-health profile of the physical types, which is what the joint model is for; on its own, the mental score is too noisy to carry a level typology.

[**TODO:** Expanding windows are currently only on the K-means - will take a bit of time to run them on the Bayesian]

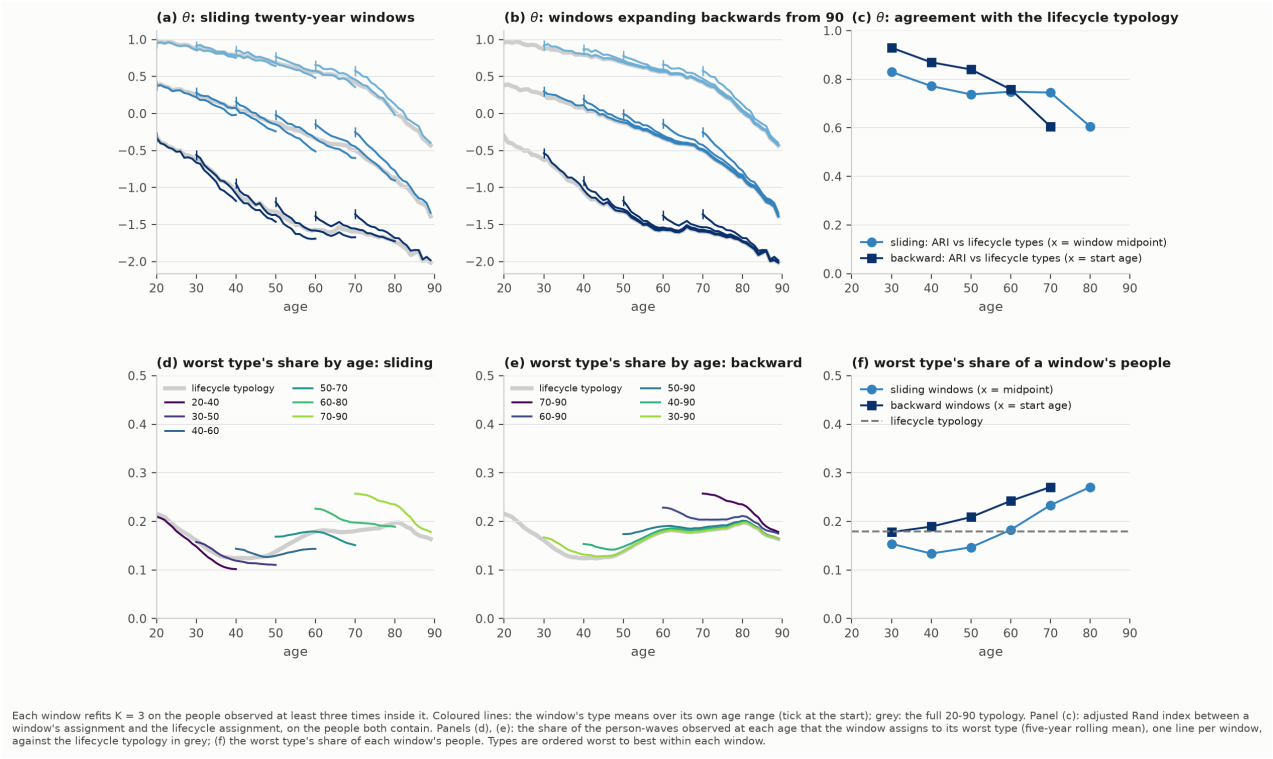


Figure 10: Partial K-means types on θ over sliding twenty-year windows (a) and windows expanding backwards from 90 (b), against the lifecycle typology in grey; (c) agreement with the lifecycle assignment. Lower row: the share of the person-waves observed at each age that each window assigns to its worst type, for the sliding (d) and backward (e) windows, against the lifecycle typology in grey; (f) the worst type's share of each window's people. The same figure on h is in Appendix C.

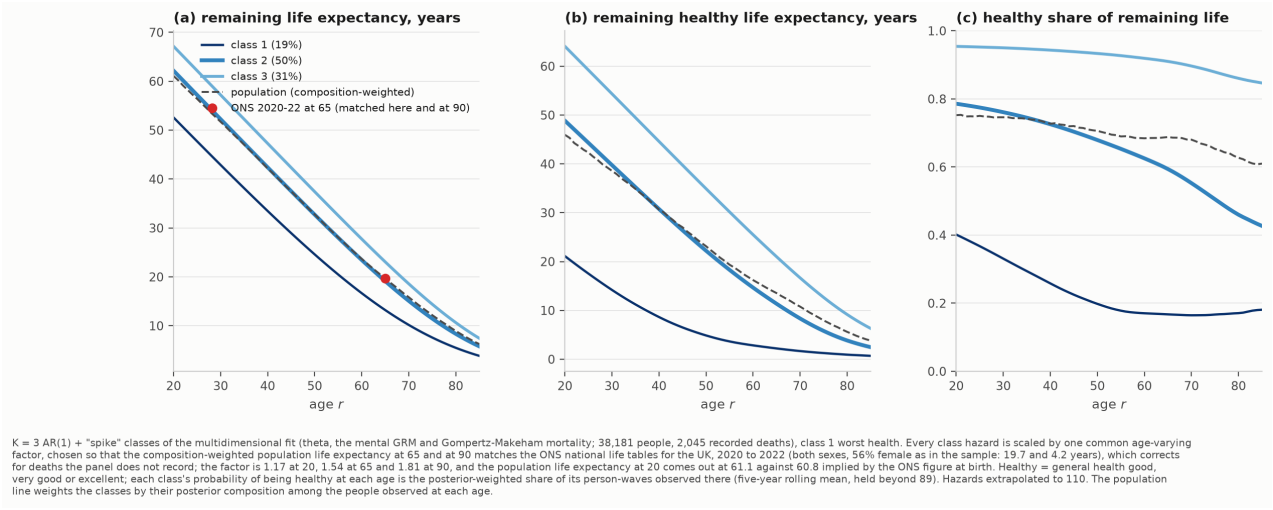


Figure 11: Remaining life expectancy and healthy life expectancy by class, $K = 3$ AR(1) plus "spike" (the multidimensional fit: θ, the mental GRM and Gompertz–Makeham mortality). (a) Remaining life expectancy from each age, by class and for the population; (b) remaining healthy life expectancy, healthy meaning general health reported good or better; (c) the healthy share of remaining life. Every class hazard is scaled by one common age-varying factor so that the population life expectancy at 65 and at 90 matches the ONS national life tables for the UK, 2020 to 2022, which corrects for the deaths the panel does not record; the class differences are the fit's, the level is the UK's.

	from age 20		from age 40		from age 65		from age 80	
	LE	HLE	LE	HLE	LE	HLE	LE	HLE
class 1 (19class 2 (50class 3 (31population	61.1	46.0	42.2	30.7	19.7	13.5	8.9	5.6

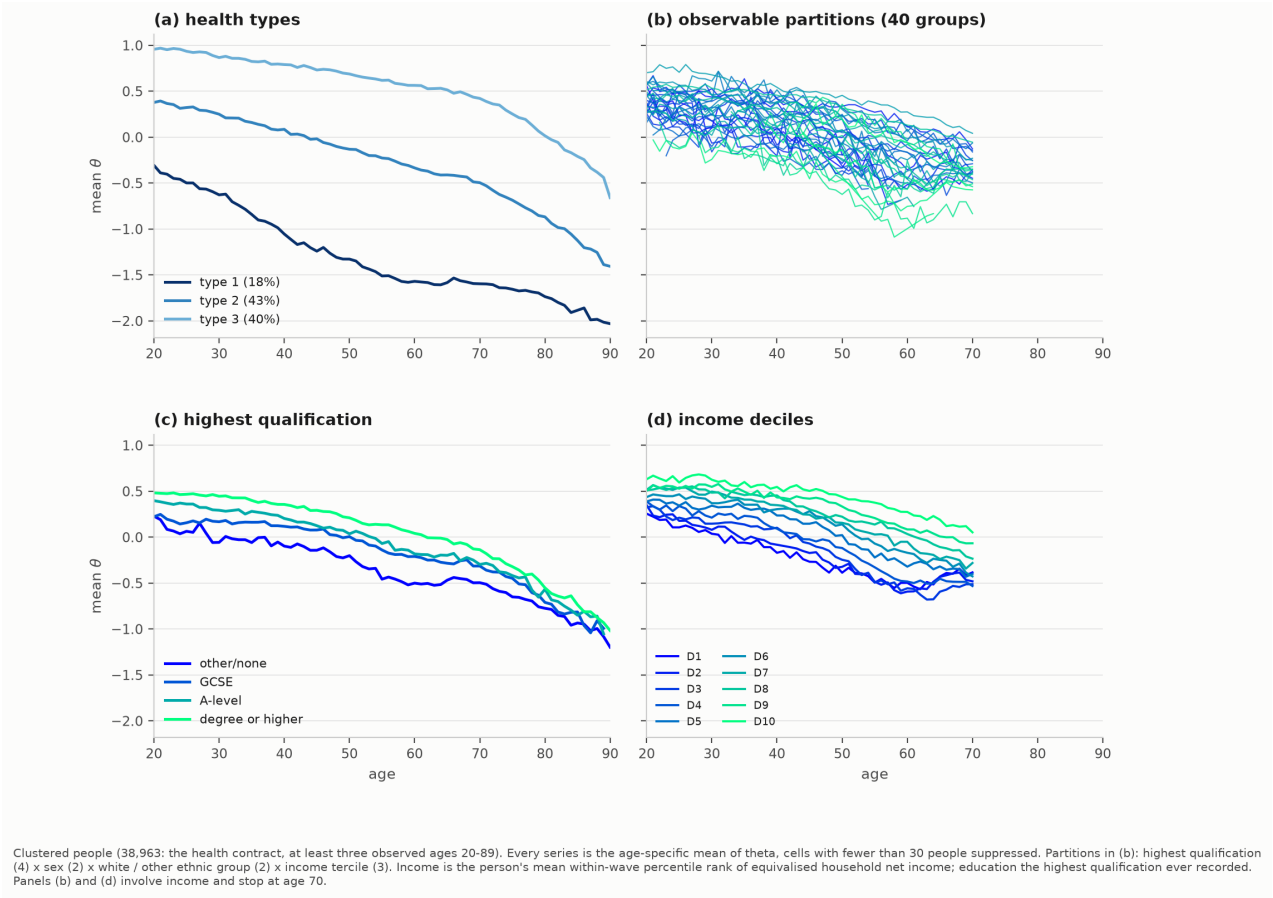
Table 2: Remaining life expectancy (LE) and healthy life expectancy (HLE), in years, by class and for the population, from ages 20, 40, 65 and 80; Figure 11.

- **Life expectancy by class** (Figure 11, Table 2; `descriptives/44_paper_life_expectancy.py`). The panel records a death only when the survey learns of it, so the fitted hazards understate mortality: left alone they give a population life expectancy of 23.5 years at 65 against the ONS figure of 19.7 (both sexes, at the sample's 56% female). One common factor on every class hazard, rising from 1.17 at 20 to 1.54 at 65 and 1.81 at 90, matches the ONS values at 65 and at 90 exactly, and then gives 61.1 years at 20 against the 60.8 implied by the ONS figure at birth, which it was not fitted to. The class *differences* are the fit's; only the level is calibrated.
- From age 20 the three classes expect 52.6, 62.2 and 67.2 more years: a gap of 14.6 years between the worst and the best class, which narrows to 9.9 years at 65 (13.2 against 23.1) and 5.2 at 80.
- Healthy life expectancy separates the classes far more. From 20 the worst class expects 21.1 healthy years of its 52.6, the best 64.1 of 67.2: the worst class spends 60% of its remaining life in less than good health, the best 5%. From 65 the worst class has 2.2 healthy years left of 13.2, the middle class 11.4 of 19.1, the best 21.0 of 23.1. Most of what separates the types is years of ill health lived, not years of life.
- The population line weights the classes by their posterior composition among the people observed at each age, so it carries the selection of Figure 8. Its healthy life expectancy at 65 (13.5 years) is above the ONS healthy life expectancy, which uses the same self-rated definition, because the

panel's survivors at each age are healthier than the population's; the mortality calibration corrects the length of life but not the health of those observed. [**TODO:** Decide whether to calibrate the healthy share as well, against the ONS health state life expectancies.]

4.4 (C) Types against observables

- Figure 12: (a) the three types; (b) every cell of highest qualification × sex × white/other ethnic group × income tercile, 40 groups with at least 30 people at an age; (c) education; (d) deciles of the person's mean within-wave rank of equivalised household net income (both built from the raw files, `data_cleaning/06_build_observables.py`). Region is left out of (b) because twelve regions empty most cells; add it if the point needs making with more lines.
- Gradients exist and are in the expected order: at 50, θ is 0.21 for degree holders against −0.20 for no qualification, and 0.44 for the top income decile against −0.39 for the bottom, with the income gap widest at 55 (0.87 standard deviations).
- But even the 40-way partition spans a band of 0.9 to 1.1 standard deviations across ages 30 to 70, against 1.5 to 2.0 between the types.
- The between-group share of the cross-sectional variance is 3% for education, 5–8% for income deciles and 6–9% for the 40 cells, against 48–65% for the types; the observable partitions are essentially independent of the types (ARI 0.01–0.03).
- *[The between-share comparison is partly mechanical (the types maximise it); the fairer statements are the ARI, the type composition of each observable group, and the predictive comparison below.]*



Clustered people (38,963: the health contract, at least three observed ages 20-89). Every series is the age-specific mean of theta, cells with fewer than 30 people suppressed. Partitions in (b): highest qualification (4) x sex (2) x white / other ethnic group (2) x income tercile (3). Income is the person's mean within-wave percentile rank of equivalised household net income; education the highest qualification ever recorded. Panels (b) and (d) involve income and stop at age 70.

Figure 12: Lifecycle trajectories of θ by (a) health type, (b) 40 observable partitions, (c) education and (d) income decile, on the clustered people; cells with fewer than 30 people suppressed. The same figure on h is in Appendix C.

Predictive value.

- Rebuilt on the held-out fits (`descriptives/33_paper_prediction.py`): every Bayesian fit was refitted with each person's last two observed rows held out (the 29,901 people with at least five rows on the health contract; 26,664 people are in both the single-channel and the multidimensional held-out samples and the exercise runs on them). One prediction per person and horizon from the last fitted row t, with outcomes at the two held rows $t+1$ and $t+2$: the measure, the cost proxy in pounds (the flat-cost index capped at its 99th percentile, waves 7–15), and employment (employed or self-employed, people aged 20–64 at t). Class posteriors and the models' own forecasts are computed from the fitted window alone and carried to the outcome's age, so nothing after t informs them. People are split 70/30, fitted on the 70 and scored on the 30 (Figure 13, Table 3).
- **Observables predict almost nothing about health or cost, but more than health about employment.** Education, income and sex add 0.06 of R^2 for θ_{t+1} and 0.01 for cost over age, against 0.48–0.54 and 0.08–0.09 for the held-out classes. For employment they lift the AUC from 0.64 to 0.78, against 0.74–0.76 for any health summary, and the two together reach 0.82. Employment at t alone reaches 0.90: once current status is known, health adds little to the labour-market outcome.
- **The current score and the classes carry about the same information, and the model's**

forecast beats both. θ_t alone reaches $R^2 = 0.69$ for θ_{t+1} and 0.14 for cost, against 0.68 and 0.12 for the AR(1) plus "spike" classes, and adding the classes to the score gains 0.03 for health and nothing for cost. The model's own forecast of θ_{t+1}, the class-weighted path plus the filtered state carried forward, one number per person, does better than the score plus its lag ($R^2 = 0.74$ against 0.73): the classes summarise the history, and the history is what matters.

- **Two waves ahead the ranking is unchanged.** R^2 for θ_{t+2} is 0.61 for the iid classes, 0.66 for the AR(1) plus "spike" classes, 0.66 for θ_t and 0.71 for the model forecast, against 0.62, 0.68, 0.69 and 0.74 one wave ahead; for cost the score falls from 0.14 to 0.11 while the classes hold at 0.12. The classes lose less over the extra wave than the current score does and draw level with it at $t + 2$, but only the forecast, which carries the filtered state forward, stays clear.
- The multidimensional classes and forecast match the single-channel ones (R^2 0.61 and 0.74 for θ_{t+1} against 0.68 and 0.74; 0.11 for cost against 0.12): the mental channel and the mortality hazard neither help nor hurt the physical forecast. The same holds on h (0.63 and 0.77 against 0.68 and 0.78; Figure C.17).

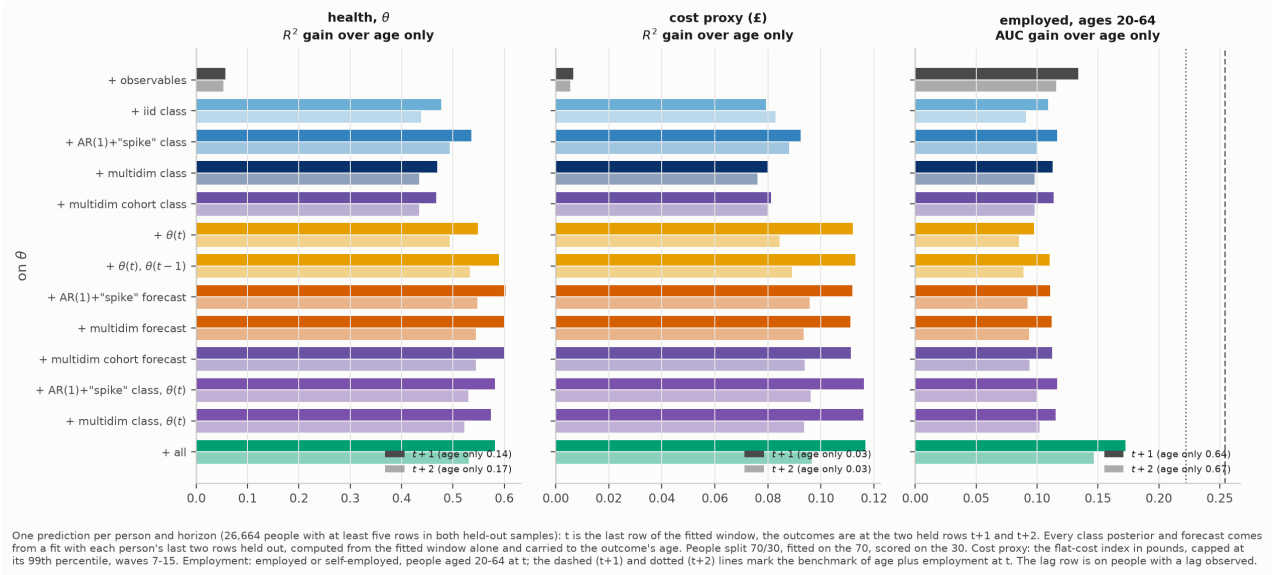


Figure 13: Out-of-sample gain over an age-only quadratic in predicting the measure, the cost proxy in pounds and employment (ages 20–64) one and two waves ahead, from each set of predictors known at t, on θ, with every class posterior and forecast from a fit that held out the two target rows; the cohort rows use the multidimensional fit with birth-decade shifts. In the employment panel the dashed $(t+1)$ and dotted $(t+2)$ lines mark age plus employment at t. The same figure on h is in Appendix C.

predictors at t	health, measure (R^2)		cost proxy (£) (R^2)		employed, ages 20-64 (AUC)	
	$t+1$	$t+2$	$t+1$	$t+2$	$t+1$	$t+2$
on θ						
age	0.140	0.167	0.032	0.027	0.643	0.666
+ observables	0.197	0.219	0.039	0.033	0.777	0.782
+ iid class	0.618	0.605	0.111	0.110	0.753	0.758
+ AR(1)+"spike" class	0.677	0.661	0.124	0.115	0.760	0.766
+ multidim class	0.609	0.601	0.112	0.103	0.756	0.764
+ multidim cohort class	0.608	0.601	0.113	0.107	0.757	0.764
+ theta(t)	0.689	0.661	0.144	0.112	0.741	0.752
+ theta(t), theta(t-1)	0.730	0.699	0.145	0.116	0.754	0.755
+ AR(1)+"spike" forecast	0.743	0.714	0.144	0.123	0.754	0.759
+ multidim forecast	0.740	0.711	0.143	0.121	0.755	0.760
+ multidim cohort forecast	0.741	0.711	0.143	0.121	0.756	0.760
+ AR(1)+"spike" class, theta(t)	0.722	0.697	0.148	0.123	0.760	0.767
+ multidim class, theta(t)	0.715	0.689	0.148	0.121	0.759	0.768
+ all	0.723	0.697	0.148	0.124	0.816	0.813
benchmark: age, employed(t)					0.897	0.889

Table 3: Out-of-sample scores by predictor set, outcome and horizon, on the held-out fits on θ; R^2 for the continuous outcomes, AUC for employment. The same table on h is in Appendix C.

Predicting death.

- The held-out design cannot predict death at a fixed horizon: everyone at t has two more rows, and $t+2$ is by construction everyone's last interview. So mortality is shown two ways (Figure 14, `descriptives/41_paper_mortality_prediction.py`). On the full-sample fits, death before the next wave at every person-wave at risk (308,949 person-waves, 1,267 deaths): the single-channel classes never saw the death record, and the multidimensional class is taken from the physical and mental channels only, with the full-likelihood version, which has seen the decedent's event, shown as an upper bound. On the held-out fits, death after the last interview against leaving the panel alive, predicted from t (13,516 people, 7% deaths), with the multidimensional posterior from survival through the fitted window only.
- **Death is the outcome health predicts least beyond age.** On θ, at every person-wave, age alone gives AUC 0.85; any class adds about 0.03 (0.88), the current score 0.05 (0.90), and the classes add nothing on top of the score. The multidimensional class from the health channels alone does as well as the single-channel classes, and so does its own class-weighted Gompertz–Makeham hazard, one number per person; letting the class see the death record buys about 0.01. On the held-out comparison the ordering is the same, from 0.80 for age to 0.83 for the classes or the score. Utilisation is the opposite case: health does nearly all the work there, so prevention targeted on these measures should expect to move costs far more readily than mortality.

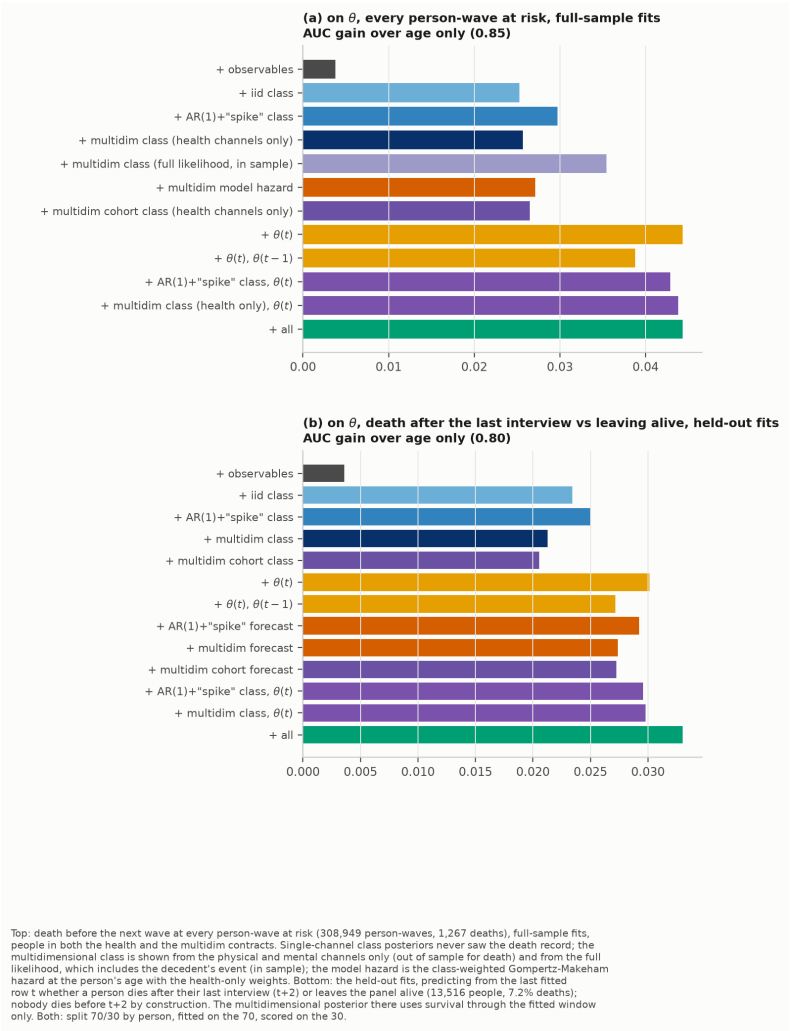


Figure 14: Predicting death, AUC gain over an age-only quadratic. (a, c) Every person-wave at risk on the full-sample fits: death before the next wave. (b, d) The held-out fits: death after the last interview against leaving the panel alive, predicted from the last fitted row. The cohort rows use the multidimensional fit with birth-decade shifts. On θ; the same figure on h is in Appendix C.



Figure 15: What identifies the type: the observables or a health reading? On θ; every typology is fitted without each person's first row (people with at least four rows, 33,880): partial K-means on the remaining rows, the independent-residuals and AR(1) plus "spike" mixtures with the first row held out, the type being the modal class of the posterior from the fitted window, and the multidimensional AR(1) plus "spike" fit with mortality likewise (on the multidimensional contract's people with at least four rows, 33,248). Each predictor set is known at the first row and sits on a quadratic in age: education, income rank, the three observables together, one reading of θ, and the reading plus the observables; multinomial logit, people split 70/30, scored on the 30. (a) Accuracy, dashed the majority-type rate; (b) macro one-vs-rest AUC. "Model posterior": the class posterior the fitted mixture itself implies from the one observation, no regression.

predictors at the first row	K-means		independent residuals		AR(1) + "spike"		multidimensional	
	accuracy	AUC	accuracy	AUC	accuracy	AUC	accuracy	AUC
majority type	0.420		0.432		0.585		0.524	
age(0)	0.431	0.541	0.438	0.528	0.585	0.594	0.524	0.532
+ education	0.458	0.591	0.468	0.585	0.585	0.629	0.524	0.575
+ income rank	0.469	0.618	0.488	0.616	0.591	0.649	0.527	0.604
+ education, income, sex	0.471	0.628	0.490	0.626	0.593	0.658	0.530	0.618
+ $\theta(0)$	0.645	0.808	0.658	0.814	0.695	0.814	0.638	0.781
+ $\theta(0)$, education, income, sex	0.654	0.815	0.666	0.821	0.698	0.820	0.640	0.789
+ $\theta(0)$, mental $\theta(0)$							0.680	0.823
model posterior from the first row			0.650	0.813	0.693	0.797	0.665	0.821

Table 4: Accuracy and macro one-vs-rest AUC on the test 30% for each predictor set and typology, Figure 15.

- **What identifies the type: a health reading, or education and income?** (Figure 15, Table 4; `descriptives/43_paper_type_from_initial.py`). The types are refitted without each person's first row, so that the first observation can be used to predict them without having shaped them, and the observables are set against a single reading of θ at that row.
- **The observables say little.** On the K-means types, age alone matches the majority rate (0.43

against 0.42, AUC 0.54); education takes the AUC to 0.59, income rank to 0.62, and all three observables together to 0.63 with accuracy 0.47. The same holds on every typology: the observables add 0.08–0.10 of AUC over age and 0–5 points of accuracy.

- **One health reading says a lot.** θ at the first row takes the AUC to 0.81 on every single-channel typology and the accuracy to 0.65–0.70, two people in three. On top of the reading the observables add one point of accuracy and 0.01 of AUC: nearly everything education and income know about the type, the health reading already knows.
- **The fitted model is the right classifier.** The mixture's own posterior from the one observation, which discounts it for the "spike", does as well as the regression (0.69 against 0.70 on the AR(1) plus "spike" classes; 0.65 against 0.66 on the independent-residuals classes). On the AR(1) plus "spike" classes the majority rate is 0.59 because the posterior concentrates its modal assignments in the middle class (59% modal against a 48% share), so the accuracy gain is smaller there while the AUC is the same.
- On the multidimensional classes, θ at the first row gives AUC 0.78; adding the mental score at the first row gives 0.82, and the model's own posterior from the two first-row scores 0.82. One reading of each channel identifies the joint type about as well as one reading of θ identifies the physical type; the observables again add 0.01.
- A reading at 20–34 classifies the K-means and independent-residuals types worse (0.60–0.61) than one at 35 or later (0.66–0.69). Holding out the first two rows instead, so that the type is defined from the third row on, changes nothing (0.64 from one reading on the K-means types, against 0.65); a second reading adds five points (0.69, AUC 0.85), the gain from averaging out one "spike". [**TODO:** The first-two-rows and h versions, with their mixture rows, go to the appendix when the fits land.]

5 Prevention exercises

We price the returns of Section 2 on the $K = 3$ AR(1) plus "spike" classes on θ (Figure 6(b)), with the independent-residuals classes alongside for comparison. The cost function is the curve estimated in pounds on θ (Figure 1), a cubic fitted to the uncapped cost index over waves 7–15, and the returns take the general-cost form of Section 2.2. Class shares and paths are the fitted mixture at its posterior mean; $\mathbb{E}[-c'(\theta) \mid j]$ at each age is the posterior-weighted mean over the person-waves observed there, and the bill $C(r)$ and its three parts are the observed cross-section priced by the same curve. Returns are pounds of treatment cost to 90 avoided per unit of μ, with no survival weighting and no discounting. The K-means version and the capped cost curve are in Appendix G; the same exercise on h is in Appendix C.

- **The bill and what prevention avoids** (Figure 16, Table 5). The treatment bill from 30 is £38,000 per person to 90, £30,000 from 50 and £18,000 from 70. On the AR(1) plus "spike" classes about 71% of it is the cost along the mean path, 13% is what the type differences add and 16% is realised deviations, the part no prevention reaches (15% from 50, 13% from 70). Uniform prevention from 30 avoids £35,000 of it per unit of μ, so $P(r)/C(r)$ is 0.91 at 30, falling to 0.67 at 50 and 0.39 at 70: the later the start, the larger the share of the remaining bill that is already determined. On the independent-residuals classes the shares are 0.68, 0.47 and 0.26, since more of the decline is read as noise.
- **Timing.** On the AR(1) plus "spike" classes the between-class channel of $P(r)$ is positive at every start age: +35% of $P(30)$, +25% of $P(50)$ and +7% of $P(70)$. On the independent-residuals classes it is +13% at 30 and turns negative by 50 (−8%), reaching −42% at 70: the sign change

of Figure 2 priced. Once the one-period noise is absorbed, the classes fan out for longer, and divergence keeps working for uniform prevention into old age.

- **Targeting.** From 30 the worst class's return is £72,000, against £35,000 uniform and £5,600 for the best class, so class targeting collects about twice the uniform return (a gain of 107%). From 50 the worst class still has the highest return and the gain is 32%; from 70 the middle class does, and the gain is 50%. On the independent-residuals classes the gains are 151%, 49% and 50%.
- Skeleton for the full version:
  - posterior uncertainty, from the draws;
  - survival-weighted sums $\sum_{a \geq r} S(a \mid r)(\cdot)$ using the Gompertz–Makeham class hazards of the multidimensional model, both ways (the sickest die, prevention raises survival);
  - targeting on observables versus on early health history versus on the class, priced by the targeting identity of Section 2, using the Section 4.4 partitions;
  - the additive-prevention robustness check (the multiplicative $(1 - \mu)d$ favours decliners by construction), and prevention as moving a person to a better class;
  - cost sides: the returns are pounds of treatment avoided per unit of μ, so a policy's value still needs the cost of delivering a unit of μ; and when the savings come, since prevention is paid at r and the avoided treatment falls decades later.

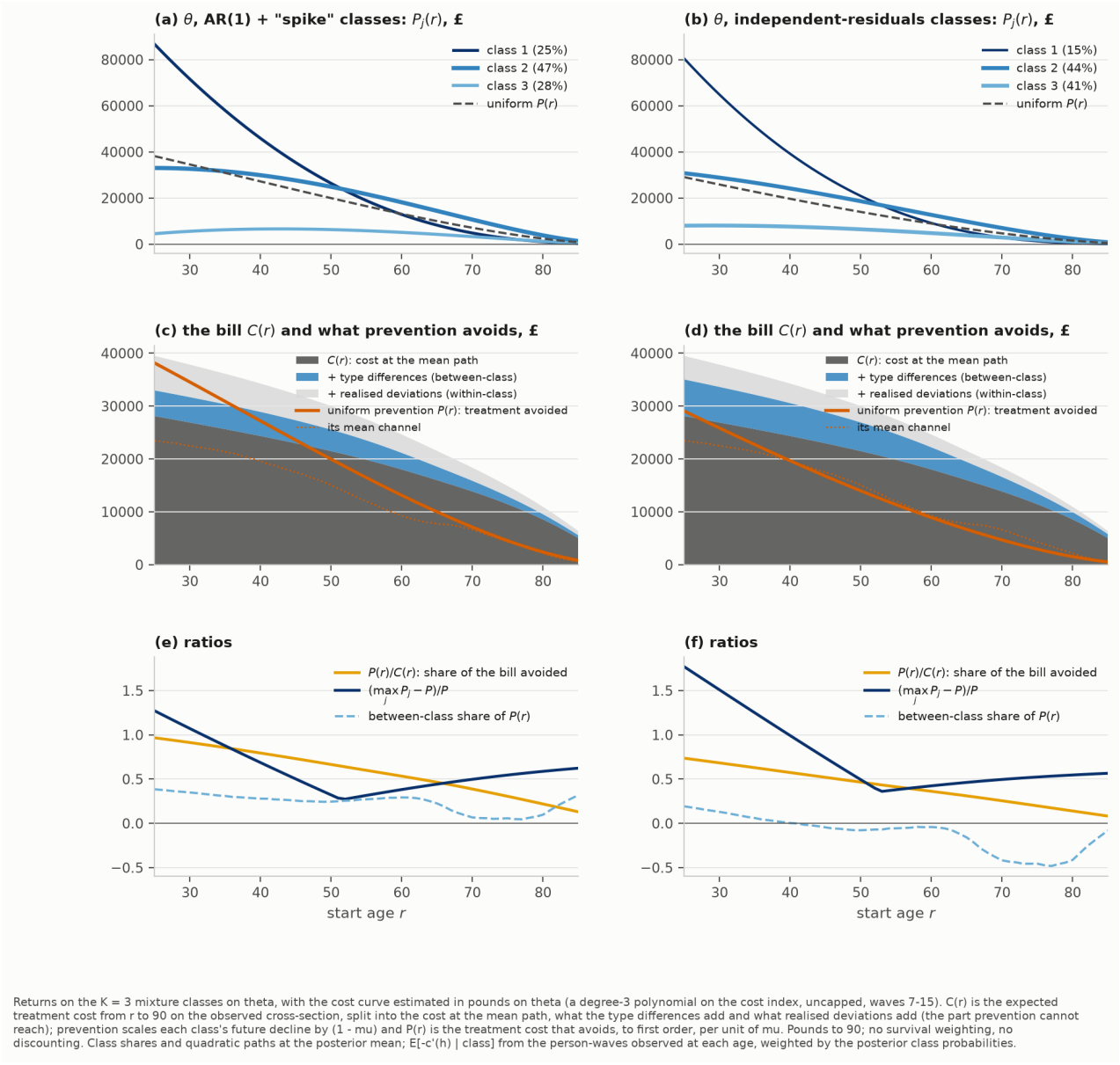


Figure 16: The treatment bill and the return to prevention by start age r on θ, in pounds under the estimated cost curve: (left) the AR(1) plus "spike" classes, (right) the independent-residuals classes. Top: $P_j(r)$ by class and the uniform $P(r)$, the treatment cost avoided per unit of μ. Middle: the bill $C(r)$ stacked into the cost at the mean path, what the type differences add and what realised deviations add, with $P(r)$ and its mean channel. Bottom: P/C, the share of the bill avoided; the gain from class targeting relative to the uniform return; and the between-class share of $P(r)$.

classes	r	$C(r)$	within	$P(r)$	P/C	between	P_1	P_2	P_3	gain
AR(1) + "spike" classes	30	37,834	16%	34,542	0.91	+35%	71,536	32,653	5,580	1.07
AR(1) + "spike" classes	50	30,005	15%	19,953	0.66	+25%	26,356	24,866	6,314	0.32
AR(1) + "spike" classes	70	18,339	13%	7,111	0.39	+7%	4,828	10,650	3,265	0.50
independent-residuals classes	30	37,834	11%	25,840	0.68	+13%	64,804	28,832	8,076	1.51
independent-residuals classes	50	30,005	10%	13,972	0.47	-8%	20,869	18,613	6,443	0.49
independent-residuals classes	70	18,339	9%	4,679	0.26	-42%	2,872	7,001	2,879	0.50

Table 5: The bill and the returns at three start ages on θ, in pounds under the estimated cost curve. $C(r)$ is the treatment bill still to come and "within" the share of it that realised deviations add, the part prevention cannot reach; $P(r)$ is the treatment cost uniform prevention avoids per unit of μ and P/C its share of the bill; "between" is the between-class channel of $P(r)$ as a share of $P(r)$; P_j the return by class; the gain is $(\max_j P_j - P)/P$.

Appendices

A Proofs and additional material for conceptual framework

[Include proofs for any propositions and perhaps some additional illustrative figures here.]

B Health measure and cost proxy

B.1 Health variables

Six SF-12 items enter as four testlets: the two physical-function items sum into PF, the two role-physical items sum into RP, and general health and pain enter alone. *[add explanation of relationship to PCS here.]* Every item is coded so category 1 is worst health. In addition to the SF-12 items, UKHLS asks which of a list of areas a person's health limits them in. Eight of these are physical and used here, in three groups:

1. LIM_FL (0-4): mobility (`disdif1`), lifting (`disdif2`), dexterity (`disdif3`), co-ordination (`disdif10`
2. LIM_SELF (0-2): continence (`disdif4`), personal care (`disdif11`)
3. LIM_SENS (0-2): hearing (`disdif5`), sight (`disdif6`)

In the early waves the battery was asked only of people reporting a long-standing illness. Someone who reports none therefore has no answer, not a zero. We count them as having no limitation, in every wave, so that the variable means the same thing across the panel; someone who does report a long-standing illness but has no usable answer is missing. Two areas are deliberately left out. "Recognising when in physical danger" is included with the mental health items and "Other health problem or disability" is included as a robustness variant and changes nothing. COND counts the sixteen ever-diagnosed physical conditions in the UKHLS inventory, 0 to 5 and 6 or more, the top category being the highest count whose share of person-waves is at least 0.5% (6 or more holds 0.86%).

Table B.1 shows the unconditional distribution of health variables. In total, 7.1% of person-waves give the best answer to all eight items, while only 12 person-waves sit at the worst answer on everything.

Table B.1: Unconditional distribution by health variable

	k	share of person-waves, best category first (%)								
GH	5	13.29	33.33	31.71	15.58	6.10				
PF	5	65.13	11.66	11.82	4.60	6.79				
RP	9	52.18	6.85	12.77	4.55	11.57	2.68	4.76	1.20	3.44
BP	5	53.72	25.24	9.14	8.26	3.64				
LIM_FL	5	83.05	6.66	5.30	3.24	1.76				
LIM_SELF	3	94.15	4.74	1.12						
LIM_SENS	3	95.62	3.82	0.56						
COND	7	48.28	28.88	12.90	5.59	2.44	1.05	0.86		

B.2 Measuring Health

[Fill in more details on the GRM here] Estimated with the latent distribution identified in the pooled sample and a grid of 121 equally spaced points on $[-6, 6]$ weighted by the normal density. Fitting took 60 EM iterations. Scores are expected a posteriori under the pooled prior, so a θ means the same thing at every age — the age profile is estimated separately and does not feed back into anyone's score.

Item	k	a	thresholds							
GH	5	2.03	−2.10	−0.98	0.16	1.45				
PF	5	3.52	−1.73	−1.38	−0.80	−0.40				
RP	9	3.09	−2.20	−2.03	−1.56	−1.37	−0.81	−0.64	−0.23	−0.03
BP	5	2.31	−2.35	−1.50	−0.99	−0.09				
LIM_FL	5	3.38	−2.49	−1.94	−1.46	−1.07				
LIM_SELF	3	2.43	−2.95	−1.99						
LIM_SENS	3	1.36	−4.49	−2.85						
COND	7	1.18	−4.61	−3.90	−3.12	−2.26	−1.25	0.11		

[Explain h vs θ figure here]

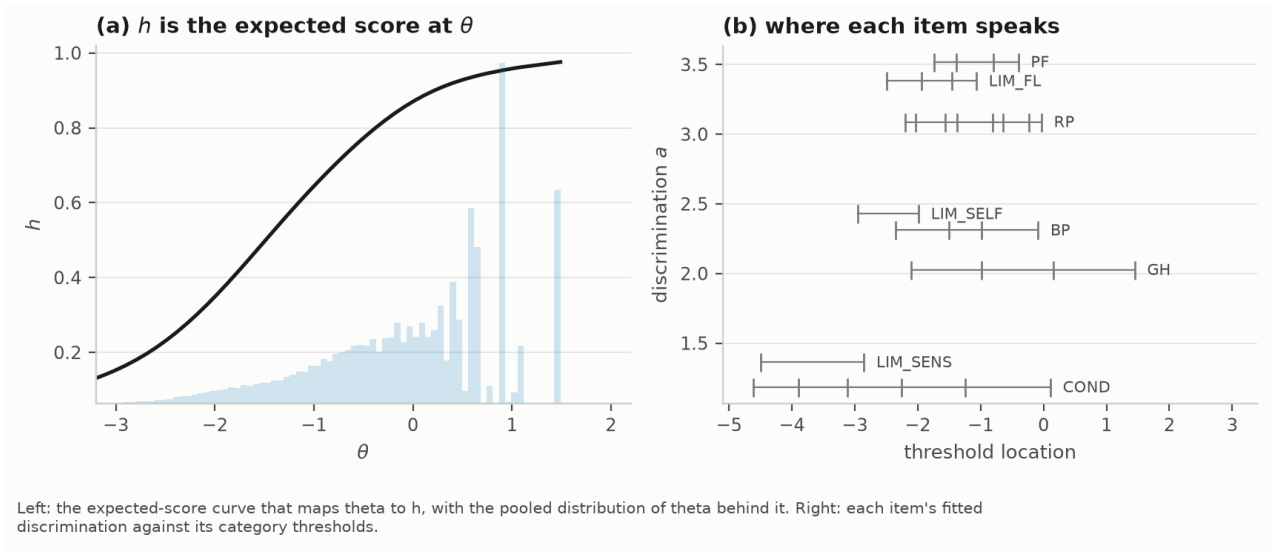


Figure B.1: The measure. (a) The expected-score curve mapping θ to h, over the pooled distribution of θ; (b) each item's discrimination against its thresholds.

	θ	h	frailty index (negated)	log frailty (negated)
UKHLS PCS	0.903	0.935	0.859	0.837
UKHLS MCS	0.274	0.266	0.232	0.228
chronic count (negated)	0.576	0.587	0.785	0.763
Spearman, vs θ	1.000	1.000	0.920	0.920

Table B.2: Correlations of θ, h, the frailty index and log frailty with the comparison metrics, 385,427 person-waves. The two frailty indices are negated, so higher is healthier on every column.

B.3 The frailty indices

The frailty index counts 31 deficits, each scored 0 to 1 and weighted equally: the six SF-12 physical items (general health, moderate activities, climbing stairs, accomplished less, limited in kind, pain), graded by response step; the eight physical impairment areas of Section B.1 (mobility, lifting, dexterity, co-ordination, continence, personal care, hearing, sight), each 0 or 1; and the 17 ever-diagnosed conditions of the UKHLS inventory (asthma, arthritis, congestive heart failure, coronary heart disease, angina, heart attack, stroke, emphysema, hyper- and hypothyroidism, chronic bronchitis, liver condition, cancer, diabetes, epilepsy, high blood pressure, clinical depression), each 0 or 1. The index is the share of deficits present, so higher is frailer. 7% of person-waves have none: 12% in the twenties, 6% in the fifties and under 1% past 80.

Log frailty is $\log(\text{frailty} + 1/31)$. The shift matters mainly at young ages, where the zeros are: with half the smallest positive value, 1/248, the zeros sit so far below the next group up that the variance at 25 is 1.15 and the profile peaks at 48, against 0.29 and a peak at 61 with 1/31.

The index differs from that of Hosseini et al. (2022) in three ways. They score 28 deficits, all 0/1, including high BMI, smoking and mental and cognitive impairments; they exclude self-rated health; and they model the zeros with a separate probit rather than shifting them.

B.4 A proxy measure for healtcare costs

Variables for the cost proxy are shown below

Quantity	Source	Status
Any admission	`hosp`, waves 7–15	complete for respondents
Nights	`hospd`	all admitted; median 3, mean 7.5, max 300
Maternity flag	`hospch`	12.7% of admissions, mean age 31.7
Nights by new condition	`hospdc`*i*, waves 2–15	21% of admissions
Nights by prior condition	`hospdcp`*i*, waves 11–15	11% of admissions
either		29% of admissions, **65% of nights**
Attribution shares are for wave 15, the first wave carrying both questions.		
GP visits, banded	`hl2gp`	0 / 1–2 / 3–5 / 6–10 / more than 10
Out-patient attendances, banded	`hl2hop`	same bands

UKHLS gives three tiers of contact, not two: `hl2gp` is "Visited GP in last 12 months" and `hl2hop` "Hospital or clinic out-patient in last 12 months", both banded 0–4 across waves 7–15. (An earlier draft had the two swapped; the labels were re-checked directly in waves g, j and o.) Unit costs come from the NHS National Cost Collection, which publishes national average unit costs annually: non-elective in-patient episode about £1,590, elective about £3,684, day case about £738, excess bed day about £313, out-patient attendance about £120.

Each admission is costed at the non-elective episode rate plus excess bed days beyond the average inclusive length of stay; out-patient attendances and GP visits at their flat rates, with band midpoints $0, 1.5, 4, 8, 15$. Maternity admissions are separable. At indicative unit costs the flat index averages £539 per person-year: £243 in-patient, £182 out-patient, £114 GP.

The one real modelling decision: UKHLS gives total nights but not the number of spells, so a person with 20 nights might have had one long admission or four short ones, which differ in cost by a factor of about three. The index assumes one spell.

For the nights carrying a condition attribution, the flat rate is replaced by a condition-specific one, mapping UKHLS condition code to NCC HRG chapter to average cost, at chapter level since the survey identifies a condition and not a procedure. Unattributed nights take the case-mix-weighted average of the attributed ones *within the same decile of the health measure*, never a global average, which would flatten exactly the gradient the index is for. Attribution reaches 18.0% of admissions pooled over waves 7–15, below the 29% on wave 15 alone because `hospcp`/`hospdcp` begin at wave 11. Weighting moves the mean to £566, about 6%. More useful is what it does not do: the imputed condition weight is nearly flat across deciles of physical health, 1.16 to 1.31. Sicker people are not admitted for systematically more expensive conditions; they are admitted more often and for longer. Condition weighting is a level adjustment, not a shape one.

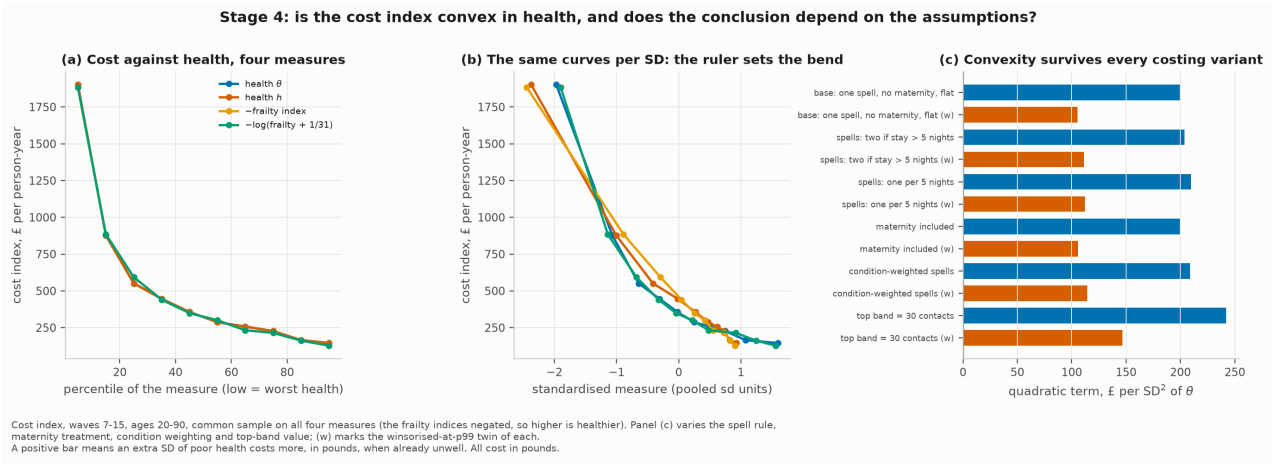


Figure B.2: Cost index in pounds against the four health measures (θ, h, and the frailty index and log frailty negated): by decile, per standard deviation of each measure, and the sensitivity of the convexity to every costing assumption the index rests on. Measures oriented so higher means better health. Panel (c) varies the spell rule, maternity treatment, condition weighting and top-band value, each with a winsorised-at-the-99th-percentile twin.

The costing assumptions do not matter, and neither does the health measure for the sign. All twelve variants of the sensitivity grid — one spell, two spells, one per five nights; maternity in or out; flat or condition-weighted; top band at 15 or 30 — are convex in pounds on the health measure's θ, with person-clustered t between 19 and 39. Winsorising at the 99th percentile halves the magnitude and changes nothing else; the sickest-to-healthiest ratio stays between 8.2× and 14.4×. The spell ambiguity, which looked like the most damaging gap in the data, is not binding. Convexity also holds on every health measure in the figure: $t = 10.3$ on h, 22.6 on log frailty and 5.8 on the frailty index in levels, against 19.5 on θ. What the horizontal measure sets is the size of the bend per standard deviation, not its sign: the quadratic term is £201 on θ against £113 on h, £237 on log frailty and £65 on the frailty index, and £106 against £35 on θ and h on capped costs.

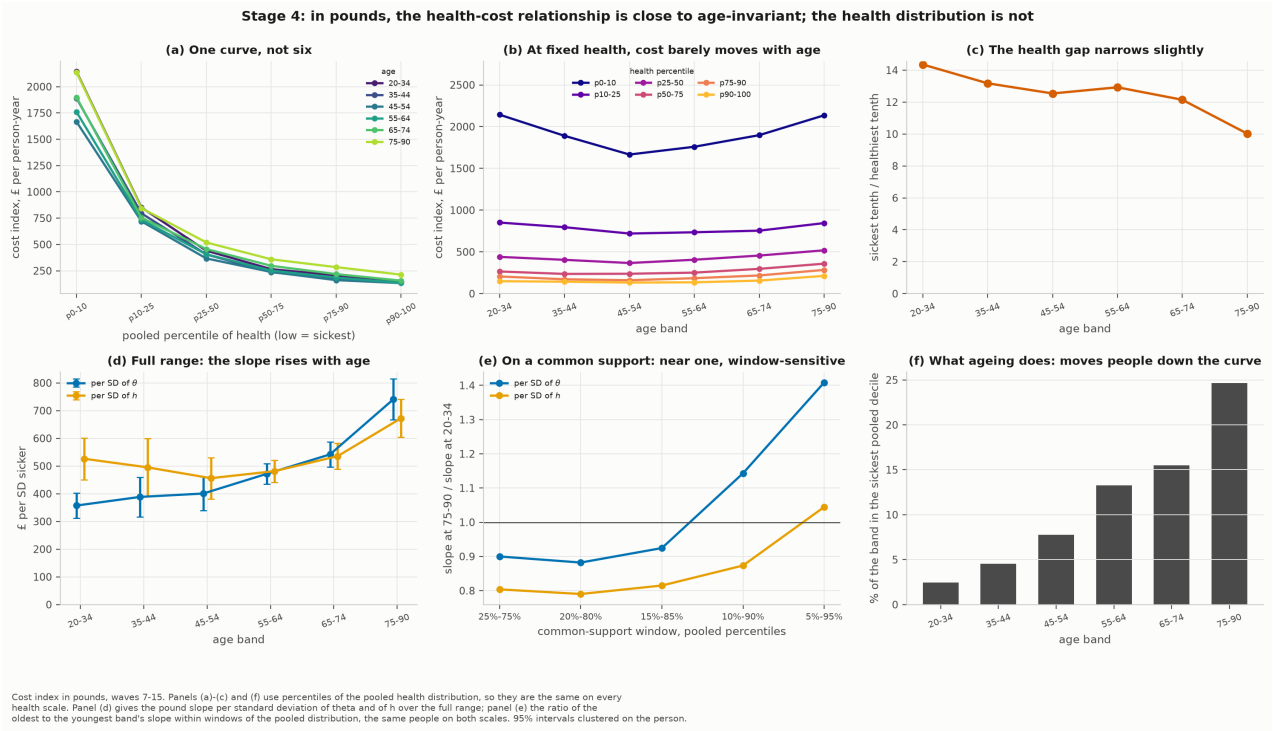


Figure B.3: The cost index in pounds against health within age bands. Panel (a) cost across pooled percentile bins of health, one line per age band; (b) the same table read down the other way; (c) the ratio of the sickest to the healthiest tenth; (d) the slope per standard deviation of θ and of h over the full range; (e) the ratio of the oldest to the youngest band's slope within windows of the pooled distribution; (f) the share of each band falling in the sickest pooled tenth. Panels (a)–(c) and (f) are the same on every health scale.

In pounds, the health–cost relationship is close to age-invariant on every measure. At a fixed level of health the index is nearly flat in age, shallow-U-shaped with a midlife minimum: the sickest tenth costs £2,143 at 20–34 and £2,136 at 75–90, and the sickest-to-healthiest ratio is 10–14× at every age. The full-range gradient does rise, £357 to £741 per standard deviation of θ and £526 to £672 per standard deviation of h, but in a pooled quadratic whose slope and curvature both interact with age, on costs capped at the 99th percentile, the curvature does not change with age ($t = 0.7$ on θ, 0.9 on h) and the slope flattens mildly ($t = -1.5$ on θ, -2.7 on h); uncapped, the 75–90 band's curvature is roughly double the others' ($t = 3.6$ and 3.0), a difference carried by the top 1% of costs. Ageing raises cost by moving people down a fixed curve, 2.5% of the 20–34 band in the sickest pooled tenth against 24.7% of the 75–90 band, not by changing its shape. Two cautions push the same way: the age profile is understated threefold, and UKHLS does not follow people into care homes, so the sickest old are under-represented in a way the sickest young are not.

B.4.1 Validation of cost proxy against published profiles

The utilisation sample is 301,390 person-waves on 59,750 people, waves 7–15; the validation restricts to ages 20–90, 283,852 person-waves. Three checks against what is independently known about NHS spending, none of which the index was fitted to.

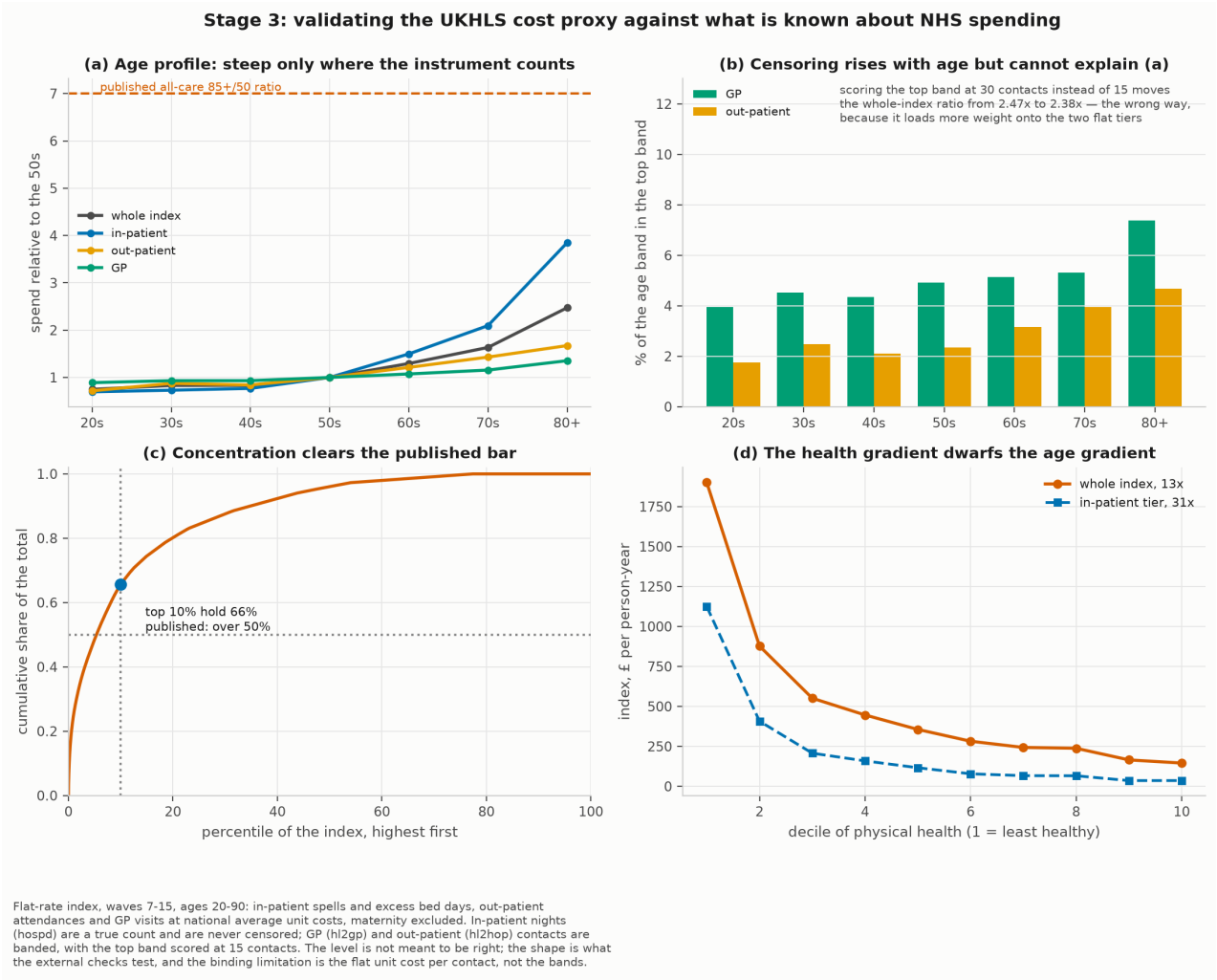


Figure B.4: Validation of the flat-rate index. Panel (a) the age profile of the whole index and of each tier, relative to the 50s, against the published all-care ratio; (b) the share of each age band top-coded at "more than ten" contacts, with the top-band sensitivity; (c) the concentration curve against the published top-decile share; (d) the gradient in physical health, the quantity the index is for.

- **Concentration passes.** The top 1% of person-waves hold 25.2% of the total, the top 5% 48.5% and the top decile 65.6%, against a published benchmark of more than half. If anything the index is over-concentrated, which is expected: 22.7% of person-waves are exactly zero on an index that carries no prescribing, A&E or community contact, the components that would give almost everyone a small positive amount.
- **Level undershoots, as predicted.** £539 per person-year against NHS spend per head near £3,000. Three tiers of contact priced at national averages cannot add to the national total; the shortfall is not informative.
- **The age profile strains.** The whole index puts the 80+ band at 2.5× the 50s, against a published all-care ratio near 7×. By tier: in-patient 3.8×, out-patient 1.7×, GP 1.4×. The one tier measured as a true count is the one with a credible gradient; the two banded tiers are nearly flat in age. The bands are not the reason: top-coding does rise with age (7.4% of the 80+ band at "more than ten" GP visits against 4.9% of the 50s), but scoring the top band at 30 contacts rather than 15 moves the whole-index ratio from 2.47× to 2.38×, the wrong way, because it loads more weight

onto the tiers that are flat in age. The binding limitation is that cost per contact is held fixed: an out-patient attendance for an 85-year-old and one for a 30-year-old enter at the same £120. That is what linkage to administrative records would fix.

The index's purpose is the gradient in *health*, and there it is strong: 13.1× between the least and most healthy decile on the whole index, 31.1× on the in-patient tier, against an age gradient of 2.5×. The caveat is that the age profile is understated by roughly a factor of three, so the index should not be used for anything whose answer depends on the age gradient: lifetime cost projections, discounting, or the timing of prevention returns. For those, report the in-patient tier alone or wait for linkage.

C The main figures on the other measures

Figures 1 to 5 are repeated here on h, the 31-deficit frailty index and $\log(\text{frailty} + 1/31)$, defined in Section 3 and Appendix B.3. The later figures are repeated on h only, since the mixtures have not yet been fitted on the frailty indices.

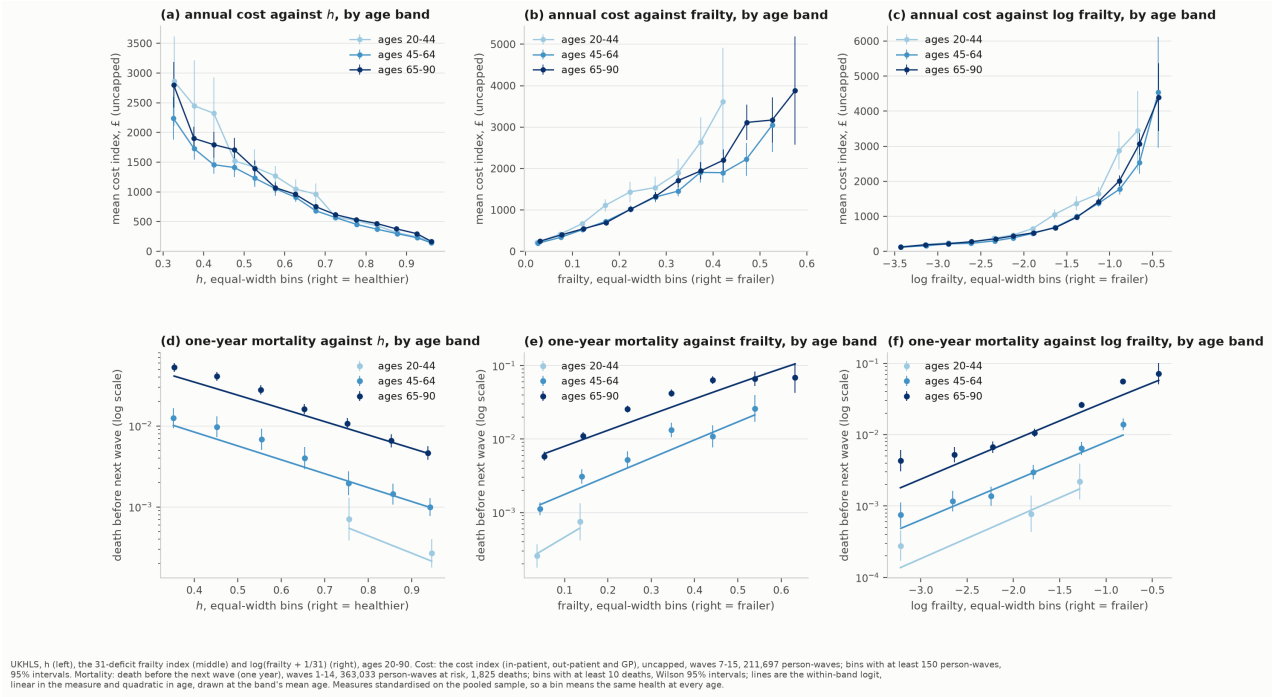


Figure C.5: As Figure 1, on h (left), the frailty index (middle) and log frailty (right). Top: mean annual cost in equal-width bins within three age bands; bottom: the one-year death rate on a log scale with the within-band logit. Higher is healthier on h and frailer on the two frailty indices.

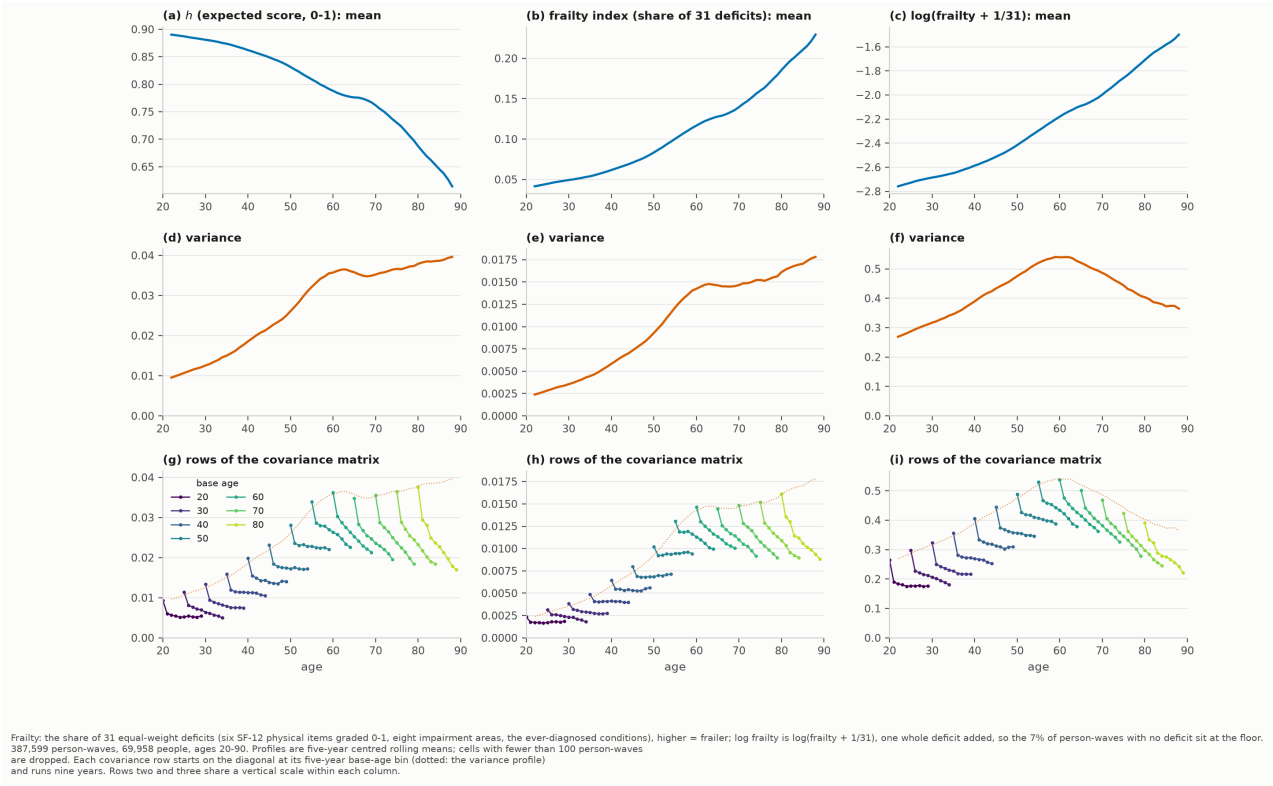


Figure C.6: As Figure 2, on h (left), the frailty index (middle) and log frailty (right): mean path, variance profile and covariance rows, model-free. Higher is healthier on h and frailer on the two frailty indices.

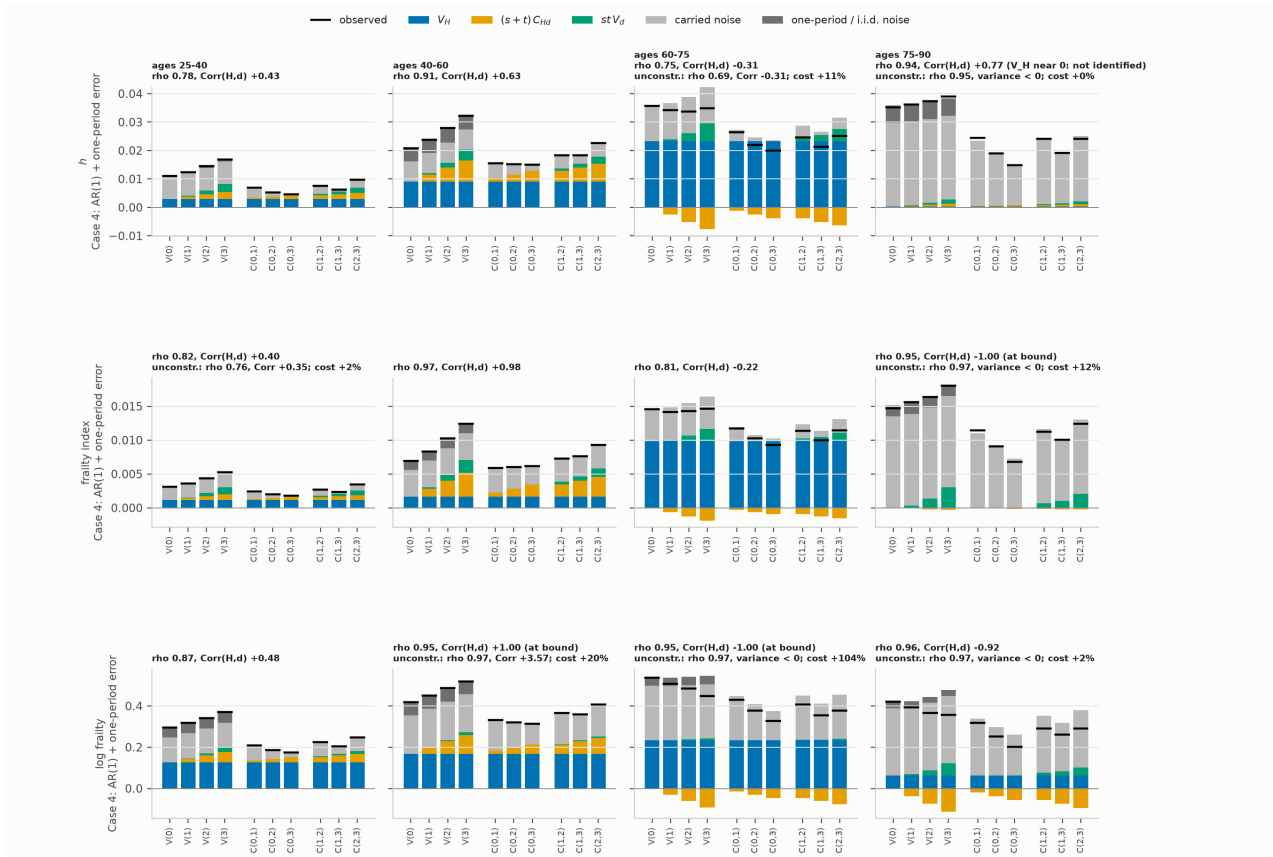


Figure C.7: As Figure 3, Case 4 on h (top), the frailty index (middle) and log frailty (bottom). Titles flag the bands where the admissibility constraint binds or V_H is below 5% of the first variance.

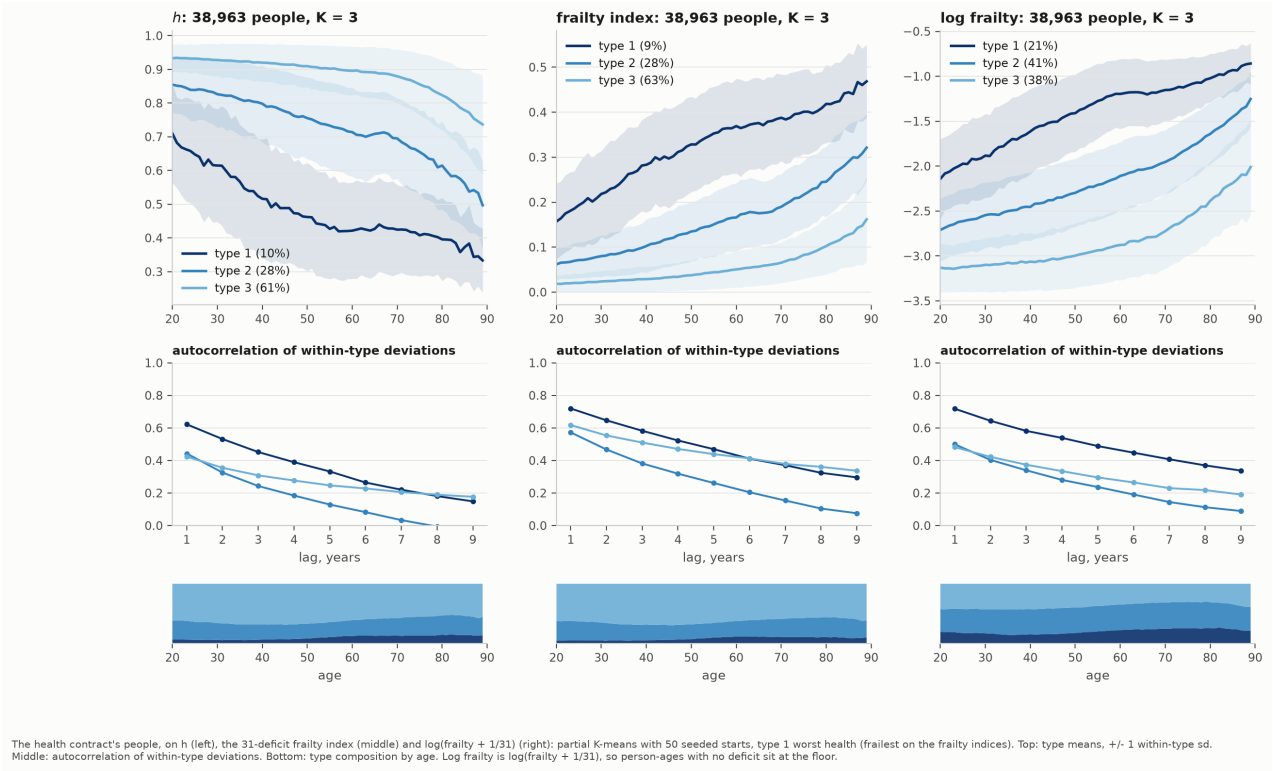


Figure C.8: As Figure 4, the partial K-means types on h (left), the frailty index (middle) and log frailty (right). Type 1 is worst health on every measure, so it is the top line on the two frailty indices.

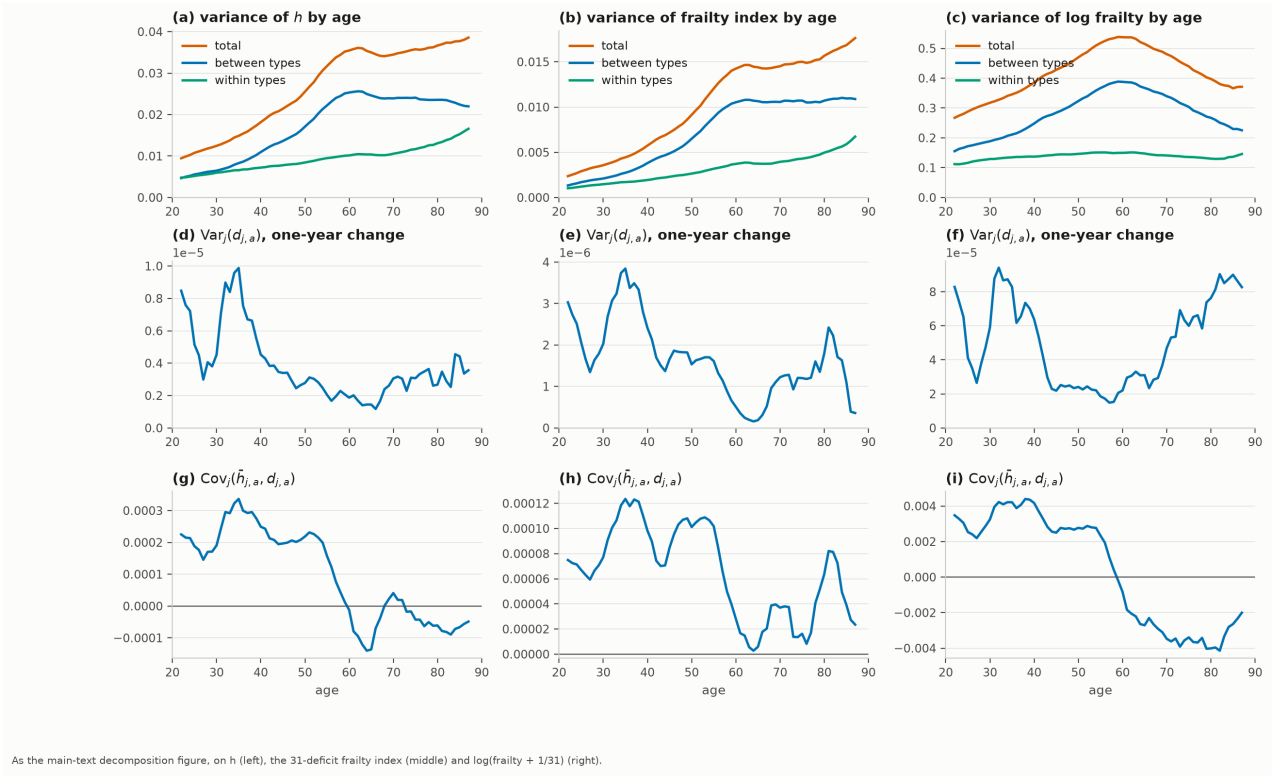
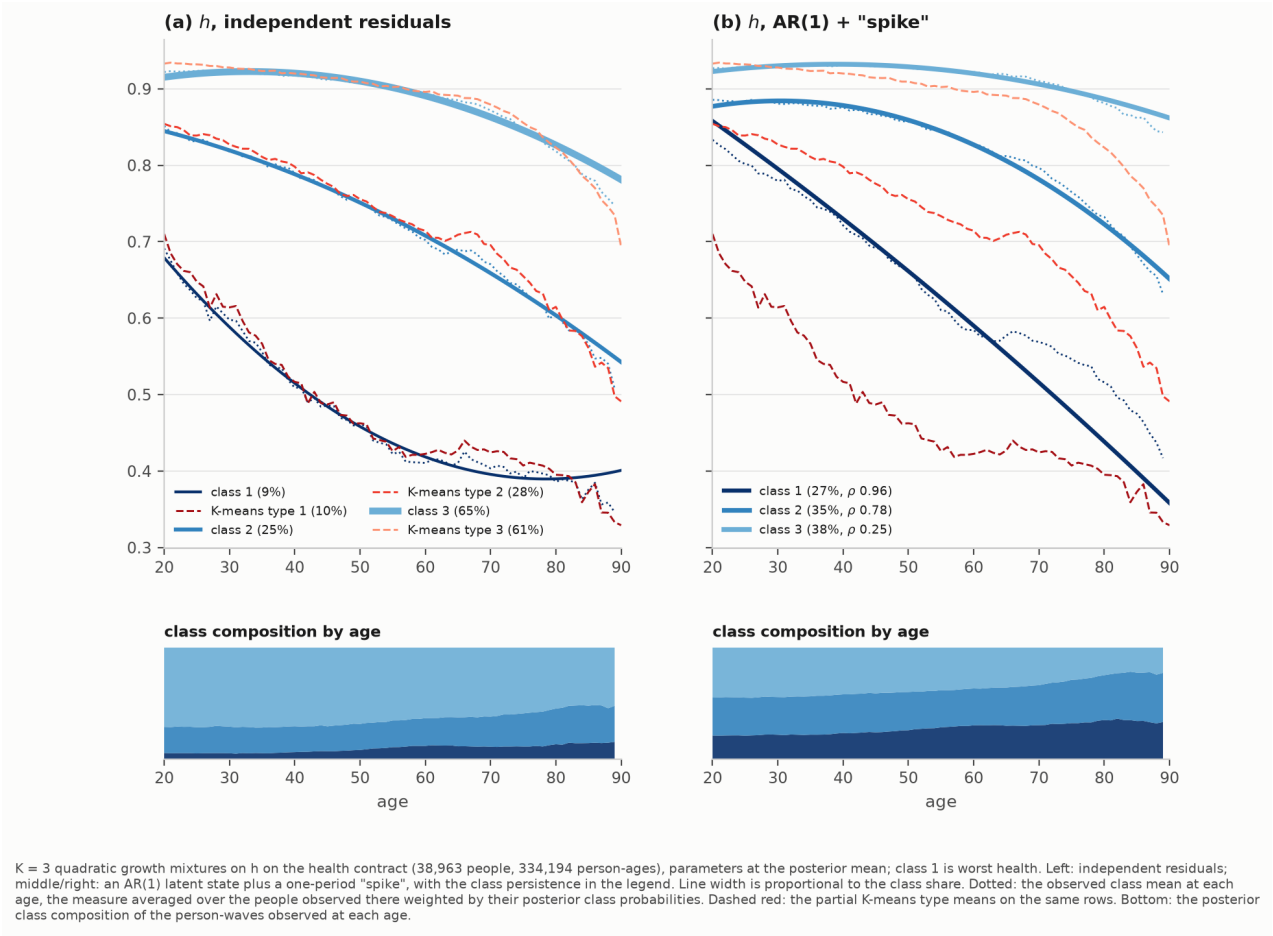
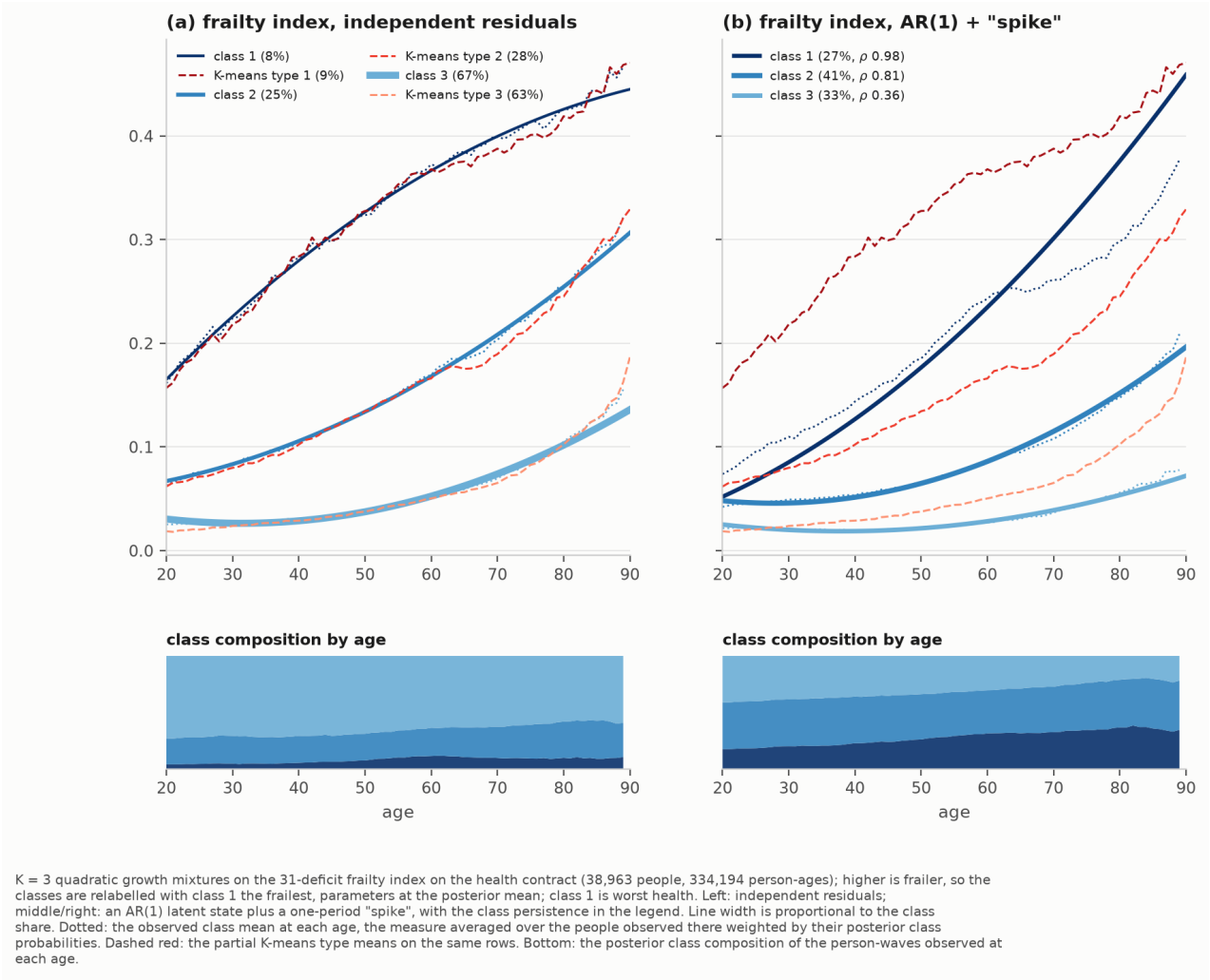


Figure C.9: As Figure 5, between versus within on h (left), the frailty index (middle) and log frailty (right).



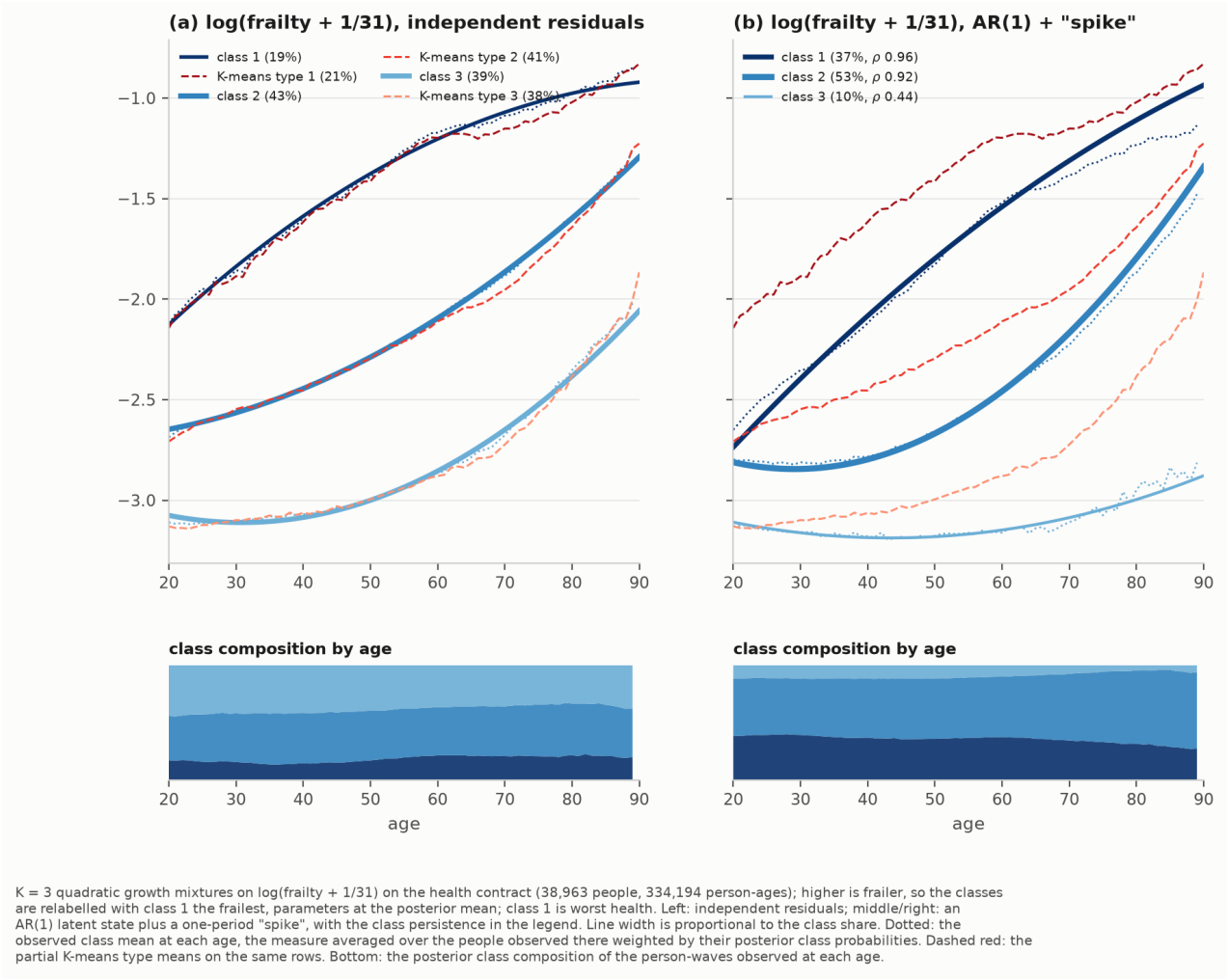
K = 3 quadratic growth mixtures on h on the health contract (38,963 people, 334,194 person-ages), parameters at the posterior mean; class 1 is worst health. Left: independent residuals; middle/right: an AR(1) latent state plus a one-period "spike", with the class persistence in the legend. Line width is proportional to the class share. Dotted: the observed class mean at each age, the measure averaged over the people observed there weighted by their posterior class probabilities. Dashed red: the partial K-means type means on the same rows. Bottom: the posterior class composition of the person-waves observed at each age.

Figure C.10: As Figure 6, the $K = 3$ mixtures on h: (a) independent residuals, (b) AR(1) plus "spike". Under (b) the healthiest class collapses to a persistence of 0.25.



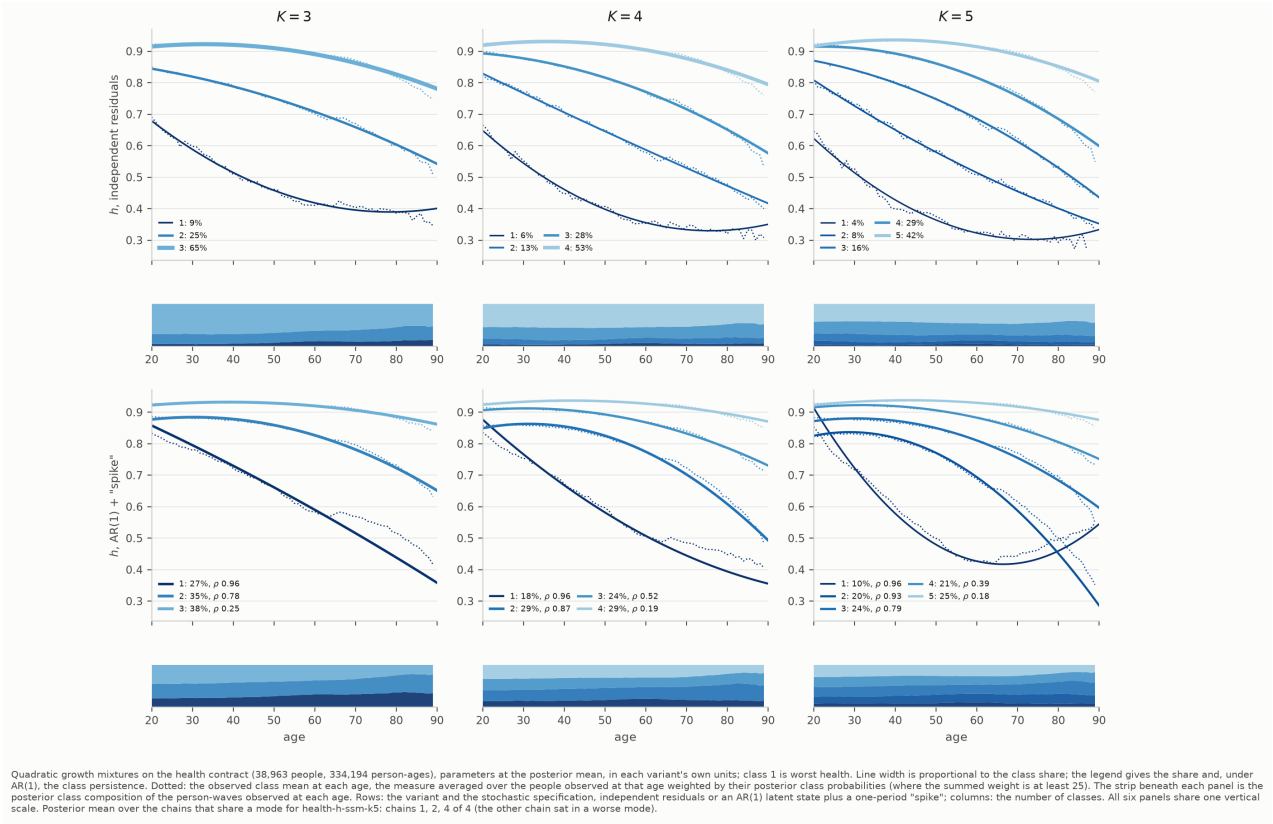
K = 3 quadratic growth mixtures on the 31-deficit frailty index on the health contract (38,963 people, 334,194 person-ages); higher is frailer, so the classes are relabelled with class 1 the frailest, parameters at the posterior mean; class 1 is worst health. Left: independent residuals; middle/right: an AR(1) latent state plus a one-period "spike", with the class persistence in the legend. Line width is proportional to the class share. Dotted: the observed class mean at each age, the measure averaged over the people observed there weighted by their posterior class probabilities. Dashed red: the partial K-means type means on the same rows. Bottom: the posterior class composition of the person-waves observed at each age.

Figure C.11: As Figure 6, the $K = 3$ mixtures on the 31-deficit frailty index, classes relabelled so that class 1 is the frailest: (a) independent residuals, (b) AR(1) plus "spike". Under (b) the healthiest class, at the floor of the index, collapses to a persistence of 0.36.



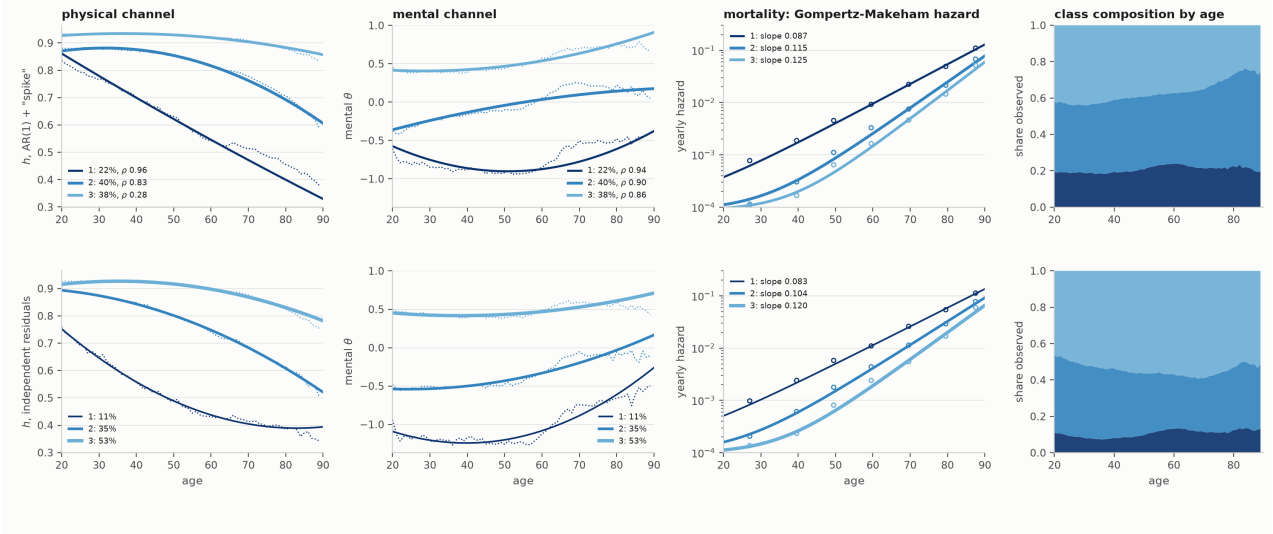
K = 3 quadratic growth mixtures on log(frailty + 1/31) on the health contract (38,963 people, 334,194 person-ages); higher is frailer, so the classes are relabelled with class 1 the frailest, parameters at the posterior mean; class 1 is worst health. Left: independent residuals; middle/right: an AR(1) latent state plus a one-period "spike", with the class persistence in the legend. Line width is proportional to the class share. Dotted: the observed class mean at each age, the measure averaged over the people observed there weighted by their posterior class probabilities. Dashed red: the partial K-means type means on the same rows. Bottom: the posterior class composition of the person-waves observed at each age.

Figure C.12: As Figure 6, the $K = 3$ mixtures on $\log(\text{frailty} + 1/31)$, classes relabelled so that class 1 is the frailest: (a) independent residuals, (b) AR(1) plus "spike". The shifted log lifts the floor only partly: under (b) a small healthiest class (10%) keeps a persistence of 0.44.



Quadratic growth mixtures on the health contract (38,963 people, 334,194 person-ages), parameters at the posterior mean, in each variant's own units; class 1 is worst health. Line width is proportional to the class share; the legend gives the share and, under AR(1), the class persistence. Dotted: the observed class mean at each age, the measure averaged over the people observed at that age weighted by their posterior class probabilities (where the summed weight is at least 25). The strip beneath each panel is the posterior class composition of the person-waves observed at each age. Rows: the variant and the stochastic specification, independent residuals or an AR(1) latent state plus a one-period "spike"; columns: the number of classes. All six panels share one vertical scale. Posterior mean over the chains that share a mode for health-h-ssm-k5: chains 1, 2, 4 of 4 (the other chain sat in a worse mode).

Figure C.13: As Figure 7, on h. The $K = 5$ fit under AR(1) plus "spike" is read over the three chains that share a mode; the fourth sat in a mode about 310 log-posterior units worse.



Multidimensional $K = 3$ mixtures on the multidim health contract (38,181 people, the health contract's people with a mental score on every row; 2,045 deaths). Rows: the physical measure and the stochastic specification. Columns: the physical and mental class paths at the posterior mean in each channel's units (dotted: the observed class mean at each age, posterior-weighted), the class-specific Gompertz-Makeham yearly hazard (circles: the posterior-weighted crude death rate by five-year band), and the posterior class composition of the person-waves observed at each age. Under AR(1) + "spike" the legend gives each class's persistence on that channel; the innovation and "spike" scales are common to the classes. The Makeham constant is below 0.0001 in every fit. Each column has one vertical scale across its rows. h + AR(1) + "spike" is read over the three chains that share a mode; the fourth sat in the mode with the class 2 and 3 intercepts pressed together on the ordering constraint, about 2,950 log-posterior units lower.

Figure C.14: As Figure 8, on h. Under AR(1) plus "spike" one of four chains sat in the mode with the class 2 and 3 intercepts pressed together on the ordering constraint, about 2,950 log-posterior units below the other three; the fit is read over the three chains that agree (split $\hat{R}$ 1.003).

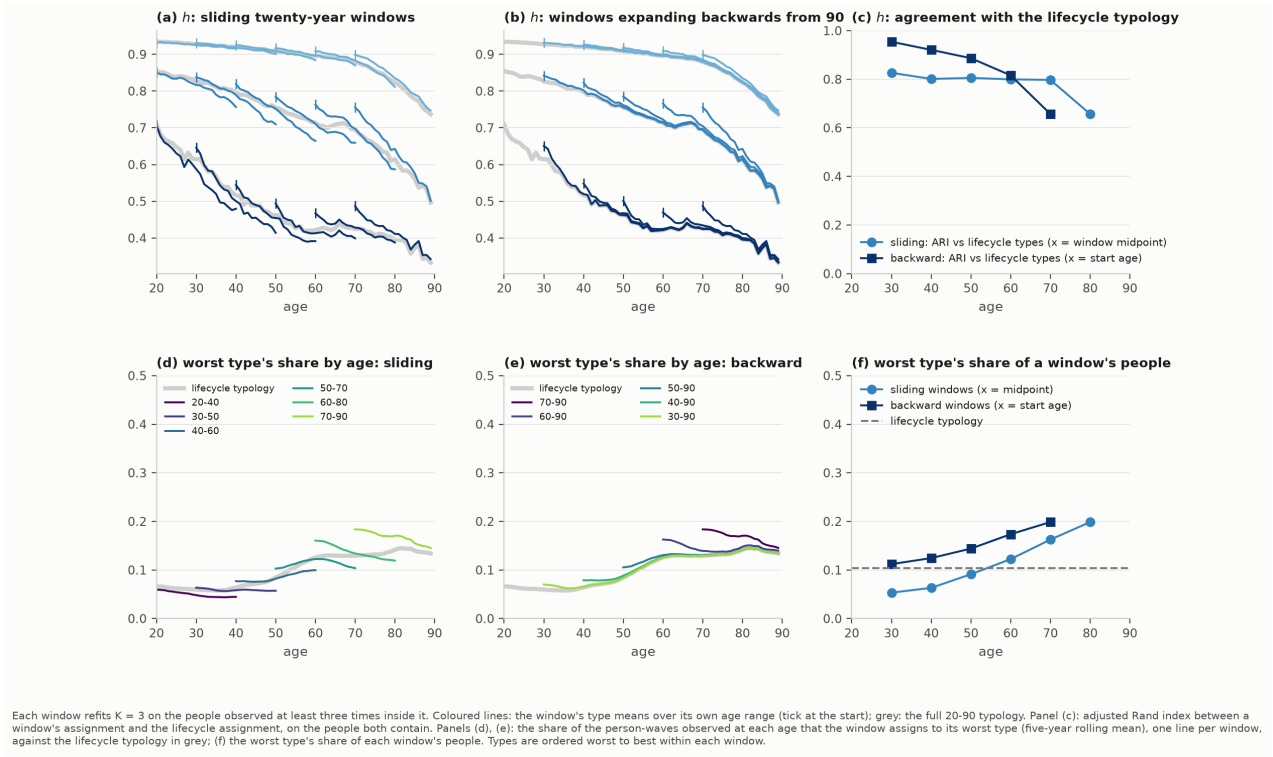
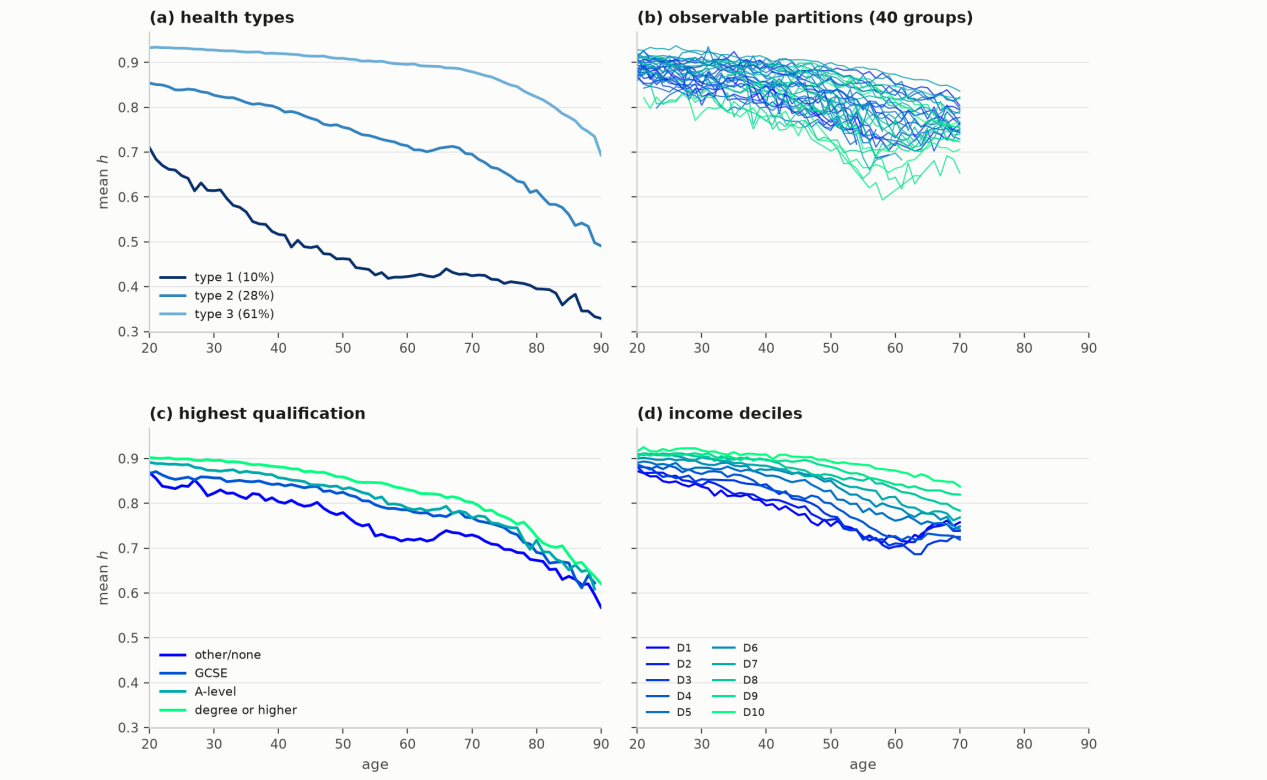
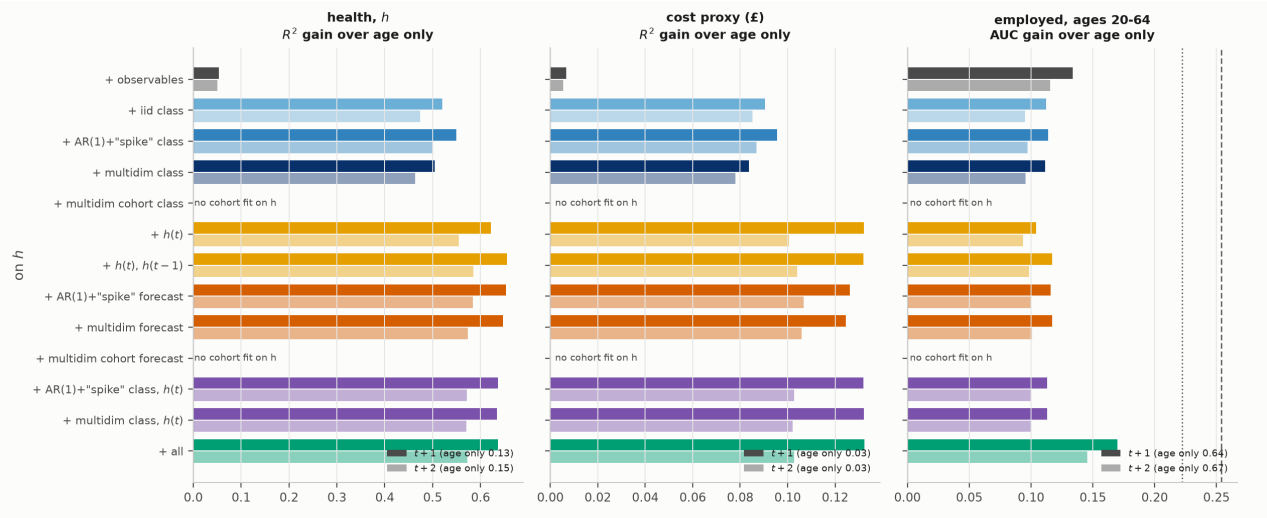


Figure C.15: As Figure 10, the partial K-means types on h over sliding and backward-expanding windows.



Clustered people (38,963: the health contract, at least three observed ages 20-89). Every series is the age-specific mean of h, cells with fewer than 30 people suppressed. Partitions in (b): highest qualification (4) x sex (2) x white / other ethnic group (2) x income tercile (3). Income is the person's mean within-wave percentile rank of equivalised household net income; education the highest qualification ever recorded. Panels (b) and (d) involve income and stop at age 70.

Figure C.16: As Figure 12, on h.

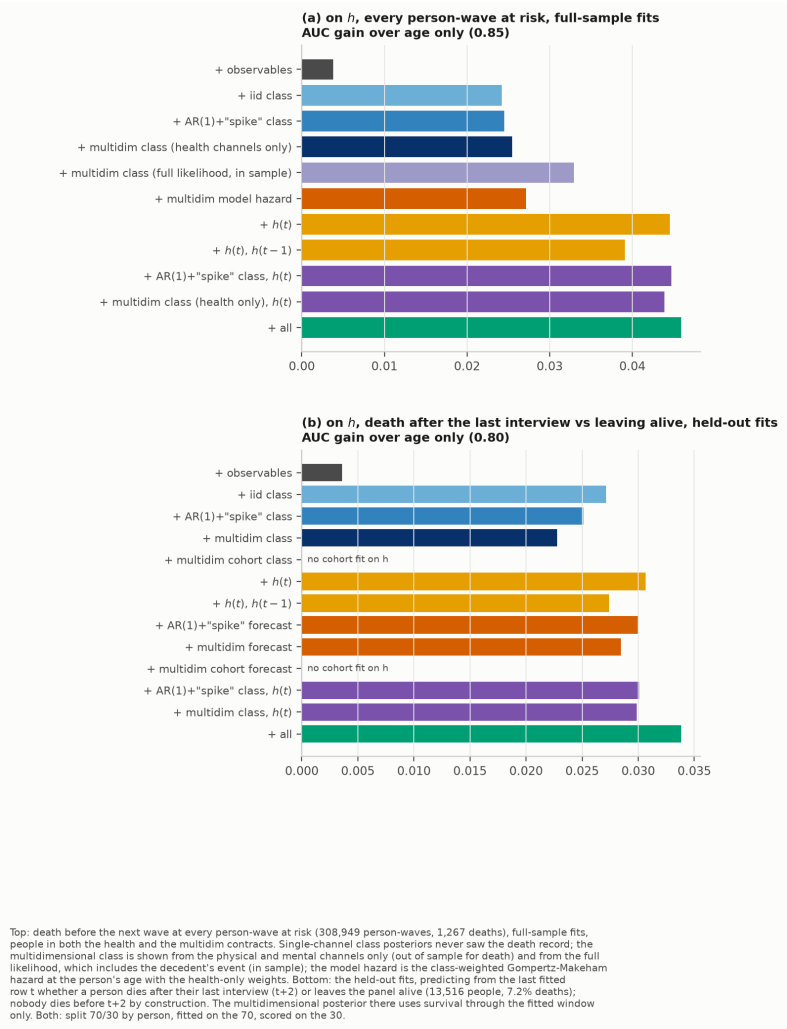


One prediction per person and horizon (26,664 people with at least five rows in both held-out samples): t is the last row of the fitted window, the outcomes are at the two held rows t+1 and t+2. Every class posterior and forecast comes from a fit with each person's last two rows held out, computed from the fitted window alone and carried to the outcome's age. People split 70/30, fitted on the 70, scored on the 30. Cost proxy: the flat-cost index in pounds, capped at its 99th percentile, waves 7-15. Employment: employed or self-employed, people aged 20-64 at t; the dashed (t+1) and dotted (t+2) lines mark the benchmark of age plus employment at t. The lag row is on people with a lag observed.

Figure C.17: As Figure 13, on h.

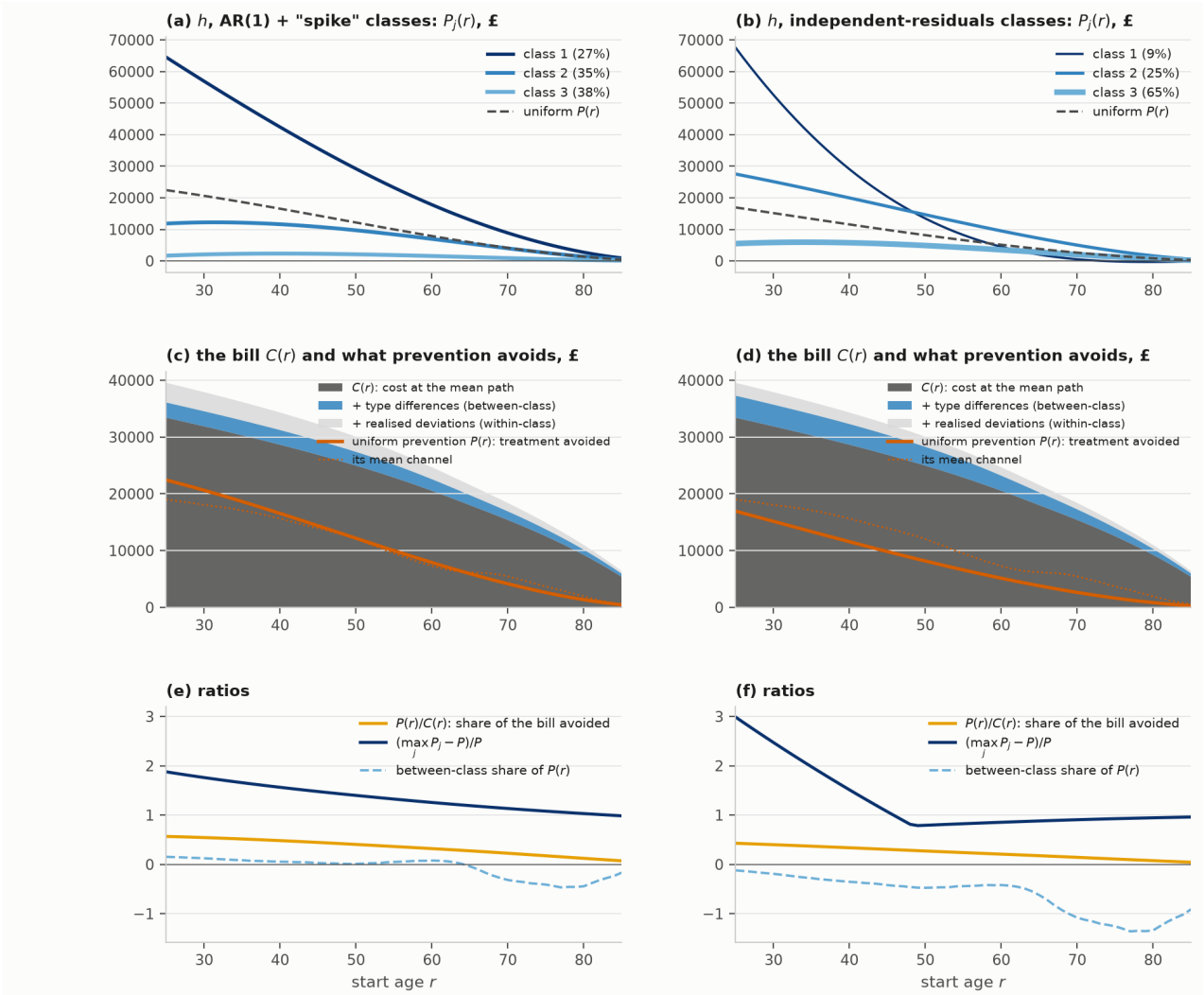
predictors at t	health, measure (R^2)		cost proxy (£) (R^2)		employed, ages 20-64 (AUC)	
	$t+1$	$t+2$	$t+1$	$t+2$	$t+1$	$t+2$
on h						
age	0.127	0.153	0.032	0.027	0.643	0.666
+ observables	0.181	0.204	0.039	0.033	0.777	0.782
+ iid class	0.647	0.627	0.122	0.112	0.756	0.762
+ AR(1)+"spike" class	0.676	0.654	0.127	0.114	0.757	0.764
+ multidim class	0.631	0.616	0.115	0.105	0.755	0.762
+ multidim cohort class						
+ h(t)	0.749	0.708	0.164	0.128	0.747	0.760
+ h(t), h(t-1)	0.783	0.738	0.164	0.131	0.761	0.765
+ AR(1)+"spike" forecast	0.781	0.738	0.158	0.134	0.759	0.766
+ multidim forecast	0.774	0.727	0.156	0.133	0.761	0.768
+ multidim cohort forecast						
+ AR(1)+"spike" class, h(t)	0.764	0.725	0.164	0.130	0.757	0.766
+ multidim class, h(t)	0.762	0.724	0.164	0.129	0.756	0.767
+ all	0.764	0.726	0.164	0.130	0.813	0.812
benchmark: age, employed(t)					0.897	0.889

Table C.3: As Table 3, on h.



Top: death before the next wave at every person-wave at risk (308,949 person-waves, 1,267 deaths), full-sample fits, people in both the health and the multidim contracts. Single-channel class posteriors never saw the death record; the multidimensional class is shown from the physical and mental channels only (out of sample for death) and from the full likelihood, which includes the decedent's event (in sample); the model hazard is the class-weighted Gompertz-Makeham hazard at the person's age with the health-only weights. Bottom: the held-out fits, predicting from the last fitted row t whether a person dies after their last interview (t+2) or leaves the panel alive (13,516 people, 7.2% deaths); nobody dies before t+2 by construction. The multidimensional posterior there uses survival through the fitted window only. Both: split 70/30 by person, fitted on the 70, scored on the 30.

Figure C.18: As Figure 14, on h.



Returns on the K = 3 mixture classes on h, with the cost curve estimated in pounds on h (a degree-3 polynomial on the cost index, uncapped, waves 7-15). C(r) is the expected treatment cost from r to 90 on the observed cross-section, split into the cost at the mean path, what the type differences add and what realised deviations add (the part prevention cannot reach); prevention scales each class's future decline by (1 - mu) and P(r) is the treatment cost that avoids, to first order, per unit of mu. Pounds to 90; no survival weighting, no discounting. Class shares and quadratic paths at the posterior mean; E[-c'(h) | class] from the person-waves observed at each age, weighted by the posterior class probabilities.

Figure C.19: As Figure 16, on h, with the cost curve estimated on h.

D Additional descriptive figures

[Here include extra figures/robustness for Sections 3.2 3.3]

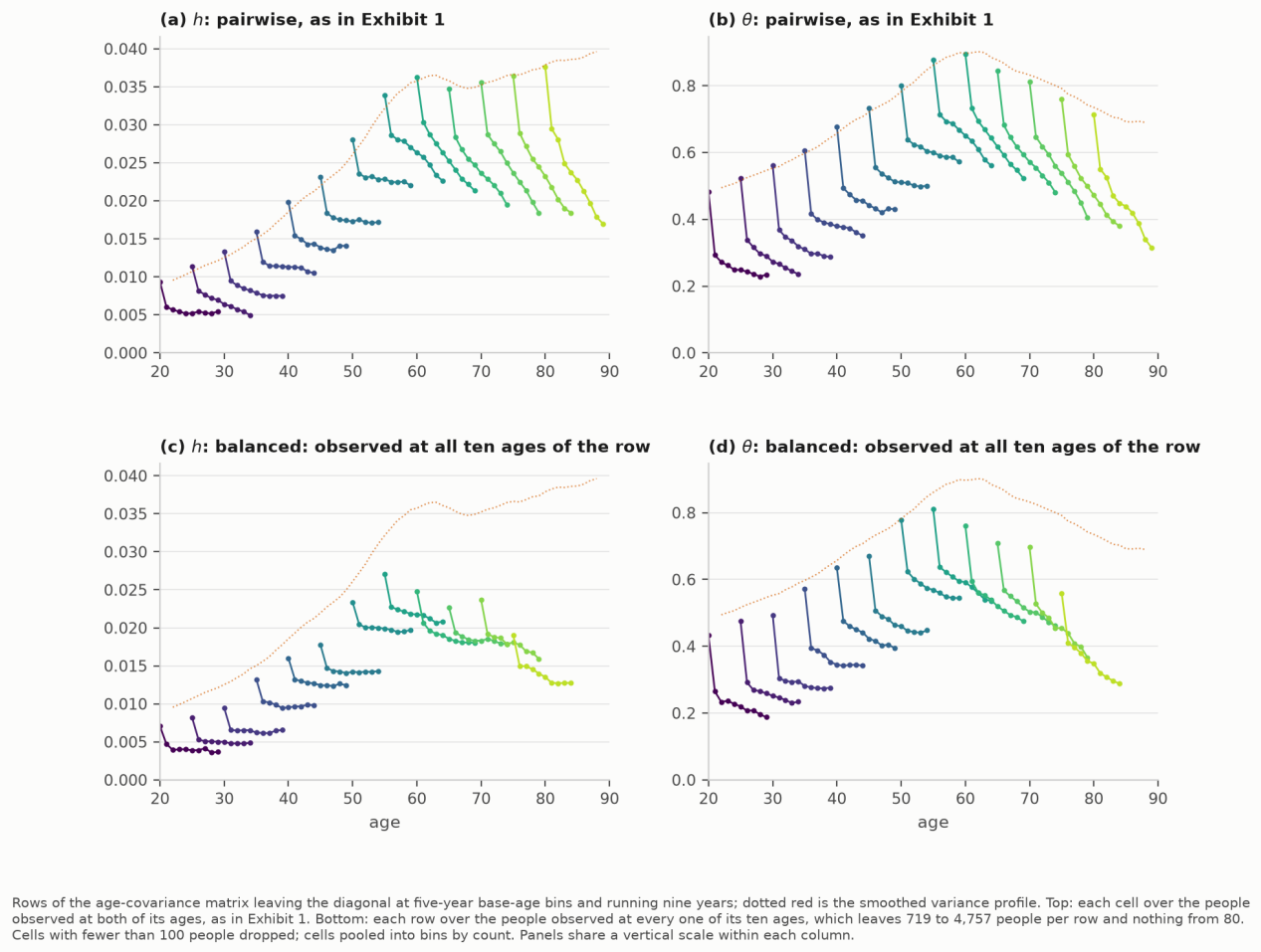


Figure D.20: The covariance rows of Figure 2 on pairwise and on balanced samples. Top: each cell over the people observed at both of its ages, as in Figure 2. Bottom: each row over the people observed at every one of its ten ages, which keeps between 719 and 4,757 people per row and nothing from age 80. Dotted red is the smoothed variance profile. The balanced rows sit below it throughout, and flatten after 60 on h: people observed ten years running are healthier and more stable than the cross-section at the same age. The one-year spike and the sign of the rows' slopes are the same on both.

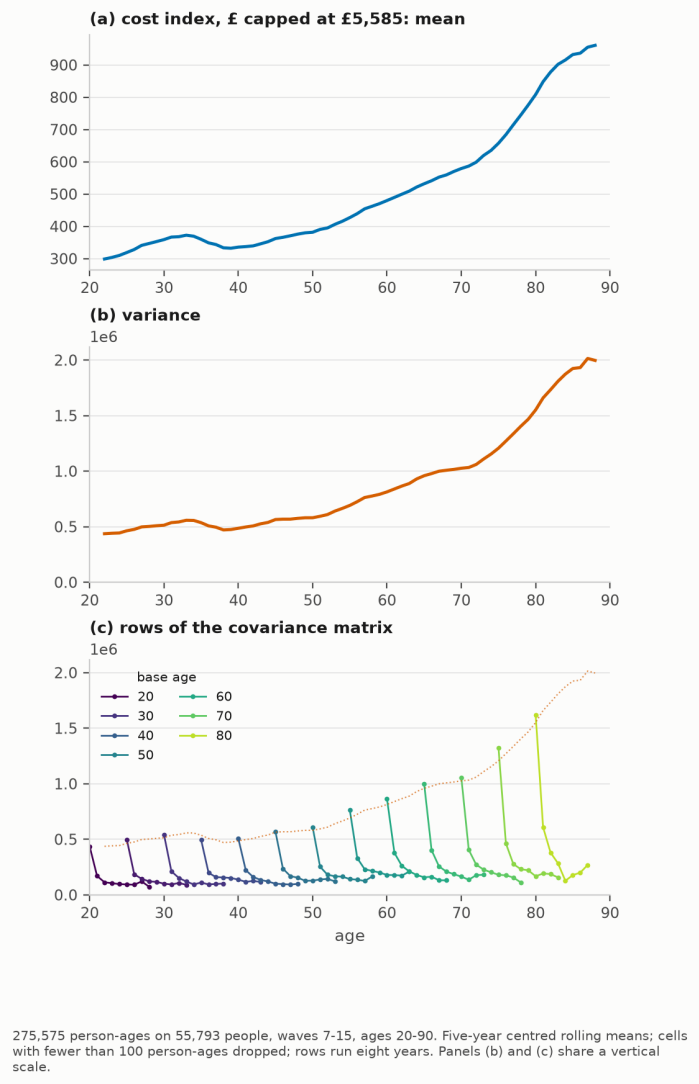


Figure D.21: The cost index through the moments of Figure 2, in pounds capped at the 99th percentile; waves 7–15, rows run eight years.

- Cost is what $c(h)$ stands for, so its mean by age is the bill $C(r)$ sums directly: the mean is flat to 40 and then rises steeply, from £330 to £960 by 88; its variance rises fourfold.
- The rows say what kind of variance it is: every row drops to about a fifth of the diagonal within a year and is flat after. Realised cost is almost entirely a one-period draw around the health state, which is exactly the within-type, in-the-moment part of the bill (Result 1) that is treated and that prevention cannot reach (Result 2).



Figure D.22: Figure 3 under Case 1, i.i.d. noise, on θ, h, the frailty index and log frailty (rows), in the design of Figure 3.

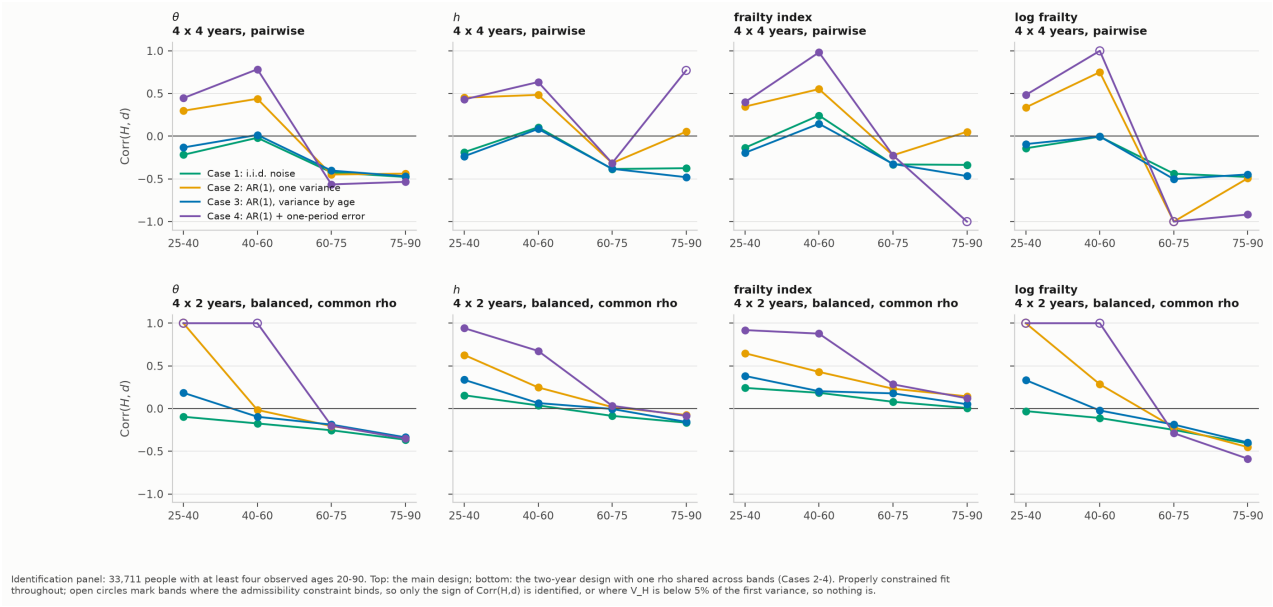


Figure D.23: Summary of Figure 3: the implied $\mathrm{Corr}(H, d)$ by age band under Cases 1 to 4, for θ, h, the frailty index and log frailty. Top: the main design, four ages four years apart on pairwise moments. Bottom: four ages two years apart on balanced moments, with one ρ shared by the four bands. Open circles mark bands where the admissibility constraint binds, or where V_H is below 5% of the first variance.

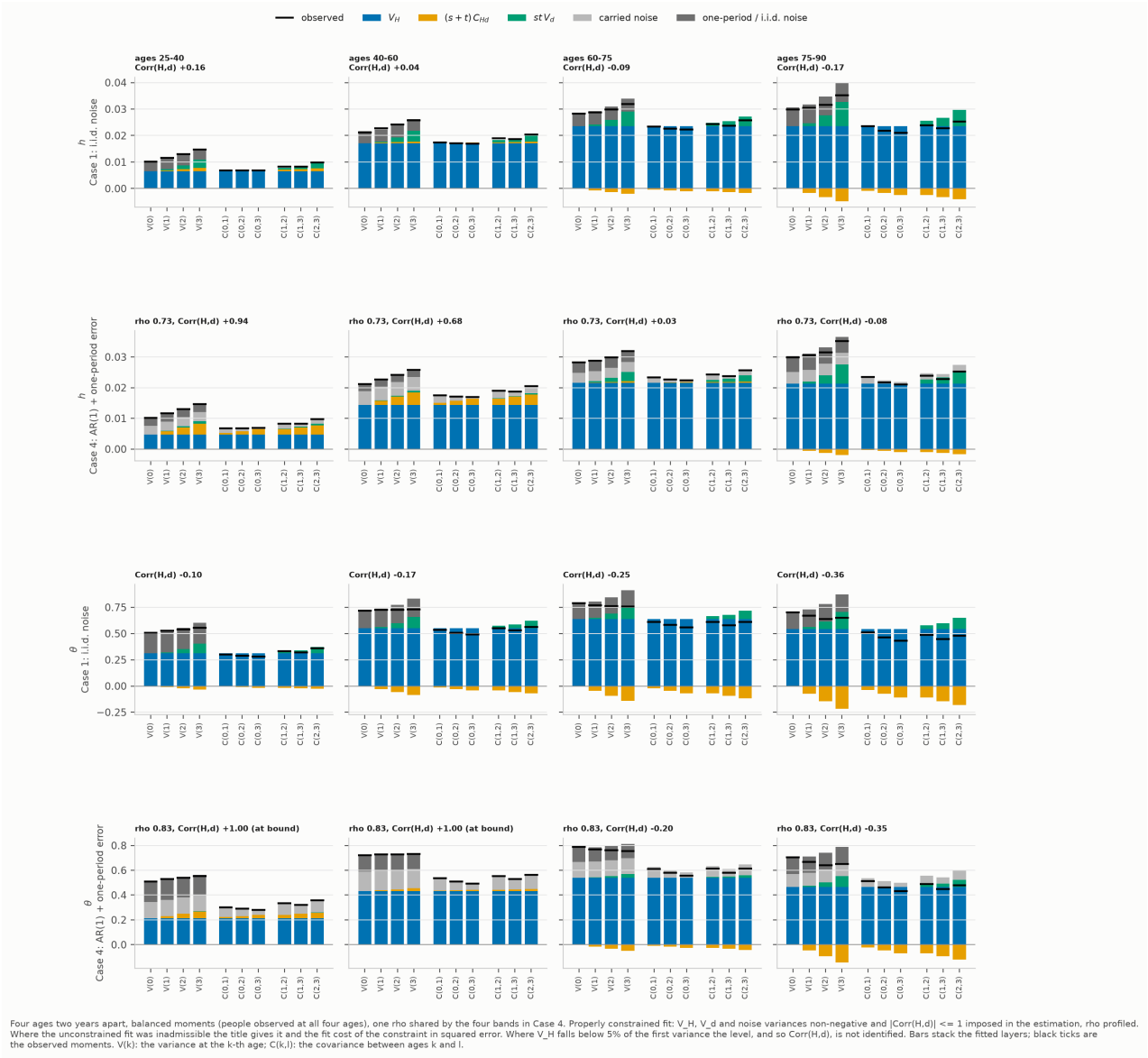


Figure D.24: Figure 3 on the two-year design: four ages two years apart, balanced moments, Cases 1 and 4 on h and θ, with one ρ shared by the four bands in Case 4. On h this is admissible in every band with nothing imposed.

E Additional K-means exercises

E.1 The types on predicted cost

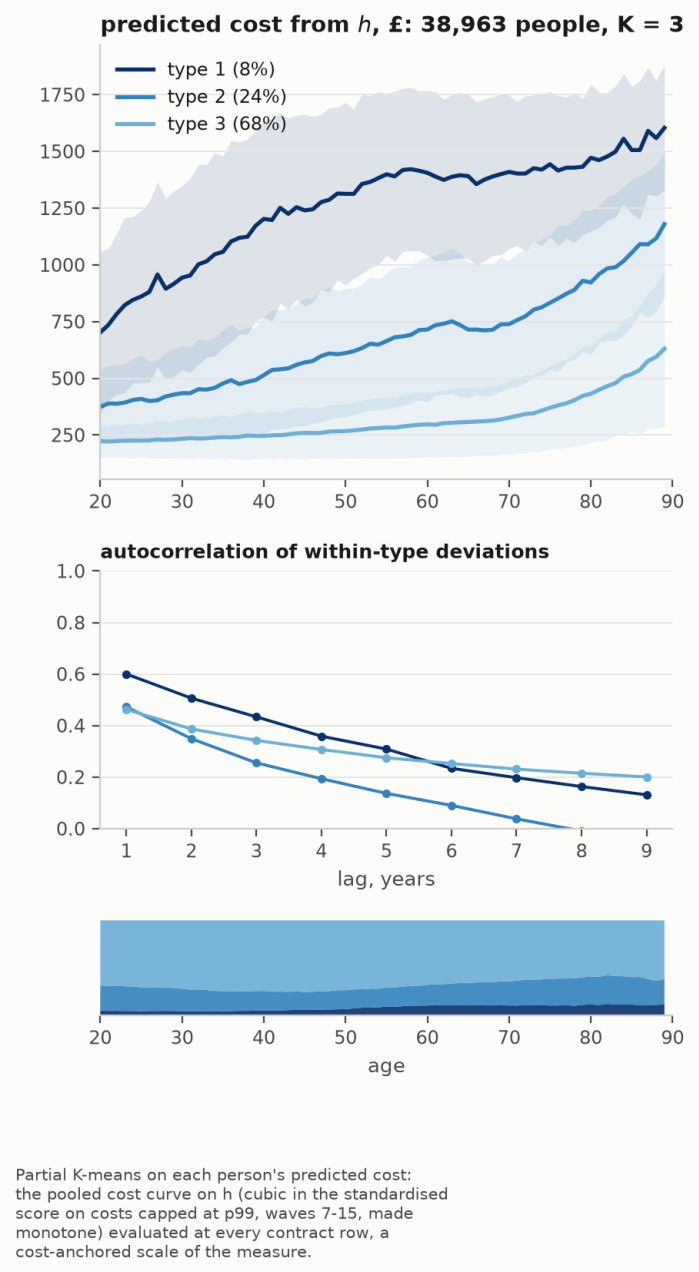


Figure E.25: As Figure 4, for partial K-means on each person's predicted cost: type means with the within-type ±1 sd band, the autocorrelation of within-type deviations by lag, and the type composition by age. Predicted cost is the pooled cost curve on h, a cubic in the standardised score fitted to costs capped at the 99th percentile over waves 7–15 and made monotone, evaluated at every contract row: a cost-anchored cardinalisation of the measure. It lands on the bounded side of the h–θ contrast: ARI 0.76 with the h types and 0.31 with the θ types.

E.2 Cohort adjustment

The partial K-means types are fitted to person-by-age profiles pooled across birth cohorts, so a cohort born later that reports worse health at a given age could be sorted into a worse type for that reason alone. The check has three steps, on the clustering sample and separately for each variant.

- Regress the score on a full set of single-year age dummies and birth-decade dummies, with the 1950s as reference and standard errors clustered on the person. With age fully flexible a decade shift is identified only from the overlap of neighbouring decades at the ages they share, which is the honest version of the mixture's quadratic-in-age-plus-decade specification in Appendix F.1.
- Subtract each person's decade shift from every one of their scores, and rerun partial K-means with $K = 3$ and 50 seeded starts on the adjusted profiles.
- Compare the adjusted typology with the unadjusted one person by person, and the two sets of decade shifts with each other.

Figure E.26 has the result. The decade shifts under flexible age are zero for every cohort born before 1960, within one standard error, and turn negative only from the 1970s on: -0.11, -0.21, -0.28 and -0.35 standard deviations of θ for the 1970s to 2000s. Those cohorts are observed only when young, so a cohort effect and a period effect cannot be told apart there. The mixture's quadratic specification attributes shifts to the earlier decades as well, which is what a quadratic does when the age profile is not quadratic. The typology is unaffected: the adjusted Rand index between the adjusted and unadjusted types is 0.903 on θ and 0.956 on h, at least 97% of people keep their type, and the type shares move by at most half a percentage point.

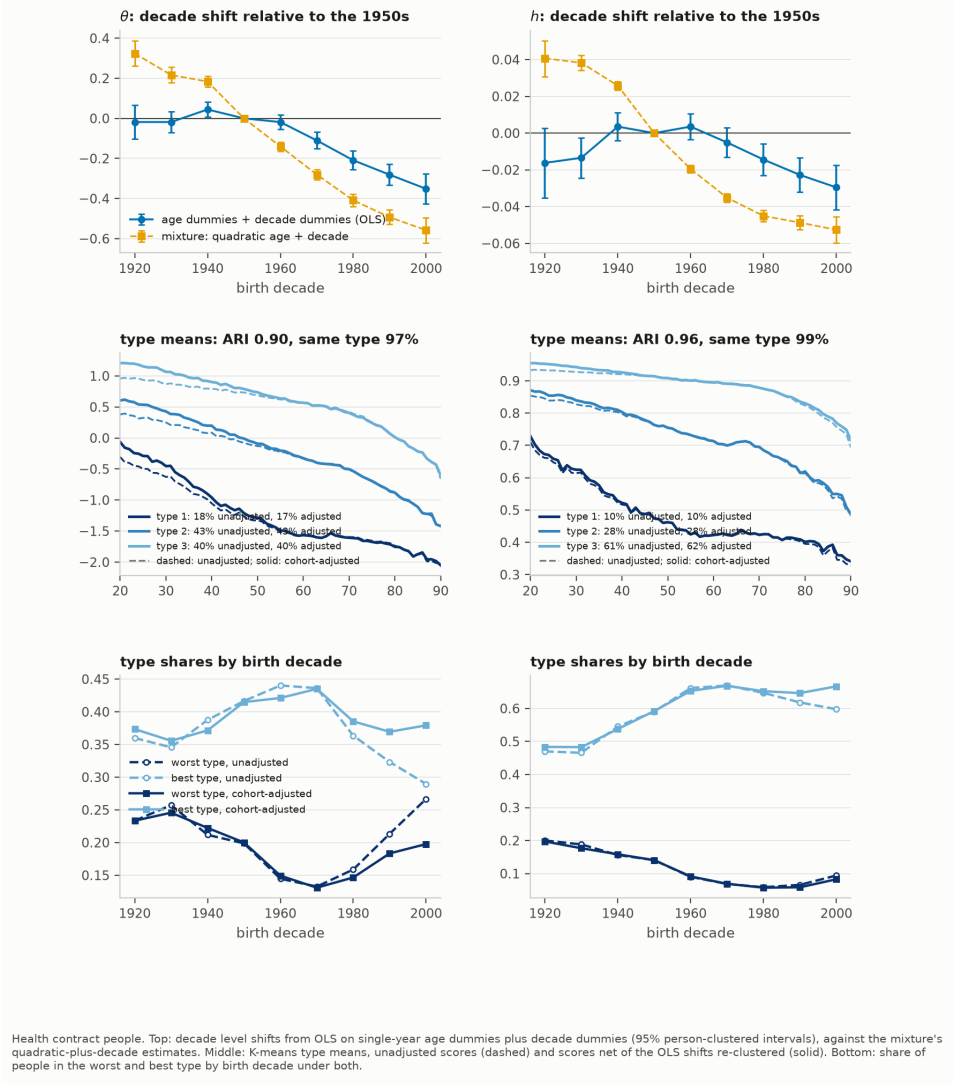


Figure E.26: Cohort adjustment within partial K-means, one column per variant. Top: birth-decade level shifts relative to the 1950s from OLS on single-year age dummies plus decade dummies, with 95% person-clustered intervals, against the mixture's quadratic-plus-decade estimates. Middle: K-means type means by age, on the unadjusted scores (dashed) and on the scores net of the OLS shifts, re-clustered (solid). Bottom: the share of people in the worst and the best type by birth decade, unadjusted (dashed, open circles) and cohort-adjusted (solid, filled squares).

F Additional Bayesian results

F.1 Birth-decade cohort effects

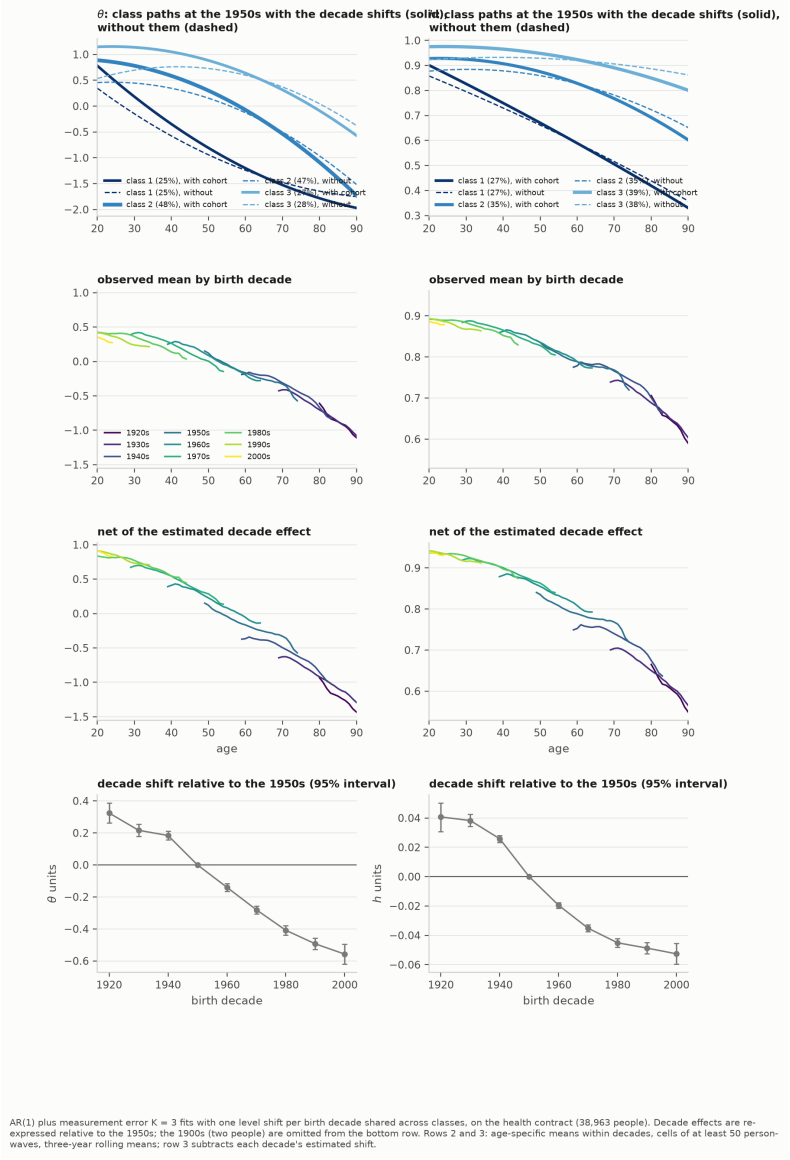
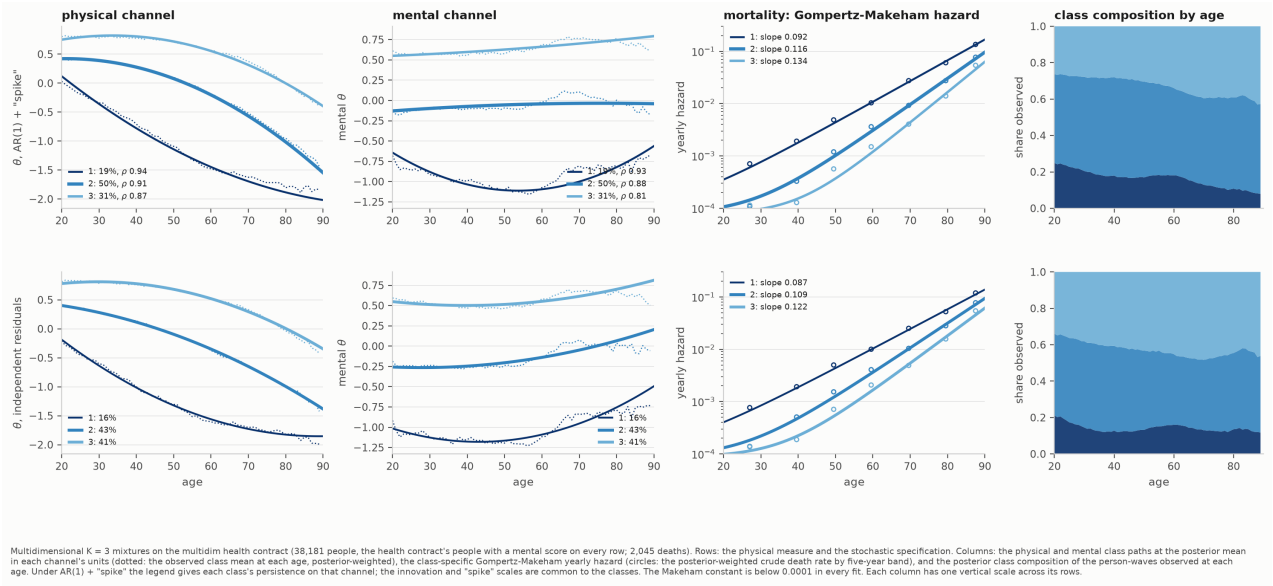


Figure F.27: The AR(1) plus measurement error fits with a shared birth-decade level shift, one column per variant. Row 1: class paths with the shifts, evaluated at the 1950s (solid), against the fits without them (dashed). Row 2: the observed mean of the variant by age within each birth decade. Row 3: the same profiles net of the estimated decade shift. Row 4: the shifts relative to the 1950s with 95% posterior intervals, in the variant's units. Contract people, cells of at least 50 person-waves.

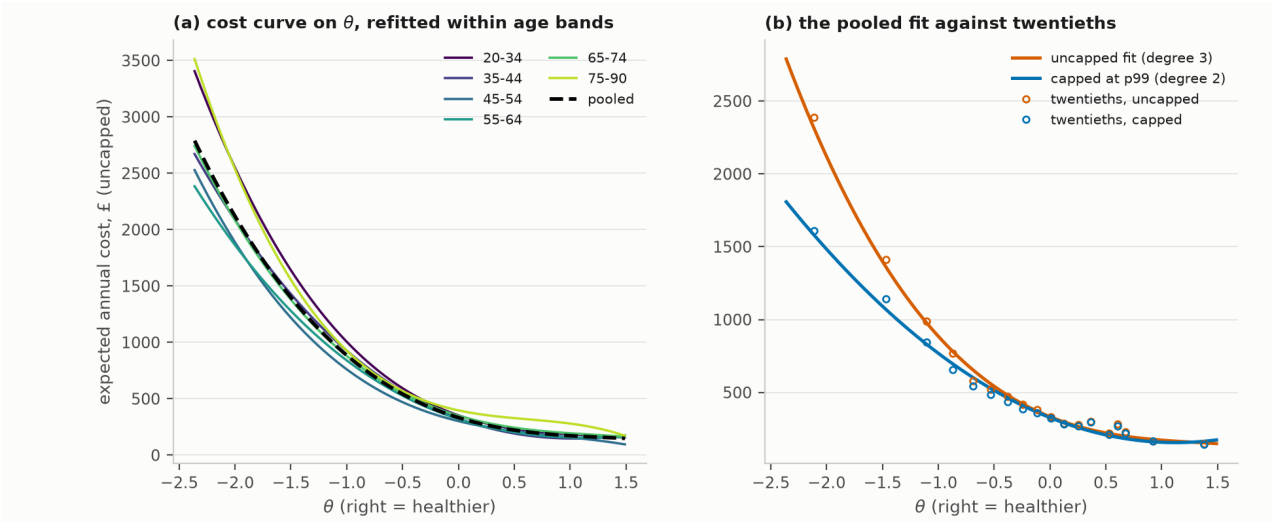


Multidimensional K = 3 mixtures on the multidim health contract (38,181 people, the health contract's people with a mental score on every row; 2,045 deaths). Rows: the physical measure and the stochastic specification. Columns: the physical and mental class paths at the posterior mean in each channel's units (dotted: the observed class mean at each age, posterior-weighted), the class-specific Gompertz-Makeham yearly hazard (circles: the posterior-weighted crude death rate by five-year band), and the posterior class composition of the person-waves observed at each age. Under AR(1) + "spike" the legend gives each class's persistence on that channel; the innovation and "spike" scales are common to the classes. The Makeham constant is below 0.0001 in every fit. Each column has one vertical scale across its rows.

Figure F.28: As Figure 8, the multidimensional $K = 3$ mixtures on θ without the birth-decade shifts: rows are AR(1) plus "spike" and independent residuals. Without the shifts the mental class paths rise with age, which the decade shifts of the main-text figure attribute to the older cohorts' higher level.

G Additional info on returns

G.1 The estimated cost curve



The cost index in pounds, waves 7-15, 211,697 person-waves, against theta: a polynomial (cubic unless that is not monotone on the 1st-99th percentile range), fitted pooled and within each age band, over the 1st-99th percentile range of theta. Panel (b): the pooled fits on uncapped costs and on costs capped at the 99th percentile (£5,536), with the mean cost in twentieths of theta.

Figure G.29: The cost curve Section 5 prices with: the cost index in pounds against θ, a cubic fitted to uncapped costs. (a) The same polynomial refitted within six age bands against the pooled fit; (b) the pooled fit on uncapped costs and on costs capped at the 99th percentile, against the mean cost in twentieths of θ.

- The estimated curve is close to age-invariant in shape. Refitted within six age bands it stays within about 12% of the pooled curve on average from 20 to 74, with a shallow U in age (lowest at 45–54, 12% below the pooled curve), and only the 75–90 band sits clearly above it, by 25%. Costing health rather than age is therefore defensible for the returns, with an understatement of late-life cost.
- Capping at the 99th percentile flattens the curve at the sick end, where the capped fit is a quadratic, and lowers every return by about a third (Appendix G.3).

G.2 K-means types under the estimated cost curve

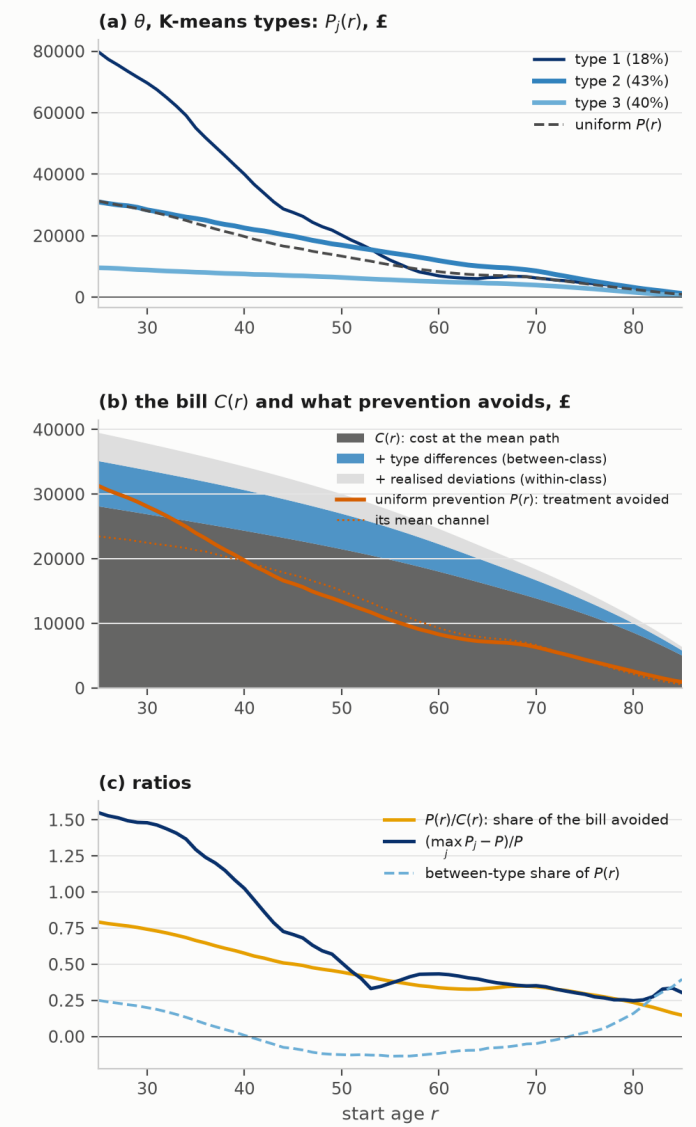


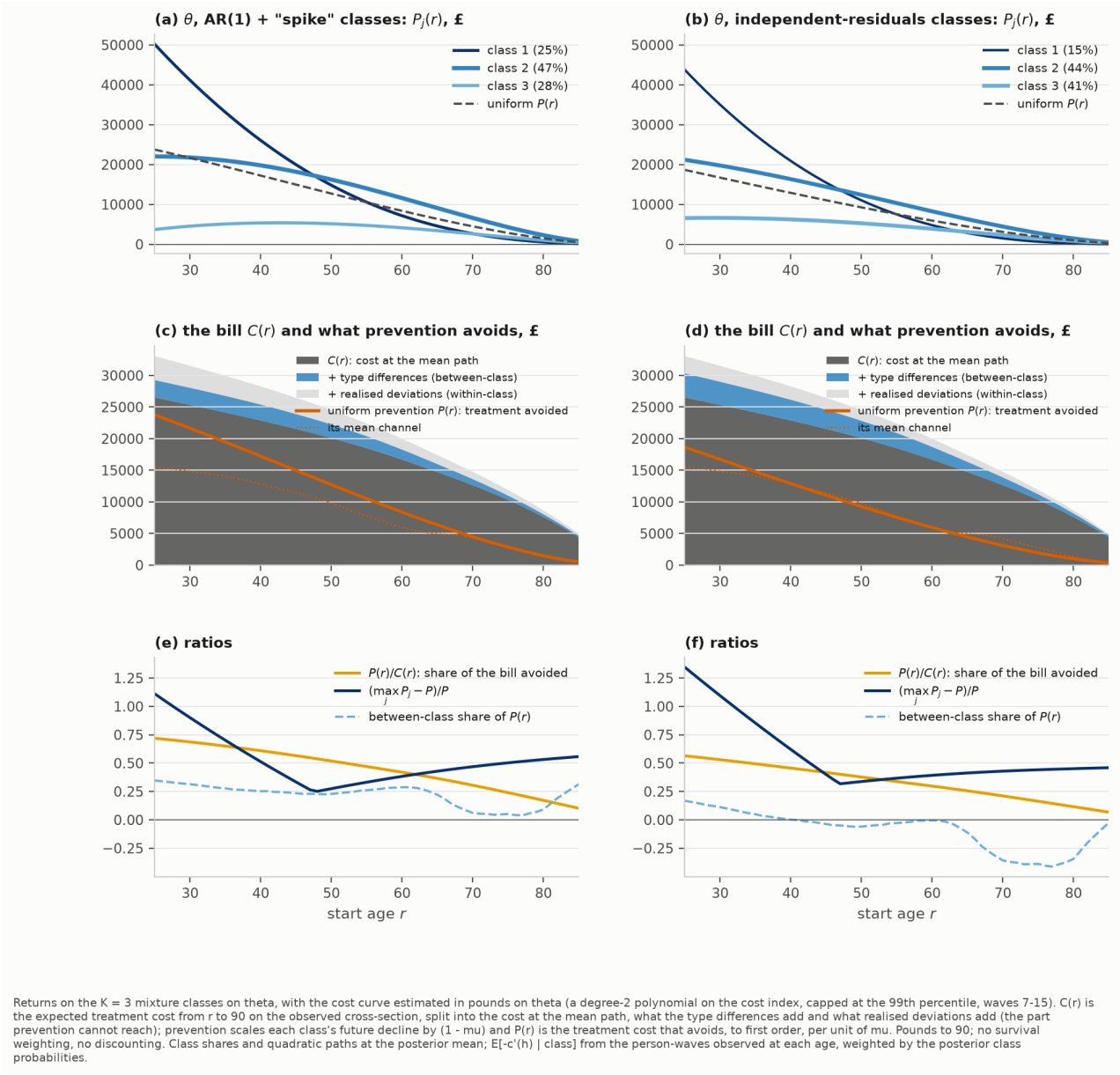
Figure G.30: As Figure 16, on the partial K-means types on θ: type shares and smoothed type means, with the same uncapped cost curve.

classes	r	$C(r)$	within	$P(r)$	P/C	between	P_1	P_2	P_3	gain
K-means types	30	37,834	11%	28,097	0.74	+20%	69,689	28,407	8,871	1.48
K-means types	50	30,005	10%	13,368	0.45	-13%	20,197	16,933	6,430	0.51
K-means types	70	18,339	9%	6,318	0.34	-5%	6,253	8,540	3,958	0.35

Table G.4: As Table 5, on the K-means types.

- On the K-means types $P(r)/C(r)$ is 0.74 at 30, 0.45 at 50 and 0.35 at 70, below the 0.91, 0.67 and 0.39 of the AR(1) plus "spike" classes; the within-type part of the bill is 11%, 10% and 9%.
- The between-type channel is +20% of $P(30)$ and turns negative from about 40 (-13% at 50, -5% at 70), as on the independent-residuals classes: the K-means types converge after midlife, where the AR(1) plus "spike" classes keep fanning out.
- Type targeting gains 148% over the uniform return from 30, 51% from 50 and 35% from 70.

G.3 The capped cost curve



Returns on the K = 3 mixture classes on theta, with the cost curve estimated in pounds on theta (a degree-2 polynomial on the cost index, capped at the 99th percentile, waves 7-15). C(r) is the expected treatment cost from r to 90 on the observed cross-section, split into the cost at the mean path, what the type differences add and what realised deviations add (the part prevention cannot reach); prevention scales each class's future decline by (1 - mu) and P(r) is the treatment cost that avoids, to first order, per unit of mu. Pounds to 90; no survival weighting, no discounting. Class shares and quadratic paths at the posterior mean; E[-c'(h) | class] from the person-waves observed at each age, weighted by the posterior class probabilities.

Figure G.31: As Figure 16, with the cost curve fitted to costs capped at the 99th percentile.

classes	r	$C(r)$	within	$P(r)$	P/C	between	P_1	P_2	P_3	gain
AR(1) + "spike" classes	30	31,519	11%	21,652	0.69	+31%	41,151	21,771	4,552	0.90
AR(1) + "spike" classes	50	24,523	9%	12,723	0.52	+23%	14,811	16,226	5,146	0.28
AR(1) + "spike" classes	70	14,751	8%	4,501	0.31	+6%	2,673	6,611	2,613	0.47
independent-residuals classes	30	31,519	8%	16,718	0.53	+11%	35,019	19,761	6,627	1.09
independent-residuals classes	50	24,523	7%	9,264	0.38	-6%	11,022	12,393	5,269	0.34
independent-residuals classes	70	14,751	6%	3,113	0.21	-36%	1,487	4,450	2,295	0.43

Table G.5: As Table 5, with the capped cost curve.

- With costs capped at the 99th percentile the bill from 30 falls from £38,000 to £32,000 and uniform prevention from £35,000 to £22,000, so P/C falls to 0.69, 0.52 and 0.31 at 30, 50 and 70 on the AR(1) plus "spike" classes (the cap removes most of the sick tail, which is where prevention's pounds are); the within-class part of the bill is 11%, 9% and 8%; the between-class shares of P are +31%, +23% and +6%, and the class-targeting gains 90%, 28% and 47%.
- The one change is at 50, where the middle class rather than the worst has the highest return under the capped curve.

References

Francesco Agostinelli and Matthew Wiswall. Estimating the technology of children's skill formation. *Journal of Political Economy*, 133(3):846–887, 2025. doi: 10.1086/733396.

Orazio Attanasio, Sarah Cattan, Emla Fitzsimons, Costas Meghir, and Marta Rubio-Codina. Estimating the production function for human capital: results from a randomized controlled trial in Colombia. *American Economic Review*, 110(1):48–85, 2020. doi: 10.1257/aer.20150183.

Richard Blundell, Jack Britton, Monica Costa Dias, and Eric French. The impact of health on labor supply near retirement. *Journal of Human Resources*, 58(1):282–334, 2023. doi: 10.3368/jhr.58.3.1217-9240R4.

Elena Capatina. Life-cycle effects of health risk. *Journal of Monetary Economics*, 74:67–88, 2015. doi: 10.1016/j.jmoneco.2015.06.002.

David Cella, William Riley, Arthur Stone, Nan Rothrock, Bryce Reeve, Susan Yount, Dagmar Amtmann, Rita Bode, Daniel Buysse, Seung Choi, Karon Cook, Robert DeVellis, Darren DeWalt, James F. Fries, Richard Gershon, Elizabeth A. Hahn, Jin-Shei Lai, Paul Pilkonis, Dennis Revicki, Matthias Rose, Kevin Weinfurt, and Ron Hays. The Patient-Reported Outcomes Measurement Information System (PROMIS) developed and tested its first wave of adult self-reported health outcome item banks: 2005–2008. *Journal of Clinical Epidemiology*, 63(11):1179–1194, 2010. doi: 10.1016/j.jclinepi.2010.04.011.

Paul Contoyannis, Andrew M. Jones, and Nigel Rice. The dynamics of health in the British Household Panel Survey. *Journal of Applied Econometrics*, 19(4):473–503, 2004. doi: 10.1002/jae.755.

Lorenzo Corsini and Irene Martelli. Measuring and detecting situations of need and deprivation using graded response models. *Applied Economics Letters*, 27(15):1289–1294, 2020. doi: 10.1080/13504851.2019.1677843.

Flavio Cunha, James J. Heckman, and Susanne M. Schennach. Estimating the technology of cognitive and noncognitive skill formation. *Econometrica*, 78(3):883–931, 2010. doi: 10.3982/ECTA6551.

Jishnu Das and Jeffrey Hammer. Which doctor? Combining vignettes and item response to measure clinical competence. *Journal of Development Economics*, 78(2):348–383, 2005. doi: 10.1016/j.jdeveco.2004.11.004.

Jishnu Das and Tristan Zajonc. India shining and Bharat drowning: comparing two Indian states to the worldwide distribution in mathematics achievement. *Journal of Development Economics*, 92(2):175–187, 2010. doi: 10.1016/j.jdeveco.2009.03.004.

Will Davis, José O. Xilau, Rusty Tchernis, and Christian A. Gregory. A flexible model of food security: estimation and implications for prediction. *American Journal of Agricultural Economics*, 108(3):954–980, 2026. doi: 10.1111/ajae.70012.

Mariacristina De Nardi, Svetlana Pashchenko, and Ponpoje Porapakkarm. The lifetime costs of bad health. *Review of Economic Studies*, 92(3):1987–2026, 2025. doi: 10.1093/restud/rdae080.

Alessio Farcomeni, Maria Grazia Pittau, Sara Viviani, and Roberto Zelli. A measurement scale for material deprivation: a model-based approach. *Review of Income and Wealth*, 71(1):e12716, 2025. doi: 10.1111/roiw.12716.

Daniel Graeber. Intergenerational health mobility in Germany. IZA Discussion Paper 16567, IZA Institute of Labor Economics, 2023. also SOEPpapers 1195, DIW Berlin.

Timothy J. Halliday. Heterogeneity, state dependence and health. *The Econometrics Journal*, 11(3): 499–516, 2008. doi: 10.1111/j.1368-423X.2008.00256.x.

Roozbeh Hosseini, Karen A. Kopecky, and Kai Zhao. The evolution of health over the life cycle. *Review of Economic Dynamics*, 45:237–263, 2022. doi: 10.1016/j.red.2021.07.001.

Brian Jacob and Jesse Rothstein. The measurement of student ability in modern assessment systems. *Journal of Economic Perspectives*, 30(3):85–108, 2016. doi: 10.1257/jep.30.3.85.

Erik Meijer, Arie Kapteyn, and Tatiana Andreyeva. Internationally comparable health indices. *Health Economics*, 20(5):600–619, 2011. doi: 10.1002/hec.1620.

Arnold B. Mitnitski, Alexander J. Mogilner, and Kenneth Rockwood. Accumulation of deficits as a proxy measure of aging. *The Scientific World Journal*, 1:323–336, 2001. doi: 10.1100/tsw.2001.58.

Robert A. Moffitt and David C. Ribar. Rasch analyses of very low food security among households and children in the Three City Study. *Southern Economic Journal*, 82(4):1123–1146, 2016. doi: 10.1002/soej.12081.

Karthik Muralidharan, Abhijeet Singh, and Alejandro J. Ganimian. Disrupting education? Experimental evidence on technology-aided instruction in India. *American Economic Review*, 109(4): 1426–1460, 2019. doi: 10.1257/aer.20171112.

James M. Poterba, Steven F. Venti, and David A. Wise. Health, education, and the postretirement evolution of household assets. *Journal of Human Capital*, 7(4):297–339, 2013. doi: 10.1086/673207.

James M. Poterba, Steven F. Venti, and David A. Wise. The asset cost of poor health. *The Journal of the Economics of Ageing*, 9:172–184, 2017. doi: 10.1016/j.jeoa.2017.02.001.

Matthias Rose, Jakob B. Bjorner, Barbara Gandek, Bonnie Bruce, James F. Fries, and John E. Ware. The PROMIS Physical Function item bank was calibrated to a standardized metric and shown to improve measurement efficiency. *Journal of Clinical Epidemiology*, 67(5):516–526, 2014. doi: 10.1016/j.jclinepi.2013.10.024.

Fumiko Samejima. Estimation of latent ability using a response pattern of graded scores. *Psychometrika*, 34(S1):1–97, 1969. doi: 10.1007/BF03372160. Psychometrika Monograph Supplement No. 17.

Samuel D. Searle, Arnold Mitnitski, Evelyne A. Gahbauer, Thomas M. Gill, and Kenneth Rockwood. A standard procedure for creating a frailty index. *BMC Geriatrics*, 8:24, 2008. doi: 10.1186/1471-2318-8-24.

Abhijeet Singh. Private school effects in urban and rural India: panel estimates at primary and secondary school ages. *Journal of Development Economics*, 113:16–32, 2015. doi: 10.1016/j.jdeveco.2014.10.004.

University of Essex, Institute for Social and Economic Research. Understanding society: Waves 1–15, 2009–2024 and harmonised BHPS: Waves 1–18, 1991–2009. UK Data Service, SN 6614, 2025.

Geoffrey L. Wallace, Robert Haveman, and Barbara Wolfe. Health status, health shocks, and asset adequacy over retirement years. *Research on Aging*, 39(1):222–248, 2017. doi: 10.1177/0164027516669567.

H Mental health

The mental GRM: the three SF-12 mental testlets and two GHQ-12 testlets, no depression diagnosis, the second channel of the multidimensional model of Section 4.3, here on its own. Higher is better; standardised on the pooled sample. The K-means and the mixtures use the multidimensional contract's rows, the health contract's people with a mental score on every row (38,181 people), so they share a sample with the multidimensional fits.

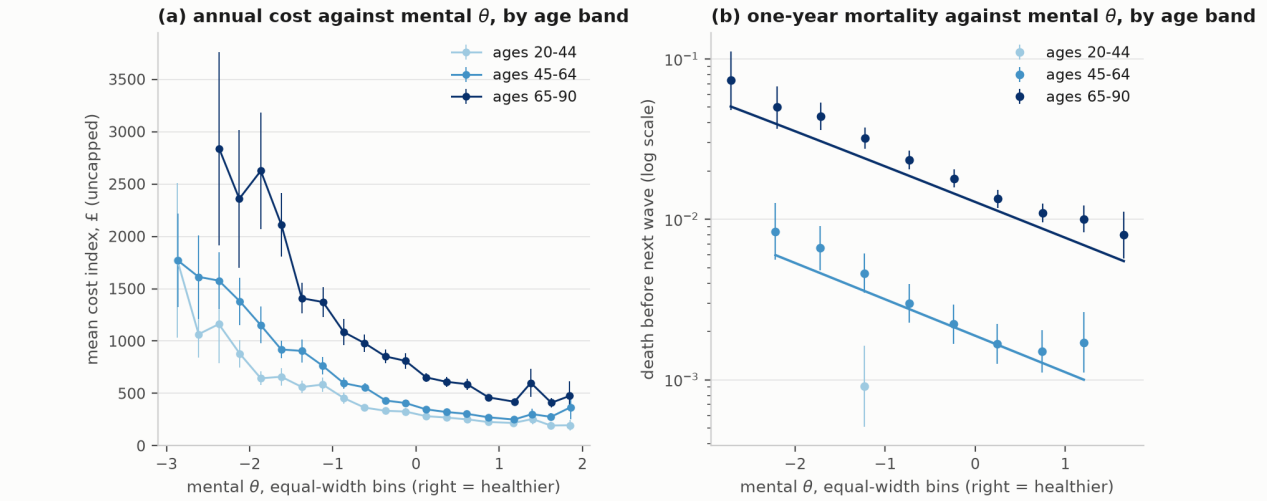


Figure H.32: As Figure 1, on the mental GRM.

- Cost is convex in mental health too: the pooled curvature has $t = 21.4$ against 36.9 on θ, and the slope is about half of θ's (£144 against £279 per sd at the mean). Unlike θ, the gradient steepens with age ($t = 22$ on the slope-by-age interaction, against -1.5 on θ): a low mental score costs little at 20–44 and a lot at 65–90.
- Mortality falls with mental health at every age, with a gradient of 0.57 log-odds per sd against 0.90 on θ, and it weakens with age ($t = -3.0$), where on θ it is stable.

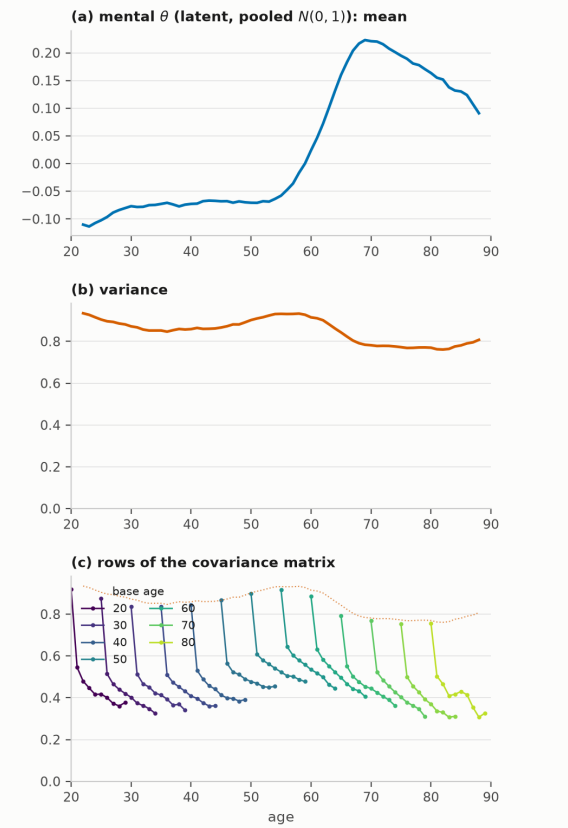
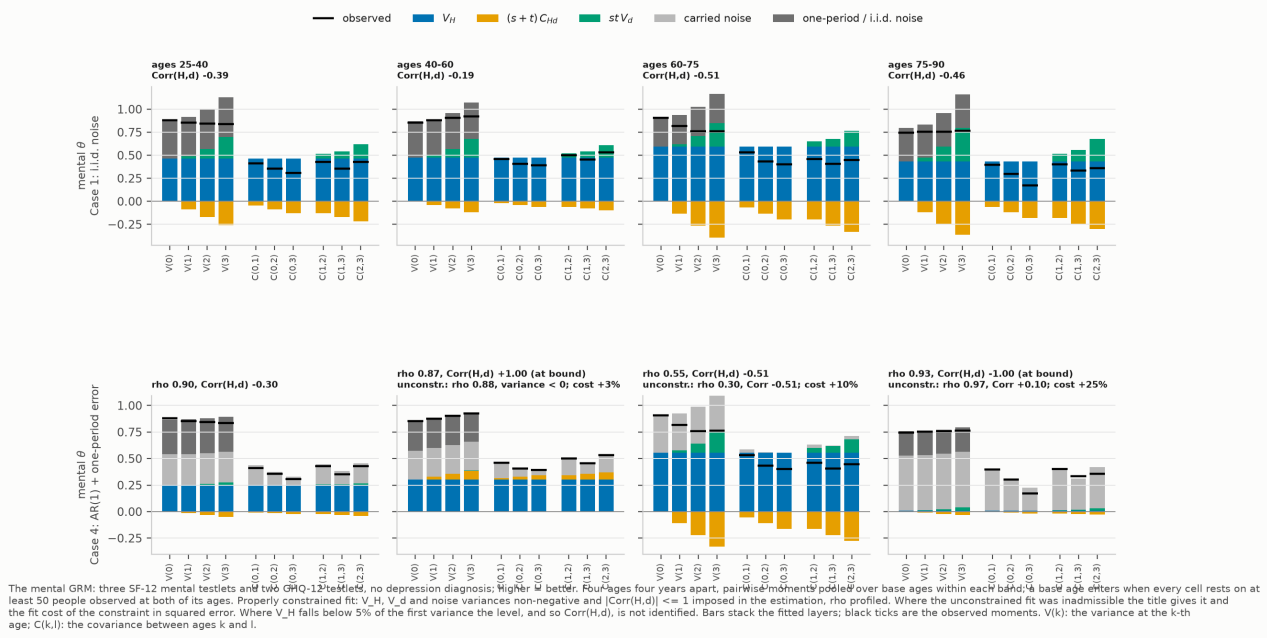


Figure H.33: As Figure 2, on the mental GRM: mean path, variance profile and covariance rows, model-free.

- The mean path is nothing like the physical one: flat at about −0.08 from 20 to 55, then up by 0.3 sd to a peak at about 70, then down again. The variance is flat at 0.85–0.93 across the whole age range, against the hump on θ.
- Roughly half of the variance is one-period: every covariance row drops from about 0.9 on the diagonal to 0.45–0.55 a year later, then decays slowly. On θ the first-year drop is about a fifth. This is the fast mental channel of the multidimensional fits, where its "spike" share is 0.34–0.55 by class.



The mental GRM: three SF-12 mental testlets and two GHQ-12 testlets, no depression diagnosis; higher = better. Four ages four years apart, pairwise moments pooled over base ages within each band; a base age enters when every cell rests on at least 50 people observed at both of its ages. Properly constrained fit: V_H, V_d and noise variances non-negative and |Corr(H,d)| <= 1 imposed in the estimation, rho profiled. Where the unconstrained fit was inadmissible the title gives it and the fit cost of the constraint in squared error. Where V_H falls below 5% of the first variance the level, and so Corr(H,d), is not identified. Bars stack the fitted layers; black ticks are the observed moments. V(k): the variance at the k-th age; C(k,l): the covariance between ages k and l.

Figure H.34: As Figure 3, Cases 1 and 4 on the mental GRM.

- Under Case 1, Corr(H, d) is negative in every band (-0.39, -0.19, -0.51, -0.46): the shocks pull back towards the level throughout, with no positive-correlation band in mid-life as on θ. Under Case 4 the fit is not clean: -0.30 at 25–40 with $\rho = 0.90$, but at 40–60 and 75–90 the constraint binds ($+1$ and -1) and at 60–75 the persistence falls to 0.55. With half the variance one-period, the four-year moments leave little to identify the persistent part from.

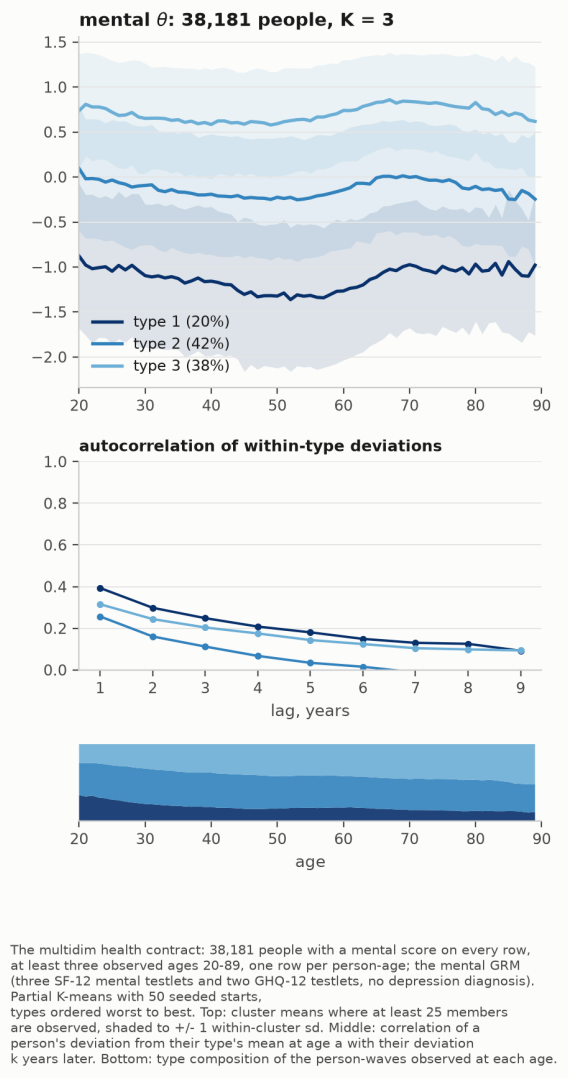


Figure H.35: As Figure 4, partial K-means types ($K = 3$) on the mental GRM, on the multidimensional contract's rows.

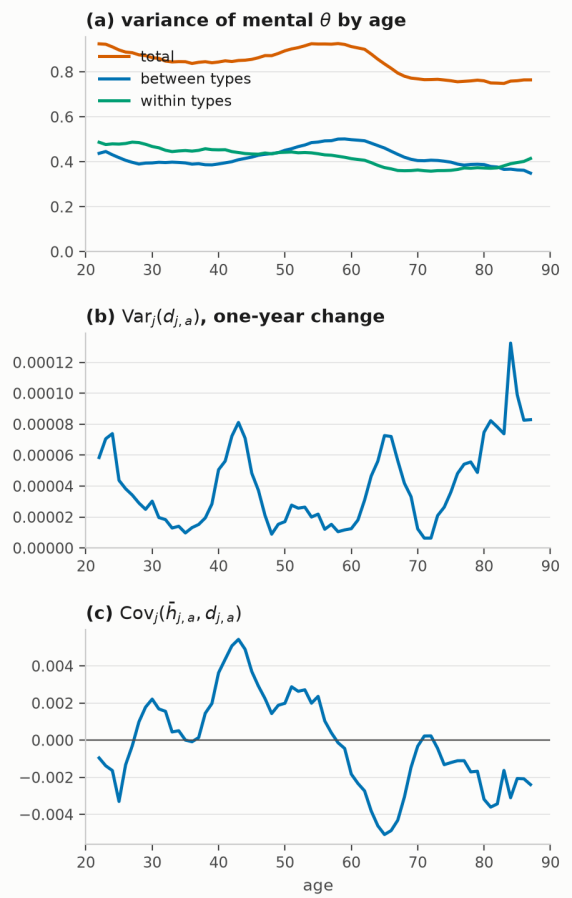
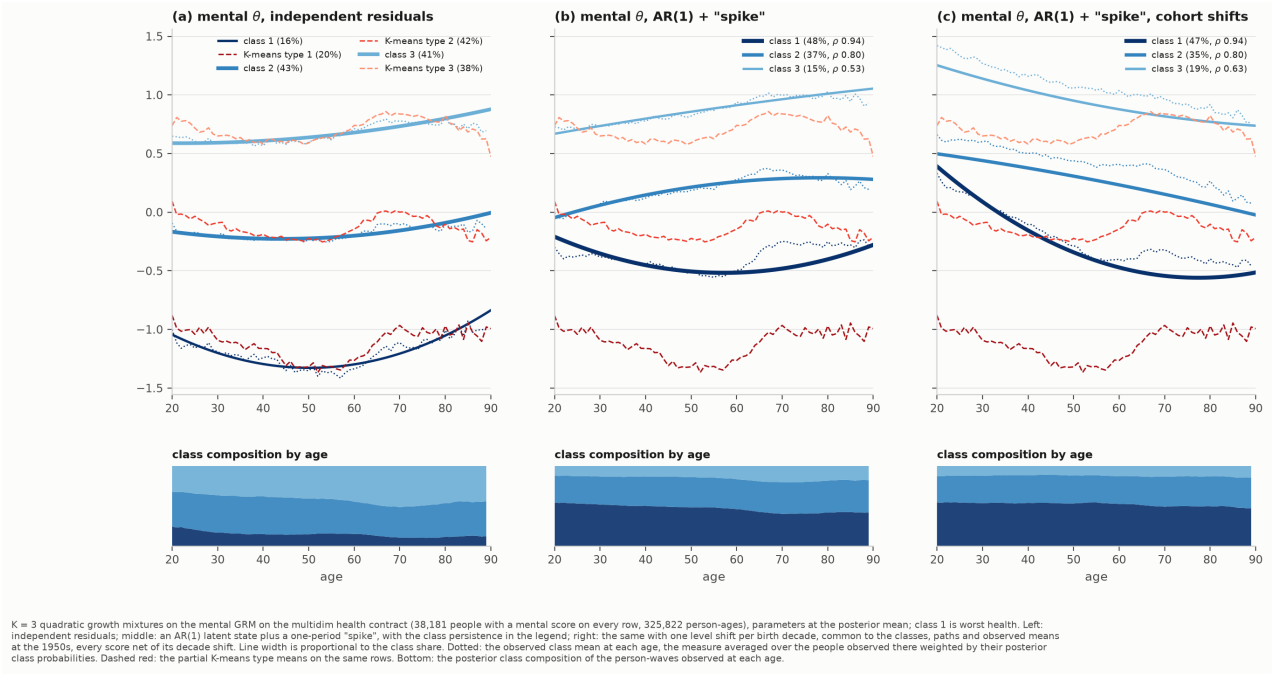


Figure H.36: As Figure 5, between versus within by age, on the mental GRM types.

- The type shares are almost those of θ (.20/.42/.38 against .18/.43/.40), but the paths are flat and parallel, at about -1.1, -0.15 and $+0.65$, each carrying the mid-life dip and the rise after 60 of the mean path; the gaps between the types hardly move with age.
- The types are different people. Agreement with the θ typology on the 38,181 shared people is ARI 0.14 with 58% in the same type (0.14 and 54% against h), against ARI 0.37–0.71 between the physical measures.
- The between-type share of the variance is 45–52% at every age, against a rise from 49% at 25 to 64–66% at 55–70 on θ. Within-type autocorrelation at one year is 0.26–0.39 against 0.37–0.64 on θ, and is near zero by six years for the middle type: the persistent part of mental health is smaller, consistent with the covariance rows of Figure H.33.
- The level-change covariance across types is positive from the late twenties to the late fifties and negative from about 58 to 70, then small: the types fan out slowly in mid-life and close up around retirement, when the mean rises.



K = 3 quadratic growth mixtures on the mental GRM on the multidim health contract (38,181 people with a mental score on every row, 325,822 person-ages), parameters at the posterior mean; class 1 is worst health. Left: independent residuals; middle: an AR(1) latent state plus a one-period "spike", with the class persistence in the legend; right: the same with one level shift per birth decade, common to the classes, paths and observed means at the 1950s, every score net of its decade shift. Line width is proportional to the class share. Dotted: the observed class mean at each age, the measure averaged over the people observed there weighted by their posterior class probabilities. Dashed red: the partial K-means type means on the same rows. Bottom: the posterior class composition of the person-waves observed at each age.

Figure H.37: As Figure 6, $K = 3$ mixtures on the mental GRM, on the multidimensional contract's rows: (a) independent residuals, (b) an AR(1) latent state plus a one-period "spike", (c) the same with one level shift per birth decade common to the classes, paths and observed class means at the 1950s with every score net of its decade shift. Dashed red: the K-means types of Figure H.35.

- Under independent residuals the mixture reproduces the K-means types: shares .16/.43/.41 against .20/.42/.38, class paths at about −1.3, −0.2 and +0.6 at 50, each with the dip to 50 and the rise after 60.
- Under AR(1) plus "spike" the classes are a different partition. The persistence is .94/.80/.53 with a "spike" scale of 0.54, so the signal shares are 0.72/0.45/0.29, and the largest class (48%) is the lowest, running from −0.2 at 20 to −0.5 at 55 and back; nothing sits near the K-means worst type at −1. With half of the variance one-period, the mixture separates people by how persistent their mental score is rather than by its level: the small top class (15%) is mostly "spike". The class composition is flat in age.
- With birth-decade shifts the shares and persistence barely move (.47/.35/.19; .94/.80/.63), but the shifts are large: relative to the 1950s, +0.2 for the 1920s–40s and −0.9 for the 2000s, 1.1 sd across the 1940s–2000s, twice the physical shift. At the 1950s every class path then declines with age, so the rise after 60 in Figure H.33 is being read as the older cohorts' higher level. The age–cohort caveat of Section 4.3 applies with more force here, since the whole post-60 rise sits inside the fifteen-year window where age and birth year are nearly collinear.